AGM 101

Fundamentals of Microbiology

Lecture-wise Theory Study Material

B.Sc. (Hons) Agriculture - First Year, First Semester

2025 Syllabus - Tamil Nadu Agricultural University (TNAU)

What this material covers

All 33 lectures of the AGM 101 syllabus in exact syllabus order, with labelled diagrams, comparison tables, fully worked numerical examples, agricultural applications, and an Exam-focus section in every lecture. A glossary and a list of standard reference texts are given at the end.

Prepared as exam-oriented theory notes. Standard ranges are given where exact values vary between textbooks.

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PART I - INTRODUCTION TO MICROBIOLOGY

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Lecture 1
Definition and Scope of Microbiology

Definition and Scope of Microbiology

1.1 What is microbiology

Microbiology is the branch of biology that deals with microorganisms - living organisms that are too small to be seen clearly by the unaided human eye and are therefore studied with the help of a microscope. The word is derived from three Greek roots: *mikros* (small), *bios* (life) and *logos* (study). Thus microbiology literally means the study of very small life forms.

Definition

Microbiology is the science that studies the structure, physiology, reproduction, genetics, classification, ecology and economic importance of microorganisms, including bacteria, fungi, algae, protozoa and viruses.

A microorganism (or microbe) is generally taken as an organism whose individual cells are of the order of one micrometre (1 um = 1/1000 mm) in size, so that a microscope is needed to observe them. Viruses are even smaller - they are measured in nanometres and are visible only under an electron microscope. Viruses are not strictly "living" in the metabolic sense; they are best described as acellular infectious agents that multiply only inside a living host cell.

1.2 The main groups of microorganisms

- **Bacteria** - prokaryotic, single-celled, no true nucleus; examples: *Bacillus*, *Rhizobium*, *Escherichia coli*.
- **Fungi** - eukaryotic; include moulds (multicellular filaments) and yeasts (single cells); examples: *Aspergillus*, *Penicillium*, *Saccharomyces*.
- **Algae** - eukaryotic, photosynthetic; include *Chlorella*, *Anabaena* (the latter is actually a cyanobacterium) and *Spirogyra*.
- **Protozoa** - eukaryotic, single-celled, mostly motile; examples: *Amoeba*, *Paramecium*, *Plasmodium*.
- **Viruses** - acellular, obligate intracellular parasites; examples: tobacco mosaic virus, bacteriophages, influenza virus.

1.3 Scope of microbiology

The scope of microbiology is very wide because microbes touch almost every sphere of human activity. The main applied branches are described below.

1.3.1 Agricultural microbiology

Deals with microbes of the soil and of plants. It studies:

- Soil fertility and the role of microbes in nutrient cycling - carbon, nitrogen, phosphorus and sulphur cycles.

- Biological nitrogen fixation by *Rhizobium* in legume root nodules, and by free-living *Azotobacter* and *Azospirillum*.
- Biofertilizers such as *Rhizobium*, *Azotobacter*, *Azospirillum*, phosphate-solubilising bacteria and blue-green algae.
- Plant pathogens - *Xanthomonas oryzae* (bacterial blight of rice), *Ralstonia solanacearum* (wilt), *Puccinia graminis* (wheat rust, a fungus), and their management.
- Biocontrol agents such as *Trichoderma*, *Bacillus subtilis* and *Pseudomonas fluorescens*.
- Composting and decomposition of organic matter, biogas production and mushroom cultivation.

1.3.2 Industrial and food microbiology

- Production of alcohol, acetic acid, citric acid, lactic acid, antibiotics (penicillin, streptomycin), enzymes and vitamins.
- Dairy microbiology - curd, butter, cheese, ghee and yoghurt; starter cultures of *Lactobacillus* and *Streptococcus*.
- Food spoilage and food preservation; detection of food-borne pathogens.

1.3.3 Medical and veterinary microbiology

- Causative agents of human diseases - tuberculosis, cholera, typhoid, tetanus, diphtheria, plague.
- Antibiotics, vaccines and antisera.
- Animal diseases of agricultural importance - anthrax, brucellosis, foot-and-mouth disease.

1.3.4 Environmental and other branches

- Environmental microbiology - sewage treatment, bioremediation of oil spills and pesticide residues, indicators of water pollution.
- Soil microbiology, marine microbiology, air microbiology, geomicrobiology.
- Immunology, microbial genetics, genetic engineering and biotechnology.
- Exomicrobiology and space microbiology - study of microbes in extreme environments.

1.4 Why microbiology is important to agriculture

Agriculture depends heavily on microbial activity. Soil is not merely an inert medium; it is a living system containing millions of microbes per gram. These microbes decompose organic residues, fix atmospheric nitrogen, solubilise insoluble phosphates, produce plant growth hormones, suppress soil-borne pathogens and help in the formation of humus. A student of agriculture must therefore understand microbiology to manage soil health, plant nutrition and plant disease.

Table 1.1 - Major groups of microorganisms and their key features

Group	Cell type	Approx. size	Nutrition	Example
Bacteria	Prokaryotic	0.5-5 um	Autotrophic or heterotrophic	*Rhizobium*
Fungi	Eukaryotic	2-10 um (hyphae longer)	Heterotrophic (absorptive)	*Aspergillus*
Algae	Eukaryotic	2-100 um	Photoautotrophic	*Chlorella*
Protozoa	Eukaryotic	10-200 um	Heterotrophic (ingestive)	*Amoeba*
Viruses	Acellular	20-300 nm	Obligate parasite	Influenza virus

Exam focus

Key points to remember

- Microbiology is the study of organisms too small to be seen with the naked eye.
- Five major groups: bacteria, fungi, algae, protozoa and viruses.
- Bacteria are prokaryotic; fungi, algae and protozoa are eukaryotic; viruses are acellular.
- Applied branches include agricultural, industrial, food, medical, veterinary and environmental microbiology.
- Agricultural microbiology covers nitrogen fixation, biofertilizers, plant pathogens and biocontrol.
- Soil contains millions of microbes per gram and their activity governs soil fertility.

Short-answer questions with model answers

1. Define microbiology and give its scope.

Microbiology is the branch of biology dealing with microorganisms - organisms too small to be seen clearly with the unaided eye and studied with a microscope. Its scope includes agricultural microbiology (soil fertility, nitrogen fixation, biofertilizers, plant diseases), industrial microbiology (antibiotics, organic acids, enzymes), food and dairy microbiology, medical and veterinary microbiology, and environmental microbiology (sewage treatment, bioremediation).

2. Why are viruses placed separately from other microorganisms?

Viruses are acellular, contain only one type of nucleic acid (DNA or RNA) enclosed in a protein coat, and have no independent metabolism. They multiply only inside a living host cell. Hence they are not considered true cells and are studied separately.

3. Name any four biofertilizers used in Indian agriculture.

Rhizobium, *Azotobacter*, *Azospirillum* and phosphate-solubilising bacteria (*Bacillus*/*Pseudomonas*). Blue-green algae are also used, especially in paddy fields.

Long-answer questions with model answer outlines

1. Discuss the scope of microbiology with special reference to agriculture.

Outline: (i) Definition and general scope. (ii) Agricultural microbiology - soil fertility, nutrient cycling, nitrogen fixation by *Rhizobium*, biofertilizers. (iii) Plant pathology - bacterial and fungal diseases and their management. (iv) Biocontrol and composting. (v) Industrial, food and dairy microbiology. (vi) Medical and

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veterinary microbiology. (vii) Environmental microbiology and bioremediation. (viii) Conclusion on the importance of microbiology in modern agriculture.

2. Classify microorganisms and describe the characteristics of each group.

Outline: (i) Basis of classification - cell type, size, nutrition. (ii) Bacteria - prokaryotic, 0.5-5 um. (iii) Fungi - eukaryotic, heterotrophic, moulds and yeasts. (iv) Algae - eukaryotic, photosynthetic. (v) Protozoa - eukaryotic, motile, ingestive. (vi) Viruses - acellular, obligate parasites. (vii) Summary table.

Lecture 2
Biogenesis and Abiogenesis Theory; Contributions of Antonie van Leeuwen...

Biogenesis and Abiogenesis Theory; Contributions of Antonie van Leeuwenhoek and Louis Pasteur

2.1 The central question: where does life come from

For centuries people believed that living organisms could arise spontaneously from non-living matter. This belief is called **abiogenesis** or **spontaneous generation**. For example, maggots were thought to arise from rotting meat, and frogs from mud. The opposite view, that living organisms arise only from pre-existing living organisms, is called **biogenesis**. The debate between these two ideas shaped the early history of microbiology.

2.2 Abiogenesis (theory of spontaneous generation)

- Abiogenesis means the origin of living things from non-living material.
- Supported in ancient times by Aristotle and later by the doctrine of *vital force* - the idea that a special life force present in air or organic matter produced life.
- Common "evidence" quoted: maggots on decaying meat, mice from wheat and dirty cloth, microorganisms appearing in broth.

2.3 Biogenesis

- Biogenesis means that life arises only from pre-existing life (*omne vivum ex vivo* - all life from life).
- The theory was established by a series of careful experiments.

2.4 Key experiments that disproved spontaneous generation

2.4.1 Francesco Redi (1668)

Placed meat in jars. Open jars developed maggots; jars covered with gauze did not, although flies gathered on the gauze. This showed maggots came from eggs laid by flies, not from the meat itself.

2.4.2 Lazzaro Spallanzani (1765)

Boiled broth in sealed flasks. No microorganisms appeared. Supporters of abiogenesis argued that boiling destroyed the "vital force" in air and that sealing prevented fresh air from entering. The argument remained unsettled.

2.4.3 Louis Pasteur (1861)

Pasteur designed the famous **swan-necked flask** experiment. Broth was boiled in a flask whose neck was drawn into a long S-shaped curve. Air could move freely into the flask, but dust particles carrying microorganisms settled in the bend of the neck and never reached the broth. The broth remained sterile indefinitely. When the neck was broken off, or the flask tilted so that broth touched the trapped dust, the broth quickly became cloudy with microorganisms. This proved conclusively that microorganisms come

from pre-existing microorganisms carried in air, and not from the broth itself.

2.5 Antonie van Leeuwenhoek (1632-1723)

- A Dutch draper and amateur lens maker, regarded as the **father of microbiology**.
- He ground his own single-lens microscopes, which could magnify about 200 to 300 times.
- In 1676 he first observed and described bacteria, which he called *animalcules*, from samples of rain water, pond water, his own teeth scrapings and pepper infusions.
- He was the first to describe protozoa, yeasts, and the movement of spermatozoa.
- He reported his findings in letters to the Royal Society of London.

2.6 Louis Pasteur (1822-1895)

- A French chemist and microbiologist, regarded as the **father of modern microbiology** and of immunology.
- Disproved spontaneous generation with the swan-neck flask experiment.
- Discovered that fermentation is caused by living microorganisms (yeasts for alcohol, bacteria for vinegar and for souring of wine). This led to the germ theory of fermentation.
- Solved the "diseases of wine" problem by heating wine briefly - the process now called **pasteurisation**, later applied to milk.
- Developed the first attenuated vaccines, for chicken cholera, anthrax in sheep and rabies in humans.
- Introduced the terms and methods of sterilisation and pure culture that became the foundation of microbiology.

Summary of the biogenesis-abiopegnesis debate

The controversy ended with Pasteur's swan-neck flask experiment, which showed that microorganisms in broth come from the air and not from the broth. The principle established is now the basis of aseptic technique: to keep a medium sterile, exclude living contaminants.

2.7 Agricultural relevance

The principle of biogenesis underlies the practice of aseptic culture in agriculture. Pure cultures of *Rhizobium* for biofertilizer production, of *Trichoderma* for biocontrol, and of mushroom spawn, must all be protected from contamination. Pasteurisation is applied to milk, fruit juices and other farm products to destroy spoilage and pathogenic microbes while preserving quality.

Table 2.1 - Abiogenesis vs biogenesis

Point	Abiogenesis	Biogenesis
Meaning	Life arises from non-living matter	Life arises only from pre-existing life
Chief supporters	Aristotle, van Helmont, Needham	Redi, Spallanzani, Pasteur
Key experiment	Appearance of maggots/microbes in open material	Redi's meat, Spallanzani's sealed broth, Pasteur's swan-neck flask
Present status	Disproved	Accepted

Exam focus

Key points to remember

- Abiogenesis = spontaneous generation; biogenesis = life from life.
- Redi (1668) used meat and gauze to disprove maggots arising from meat.
- Spallanzani (1765) boiled broth in sealed flasks.
- Pasteur (1861) gave the swan-neck flask experiment, finally disproving abiogenesis.
- Leeuwenhoek is the father of microbiology; Pasteur is the father of modern microbiology.
- Pasteur discovered that fermentation is caused by microorganisms and introduced pasteurisation.

Short-answer questions with model answers

1. Distinguish between abiogenesis and biogenesis.

Abiogenesis (spontaneous generation) is the theory that living organisms arise from non-living matter. Biogenesis is the theory that living organisms arise only from pre-existing living organisms. Abiogenesis was disproved by Pasteur's swan-neck flask experiment; biogenesis is the accepted theory.

2. Describe Pasteur's swan-neck flask experiment.

Pasteur boiled nutrient broth in a flask with a long S-shaped neck open to air. Dust and microbes settled in the bend and could not reach the broth, so the broth stayed sterile. When the neck was broken or the flask tilted so broth touched the dust, growth appeared. This proved microbes come from the air, not from the broth.

3. List the contributions of Antonie van Leeuwenhoek.

He built powerful single-lens microscopes (200-300x), first observed and described bacteria (which he called animalcules), protozoa and yeasts, and reported them to the Royal Society. He is called the father of microbiology.

Long-answer questions with model answer outlines

1. Trace the history of the abiogenesis-biogenesis controversy.

Outline: (i) Meaning of both theories. (ii) Aristotle and the vital force doctrine. (iii) Redi's meat experiment. (iv) Spallanzani's sealed-flask experiment and its criticism. (v) Pasteur's swan-neck flask experiment and its conclusiveness. (vi) Significance for aseptic technique. (vii) Conclusion.

2. Describe the contributions of Louis Pasteur to microbiology.

Outline: (i) Disproof of spontaneous generation. (ii) Germ theory of fermentation. (iii) Pasteurisation. (iv) Attenuated vaccines - chicken cholera, anthrax, rabies. (v) Pure culture and sterilisation methods. (vi) His title as father of modern microbiology. (vii) Agricultural applications of his work.

Lecture 3
Contributions of John Tyndall, Joseph Lister, Edward Jenner, Robert Koc...

Contributions of John Tyndall, Joseph Lister, Edward Jenner, Robert Koch, Alexander Fleming, Waksman, Enders and Hilleman; Germ Theory of Diseases, Fermentation and Origin of Life

3.1 John Tyndall (1820-1893)

- An Irish physicist who explained why some boiled infusions remained sterile while others did not.
- He showed that repeated boiling on successive days destroyed heat-resistant forms - the basis of **tyndallisation** (intermittent sterilisation).
- Discovered that some bacteria exist as heat-resistant **endospores**, explaining the earlier failures of Spallanzani-type experiments.
- Tyndallisation is used for media that are damaged by high temperature: the medium is steamed at 100 C for 20-30 minutes on three successive days; spores that germinate between treatments are killed on the next heating.

3.2 Joseph Lister (1827-1912)

- An English surgeon, the father of **antiseptic surgery**.
- Introduced the use of phenol (carbolic acid) to kill microbes in the operating theatre, on instruments and on wounds, greatly reducing post-operative infection.
- His work, inspired by Pasteur's germ theory, established the principle of asepsis in medicine and surgery.

3.3 Edward Jenner (1749-1823)

- An English physician, the father of **vaccination**.
- In 1796 he observed that milkmaids who had cowpox did not catch smallpox.
- He inoculated a boy with material from cowpox lesions and later exposed him to smallpox - the boy did not develop the disease.
- He introduced the term **vaccine** (from *vacca*, cow). His work laid the foundation of immunology.

3.4 Robert Koch (1843-1910)

- A German physician, one of the founders of medical microbiology and bacteriology.
- Isolated and identified the causative organisms of anthrax (*Bacillus anthracis*, 1876), tuberculosis (*Mycobacterium tuberculosis*, 1882) and cholera (*Vibrio cholerae*, 1883).
- Established **Koch's postulates**, the rules for proving that a particular microorganism causes a particular disease.

- Introduced pure culture techniques on solid media - first gelatin, then agar - and the use of the petri dish.
- Won the Nobel Prize in 1905 for his work on tuberculosis.

3.4.1 Koch's postulates

– The suspected organism must be found in all cases of the disease and its distribution must correspond to the lesions.

– The organism must be isolated and grown in pure culture in the laboratory.

– The pure culture, when inoculated into a healthy susceptible host, must produce the same disease.

– The same organism must be re-isolated from the experimentally infected host.

3.5 Alexander Fleming (1881-1955)

- A Scottish bacteriologist who discovered **penicillin** in 1928.
- He noticed that a mould contaminating a culture of *Staphylococcus* had dissolved the bacteria around it. The mould was *Penicillium notatum*.
- The active substance was named penicillin - the first antibiotic. He shared the Nobel Prize in 1945 with Florey and Chain, who made it available as a medicine.
- The discovery began the antibiotic era and saved millions of lives.

3.6 Selman Waksman (1888-1973)

- An American soil microbiologist who coined the term **antibiotic**.
- Systematically screened soil actinomycetes for antibacterial substances and discovered **streptomycin** (1943) from *Streptomyces griseus*.
- Streptomycin was the first effective drug against tuberculosis. He received the Nobel Prize in 1952.

3.7 John Franklin Enders (1897-1985)

- An American virologist who, with Weller and Robbins, developed a method to grow poliovirus in non-nervous tissue culture (1949).
- This made possible the large-scale production of polio vaccine by Salk and Sabin.
- Shared the Nobel Prize in 1954. His work advanced the whole field of cell culture and vaccine production.

3.8 Maurice Hilleman (1919-2005)

- An American microbiologist, regarded as one of the most prolific vaccine developers of the twentieth century.
- Developed more than 40 vaccines, including those for measles, mumps, rubella, hepatitis A, hepatitis B, chickenpox and meningitis.
- The combined MMR vaccine is largely his work. His vaccines save an estimated several million lives each year.

3.9 Germ theory of disease

The germ theory states that many diseases are caused by the invasion of the body by microorganisms, which then multiply and produce harmful effects. Before this theory, disease was attributed to bad air (*miasma*) or to imbalance of body humours.

- Key contributors: Pasteur (fermentation and putrefaction by microbes), Koch (specific organism for a specific disease), Lister (antisepsis), and later many others.
- The theory is proved for a particular disease by applying Koch's postulates.
- Agricultural significance: the same theory explains plant diseases - bacterial blight of rice by *Xanthomonas oryzae*, citrus canker by *Xanthomonas citri*, and crown gall by *Agrobacterium tumefaciens*.

3.10 Fermentation and the origin of life

- Pasteur showed that fermentation (conversion of sugar to alcohol or acid) is carried out by living microorganisms, and that each type of fermentation has its own microbe.
- Buchner later showed that cell-free extracts could also ferment sugar, proving that fermentation is an enzyme-mediated chemical process, not a "vital" property of the cell.
- Origin of life: the modern view is that life first arose chemically from simple inorganic molecules on the early earth - the **chemical evolution** theory associated with Oparin and Haldane, supported by the Miller-Urey experiment which produced amino acids from simple gases and electrical discharge.
- The first cells were probably anaerobic, fermenting heterotrophs; photosynthesis later released oxygen, allowing aerobic life.

Table 3.1 - Great contributors and their contributions

Scientist	Contribution
John Tyndall	Endospores; tyndallisation (intermittent sterilisation)
Joseph Lister	Antiseptic surgery using phenol
Edward Jenner	First vaccine (smallpox, using cowpox); father of vaccination
Robert Koch	Germ theory; Koch's postulates; causes of anthrax, TB, cholera
Alexander Fleming	Discovery of penicillin
Selman Waksman	Coined "antibiotic"; discovered streptomycin
J. F. Enders	Poliovirus in tissue culture; enabled polio vaccine
Maurice Hilleman	Over 40 vaccines, including MMR

Exam focus

Key points to remember

- Tyndall explained endospores and gave tyndallisation.
- Lister is the father of antiseptic surgery; Jenner, of vaccination.
- Koch gave Koch's postulates and identified the causes of anthrax, TB and cholera.
- Fleming discovered penicillin (1928); Waksman discovered streptomycin and coined "antibiotic".

- Enders grew poliovirus in tissue culture; Hilleman developed many vaccines including MMR.
- Germ theory: specific microbes cause specific diseases.

Short-answer questions with model answers

1. State Koch's postulates.

(i) The organism must be present in all cases of the disease. (ii) It must be isolated and grown in pure culture. (iii) The pure culture must reproduce the disease when inoculated into a healthy host. (iv) The same organism must be re-isolated from the experimentally infected host.

2. What is tyndallisation? Why is it used?

Tyndallisation is intermittent sterilisation by steaming at 100 C for 20-30 minutes on three successive days. It is used for media that are damaged by autoclaving; heat-resistant spores that germinate between the treatments are killed in the next heating.

3. Write a note on the discovery of antibiotics.

Fleming discovered penicillin from *Penicillium notatum* in 1928. Waksman coined the term antibiotic and discovered streptomycin from *Streptomyces griseus* in 1943. These discoveries began the antibiotic era.

Long-answer questions with model answer outlines

1. Explain the germ theory of disease and the contributions that established it.

Outline: (i) Meaning of the germ theory. (ii) Earlier miasma theory. (iii) Pasteur's evidence from fermentation. (iv) Koch's postulates and identification of specific pathogens. (v) Lister's antisepsis. (vi) Application to plant diseases. (vii) Conclusion.

2. Describe the contribution of five Nobel-winning microbiologists to medicine and agriculture.

Outline: (i) Koch - TB, postulates. (ii) Fleming - penicillin. (iii) Waksman - streptomycin. (iv) Enders - tissue culture and polio vaccine. (v) Hilleman - MMR and other vaccines. (vi) Their combined impact on human health and on the control of animal and plant diseases.

Lecture 4
Microscopy - Principles: Resolving Power and Magnification; Light Micro...

Microscopy - Principles: Resolving Power and Magnification; Light Microscopy

4.1 Introduction

Because microorganisms are very small, their study depends on the microscope. A microscope produces an enlarged image of an object so that structures not visible to the naked eye can be examined. Two ideas are central to understanding any microscope: **magnification** and **resolving power**.

4.2 Magnification

- Magnification is the ratio of the size of the image to the size of the object; it tells how many times the object is enlarged.
- The total magnification of a compound microscope = magnification of the objective lens x magnification of the eyepiece (ocular).
- Typical objectives are 4x, 10x, 40x and 100x; the eyepiece is usually 10x.
- Thus a 100x oil-immersion objective with a 10x eyepiece gives 1000x total magnification.
- Magnification alone is not useful: an image may be large but blurred. Useful magnification requires good resolution.

4.3 Resolving power (resolution)

- Resolving power is the smallest distance between two points at which they can still be seen as two separate points and not as one blurred spot.
- It is the most important property of a microscope, more important than magnification.
- The resolving power of a lens is given by the formula:

Abbe's formula for resolving power

d = 0.61 x wavelength / (n sin theta)

where d = the smallest resolvable distance, wavelength is the wavelength of the light used, n is the refractive index of the medium between the lens and the specimen, and theta is half the angle of the cone of light entering the objective. The term n sin theta is called the numerical aperture (NA) of the objective.

- To improve resolution, either decrease the wavelength of light or increase the numerical aperture.
- The best light microscope has a numerical aperture of about 1.25-1.4 and resolves about 0.2 um (200 nm) with blue light. Two points closer than about 0.2 um cannot be separated by a light microscope.
- This is why viruses (20-300 nm) cannot be seen with a light microscope but can be seen with an electron microscope, which uses electrons of much shorter wavelength.

Worked example 4.1 - Resolving power calculation

Problem: Calculate the resolving power of an objective of numerical aperture 1.25 used with green light of wavelength 550 nm. (sin theta term already included in NA.)

Step 1: Write the formula. d = 0.61 x wavelength / NA.

Step 2: Substitute values. d = 0.61 x 550 nm / 1.25.

Step 3: Compute. d = 335.5 / 1.25 = 268.4 nm = 0.268 um.

Answer: The microscope can just resolve two points about 0.27 um apart. If blue light of 450 nm is used instead, d = 0.61 x 450 / 1.25 = 219.6 nm = 0.22 um, a clear improvement - showing why blue light gives the best resolution.

4.4 Light microscopy

The ordinary compound light microscope uses visible light (about 400-700 nm) as the source of illumination and glass lenses to form the image. It is the workhorse of the microbiology laboratory.

4.4.1 Optical parts

- **Eyepiece (ocular)** - the lens nearest the eye; usually 10x or 15x; it further magnifies the image formed by the objective.
- **Objective lenses** - the lenses nearest the specimen; a nosepiece carries several objectives (4x, 10x, 40x, 100x).
- **Condenser** - a lens below the stage that concentrates light from the source onto the specimen.
- **Iris diaphragm** - controls the amount of light passing through the condenser; adjusts contrast.

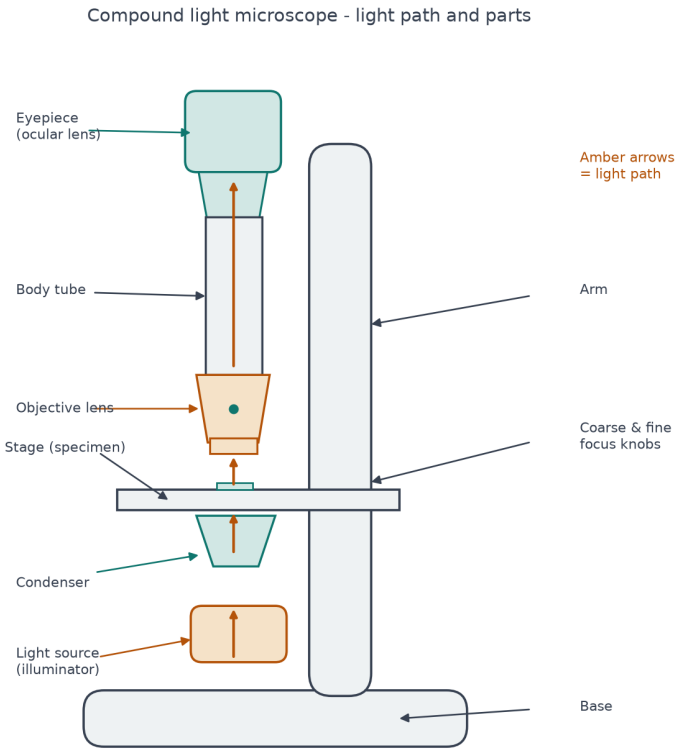


Figure 4.1 - Compound light microscope: parts and the path of light from the source, through the condenser and specimen, to the objective and eyepiece.

4.4.2 Mechanical parts

- **Base** - supports the microscope.
- **Arm** - connects the base to the body tube; used to carry the microscope.
- **Stage** - the platform on which the slide is placed; stage clips hold the slide.
- **Body tube** - holds the eyepiece and objectives at the correct distance.
- **Coarse and fine adjustment knobs** - move the tube or stage up and down for focusing.

4.4.3 Working principle

Light from the source passes through the condenser, which focuses it onto the specimen. The specimen absorbs and scatters some of the light. The objective lens collects the light and forms a magnified real image, which the eyepiece further magnifies into a virtual image seen by the eye. Total magnification is the product of the objective and eyepiece magnifications.

4.4.4 Bright-field microscopy

- The usual form of light microscopy; the specimen appears dark against a bright background.
- Requires staining of the specimen for contrast, because most microbes are nearly transparent.
- Used for examining stained smears of bacteria, fungi and other microbes.

4.4.5 Oil immersion

- For the 100x objective, a drop of immersion oil (refractive index about 1.515, same as glass) is placed between the objective and the slide.

- The oil prevents refraction of light rays away from the objective and increases the numerical aperture, giving higher resolution and a brighter image.

4.4.6 Care and use

- Always start focusing with the low-power objective and then change to higher powers.
- Use the coarse adjustment only with low power; use the fine adjustment at high power.
- Keep lenses clean with lens tissue; never use rough cloth.
- Store the microscope with the low-power objective in position and the cover on.

Table 4.1 - Magnification of common objective and eyepiece combinations

Objective	Eyepiece	Total magnification	Typical use
4x	10x	40x	Scanning, locating the specimen
10x	10x	100x	Low-power examination
40x	10x	400x	High-power study of cell shape
100x (oil)	10x	1000x	Detailed study, bacteria, oil immersion

Exam focus

Key points to remember

- Magnification = objective x eyepiece.
- Resolving power is more important than magnification; d = 0.61 x wavelength / NA.
- The best light microscope resolves about 0.2 um; this is why viruses need an electron microscope.
- Numerical aperture and wavelength govern resolution.
- Oil immersion raises NA and improves resolution at 100x.
- Bright-field microscopy needs stained specimens for contrast.

Short-answer questions with model answers

1. Define resolving power. Give the formula.

Resolving power is the smallest distance at which two points can be distinguished as separate. It is given by d = 0.61 x wavelength / (n sin theta), where n sin theta is the numerical aperture.

2. Why is oil used with the 100x objective?

Immersion oil has the same refractive index as glass, so it prevents light rays from being refracted away from the objective. This increases the numerical aperture, giving higher resolution and a brighter image.

3. Distinguish magnification from resolution.

Magnification is how many times the object is enlarged; resolution is the ability to show two close points as separate. A large magnification without good resolution gives a blurred image; hence resolution is more important.

Long-answer questions with model answer outlines

1. Describe the compound light microscope with a labelled diagram.

Outline: (i) Definition and use. (ii) Optical parts - eyepiece, objectives, condenser, diaphragm. (iii) Mechanical parts - base, arm, stage, body tube, adjustment knobs. (iv) Path of light. (v) Total magnification. (vi) Bright-field and oil immersion. (vii) Diagram (Figure 4.1).

2. Explain the principles of magnification and resolving power in microscopy.

Outline: (i) Magnification definition and calculation. (ii) Resolving power definition. (iii) Abbe's formula. (iv) Numerical aperture and wavelength. (v) Limiting resolution of light microscope (0.2 um). (vi) Worked example. (vii) Why electron microscopy is needed for viruses.

Lecture 5
Different Types of Microscopes - UV, Dark Field, Phase Contrast and Flu...

Different Types of Microscopes - UV, Dark Field, Phase Contrast and Fluorescence

5.1 Need for special microscopes

The ordinary bright-field microscope shows stained, killed specimens. But many microbes are best studied alive, unstained and in their natural state. Special microscopes have therefore been developed to increase contrast or resolution by means other than simple staining.

5.2 Ultraviolet (UV) microscope

- Uses ultraviolet light (about 200-400 nm) instead of visible light.
- Because resolution depends on wavelength (d = 0.61 x wavelength / NA), the shorter wavelength of UV light roughly doubles the resolution compared with visible light.
- Ordinary glass absorbs UV, so quartz lenses and quartz slides are used.
- The image is invisible to the human eye; it is recorded on photographic film or by a detector.
- Uses: study of fine cell structure, nucleic acids (which absorb UV strongly at 260 nm), and detection of some microorganisms.

5.3 Dark-field (dark-ground) microscope

- The specimen is brightly illuminated against a dark background - the reverse of bright field.
- A special condenser (cardioid or paraboloid) with an opaque stop allows only a hollow cone of light to strike the specimen; direct light is blocked and only light scattered by the specimen reaches the objective.
- The specimen appears self-luminous (bright) on a black background.
- No staining is required; living cells can be examined.
- Uses: observation of living spirochaetes such as *Treponema*, motility of bacteria, flagella, and unstained cells in wet mounts. Also used to detect *Leptospira* in veterinary samples.

5.4 Phase-contrast microscope

- Converts differences in refractive index (optical density) within the specimen into differences in brightness (contrast).
- Living, unstained cells have slightly different refractive indices in different parts; light passing through them is retarded by about a quarter of a wavelength.
- An annular diaphragm in the condenser and a matching phase plate in the objective shift the phase of the light so that these small differences appear as light and dark contrast.
- The specimen appears with internal detail - cell walls, vacuoles, spores, nuclei - without staining.

- Uses: study of living microorganisms, their growth, division, sporulation, and the effect of drugs on living cells. Very useful for examining soil bacteria and fungi directly.

5.5 Fluorescence microscope

- Uses the phenomenon of fluorescence: certain substances absorb light of one (shorter) wavelength and emit light of a longer wavelength.
- The specimen is stained with a fluorescent dye (fluorochrome) such as fluorescein isothiocyanate (FITC) or auramine, or the specimen is naturally fluorescent.
- UV or blue light excites the dye; the emitted visible light forms the image. Filters block the exciting light so that only the fluorescent image is seen.
- The object appears as a bright, glowing structure against a dark background.
- Uses: fluorescent antibody (immunofluorescence) techniques for rapid identification of pathogens; auramine stain for *Mycobacterium tuberculosis*; study of nucleic acids and chromosomes; detection of *Rhizobium* in root nodules; viability staining in soil and food microbiology.

5.6 Comparison of light microscope types

Table 5.1 - Comparison of microscope types

Microscope	Source of light	Contrast mechanism	Live or stained	Main use
Bright field	Visible light	Absorption by stained specimen	Usually stained, dead	General examination
Dark field	Visible light	Scattered light	Live, unstained	Spirochaetes, motility
Phase contrast	Visible light	Refractive index difference	Live, unstained	Living cell structure, growth
UV	Ultraviolet	Absorption; shorter wavelength	Stained	Fine structure, nucleic acids
Fluorescence	UV/blue light	Fluorescence of dye	Stained with fluorochrome	Rapid identification, immunofluorescence

5.7 Agricultural applications

- Dark-field microscopy helps in the diagnosis of animal diseases caused by spirochaetes such as leptospirosis.
- Fluorescence microscopy is used with fluorescent antibodies to detect plant pathogens such as *Xanthomonas* and *Ralstonia* in seed and soil samples, and to count *Rhizobium* in inoculants.
- Phase contrast allows direct observation of living soil bacteria and their behaviour on roots.

Exam focus

Key points to remember

- UV microscope uses quartz optics and doubles resolution.
- Dark-field microscopy shows bright objects on a dark background; used for spirochaetes and motility.

- Phase contrast converts refractive index differences into brightness differences; used for living cells.
- Fluorescence microscopy uses fluorochromes and emitted light; used in immunofluorescence.
- None of these except UV changes resolution fundamentally; all improve contrast.

Short-answer questions with model answers

1. What is the principle of dark-field microscopy?

A special condenser with an opaque stop sends only a hollow cone of light so that direct light misses the objective. Only light scattered by the specimen enters the objective, so the specimen appears bright on a dark background. It is used to see living, unstained organisms such as spirochaetes.

2. How does phase-contrast microscopy produce contrast?

Living cells have small differences in refractive index. Light passing through them is retarded in phase. An annular diaphragm and a phase plate shift this phase so that the small differences appear as visible light and dark contrast, allowing unstained living cells to be studied.

3. State two uses of the fluorescence microscope in agriculture.

Fluorescent antibody techniques detect plant pathogens such as *Xanthomonas* in seed and soil; auramine staining helps detect mycobacteria. It is also used to count *Rhizobium* in biofertilizer inoculants.

Long-answer questions with model answer outlines

1. Describe the special types of light microscopes and their uses.

Outline: (i) Need for special microscopes. (ii) UV microscope - principle, quartz optics, uses. (iii) Dark-field - condenser with stop, uses. (iv) Phase contrast - annular diaphragm and phase plate, uses. (v) Fluorescence - fluorochromes, filters, uses. (vi) Comparison table. (vii) Agricultural applications.

2. Explain how contrast is produced in dark-field, phase-contrast and fluorescence microscopy.

Outline: (i) Contrast problem in bright field. (ii) Dark field - scattered light only. (iii) Phase contrast - refractive index to brightness. (iv) Fluorescence - excitation and emission of light by dyes. (v) Comparison and applications.

Lecture 6
Electron Microscopes; Atomic and Confocal Scanning Laser Microscopy

Electron Microscopes; Atomic and Confocal Scanning Laser Microscopy

6.1 The electron microscope

The electron microscope uses a beam of electrons instead of light. Because the wavelength of electrons is very much shorter than that of visible light (of the order of 0.005 nm at high accelerating voltage), the resolving power is enormously improved - about 0.5 nm or better. This makes it possible to see viruses, macromolecules and the fine structure of cells.

- Lenses are not glass but electromagnetic coils that focus the electron beam.
- The whole column is kept under a high vacuum, because electrons are absorbed by air.
- The specimen must be very thin and dry; it must be fixed, dehydrated and often stained with heavy metals (osmium, uranium, lead) to scatter electrons.
- The image is formed on a fluorescent screen or recorded on film or by a digital detector. The image is always black and white (grey scale).
- Living specimens cannot be examined because of the vacuum and the electron beam.

6.2 Transmission electron microscope (TEM)

- The electron beam passes **through** the very thin specimen (about 50-100 nm thick).
- Electrons that pass through form the image; dense areas scatter electrons and appear dark.
- Resolution up to about 0.2-0.5 nm; magnification up to about 500,000x or more.
- Uses: internal structure of cells, cell organelles, virus morphology, bacterial cell wall layers, flagella, bacteriophages.
- Limitation: specimens must be extremely thin, and three-dimensional structure is lost.

6.3 Scanning electron microscope (SEM)

- The electron beam is focused onto the **surface** of the specimen and scans it point by point.
- Electrons reflected or emitted from the surface are collected by a detector and used to build the image.
- The image shows the three-dimensional surface topography with great depth of field.
- Resolution about 5-10 nm; magnification up to about 100,000x.
- Uses: surface features of bacteria, spores, fungal hyphae, insect surfaces, plant-microbe interactions, biofilm structure, pollen grains.
- Limitation: only surface detail; the specimen must be coated with a thin metal film (gold or gold-palladium).

6.4 Atomic force microscope (AFM)

- A scanning probe microscope that does not use lenses or electron beams at all.
- A very sharp probe (tip) mounted on a flexible cantilever is moved across the specimen surface.
- Forces between the tip and the surface deflect the cantilever; a laser and detector measure this deflection and build a three-dimensional map of the surface.
- Resolution down to the atomic or molecular level (nanometres).
- Can be used in air or liquid, so living cells and biological molecules in near-natural conditions can be studied.
- Uses: imaging DNA, proteins, bacterial surfaces, cell membranes, and measuring molecular forces.

6.5 Confocal scanning laser microscope (CSLM)

- Uses a laser beam to scan the specimen and a pinhole aperture to exclude out-of-focus light.
- Only light from a single thin plane (the focal plane) reaches the detector, so the image is very sharp and free from blur.
- By moving the focal plane step by step, a series of optical sections is obtained and combined by computer into a three-dimensional image.
- Fluorescent dyes are usually used; the specimen can be living.
- Uses: three-dimensional structure of biofilms, microbial communities on roots and in soil, location of specific molecules within cells, study of plant-microbe interactions.

6.6 Comparison of electron and other advanced microscopes

Table 6.1 - Comparison of advanced microscopes

Microscope	Probe	Resolution	Specimen	Main use
Light microscope	Visible light	~200 nm	Living or fixed	Routine examination
TEM	Electrons (through)	0.2-0.5 nm	Thin, fixed, dead	Internal fine structure
SEM	Electrons (surface)	5-10 nm	Coated, dead	Surface topography
AFM	Physical probe	~0.1-1 nm	Living or fixed	Surface at molecular level
CSLM	Laser light	~200 nm (in x-y)	Living or fixed	3-D sections, biofilms

6.7 Agricultural and research applications

- TEM is used to study virus particles causing plant diseases and bacteriophages used in phage typing.
- SEM is used to study the attachment of *Rhizobium* to root hairs and the structure of fungal spores and hyphae.
- CSLM is used to study biofilms on roots and the three-dimensional colonisation of soil particles by microbes.
- AFM is used to study the surface of spores and the interaction of bacteria with plant surfaces at the molecular level.

Exam focus

Key points to remember

- Electron microscopes use electrons of very short wavelength, giving resolution down to about 0.5 nm.
- TEM: beam passes through ultrathin specimen; internal structure. SEM: beam scans the surface; three-dimensional appearance.
- Electron microscope specimens must be dead, dry and in a vacuum.
- AFM uses a physical probe and can examine living specimens in air or liquid.
- CSLM uses a laser and pinhole to obtain sharp optical sections for 3-D reconstruction.

Short-answer questions with model answers

1. Why does the electron microscope have a much higher resolving power than the light microscope?

Because resolution depends on wavelength, and the wavelength of electrons (about 0.005 nm at high voltage) is far shorter than that of visible light (about 400-700 nm). Hence the electron microscope resolves down to about 0.5 nm against about 200 nm for a light microscope.

2. Distinguish TEM from SEM.

TEM passes electrons through a very thin specimen and shows internal structure at very high resolution (0.2-0.5 nm). SEM scans the surface with electrons and shows three-dimensional surface topography at lower resolution (5-10 nm).

3. What is the advantage of the atomic force microscope?

It uses a physical probe rather than light or electrons, so it can examine living specimens in air or liquid, gives three-dimensional surface images, and reaches molecular or atomic resolution.

Long-answer questions with model answer outlines

1. Describe the electron microscope and its two types with their uses.

Outline: (i) Principle - electrons and short wavelength. (ii) Electromagnetic lenses and vacuum. (iii) Specimen preparation. (iv) TEM - through the specimen, internal structure. (v) SEM - surface scanning, 3-D topography. (vi) Comparison table. (vii) Applications in agriculture and research.

2. Explain the principle and applications of confocal scanning laser and atomic force microscopy.

Outline: (i) Limitations of conventional microscopy. (ii) CSLM - laser, pinhole, optical sections, 3-D images, biofilm studies. (iii) AFM - cantilever probe, laser detection, molecular resolution, living specimens. (iv) Applications. (v) Comparison with electron microscopy.

PART II - MICROSCOPY, STAINING, STERILIZATION AND CULTURE

Mounish Raj · MouRa · Agri AI

Lecture 7
Staining Techniques - Principle and Types of Stains - Simple, Negative,...

Staining Techniques - Principle and Types of Stains - Simple, Negative, Differential and Structural Staining Methods

7.1 Why staining is needed

Living bacterial cells are almost transparent and have nearly the same refractive index as water. In an ordinary bright-field microscope they show very little contrast. Stains are coloured chemical compounds (dyes) that bind to cell structures and make them visible, and also help to differentiate between types of cells and their parts.

7.2 Principle of staining

- A stain is a coloured organic compound (a dye) that carries a charged group and can bind to cell material.
- **Basic dyes** carry a positive charge and bind to negatively charged cell components such as nucleic acids and proteins. Examples: crystal violet, methylene blue, safranin, malachite green, carbol fuchsin. Most bacterial stains are basic dyes.
- **Acidic dyes** carry a negative charge and bind to positively charged cell components. Examples: eosin, acid fuchsin, nigrosin, India ink. They are used in negative staining.
- The coloured part of the dye molecule is called the **chromophore**; the charged group that binds is the **auxochrome**.
- A **mordant** is a substance that fixes the dye in the cell and increases the intensity of the stain (for example, Gram's iodine, or phenol in carbol fuchsin).
- A **decolouriser** removes dye from certain cells and is the key step in differential staining (for example, alcohol in Gram staining).
- A **counterstain** is a second, contrasting dye applied to colour the cells that have lost the first stain (for example, safranin in Gram staining).
- A smear is first **heat-fixed** to kill the cells, make them adhere to the slide, and make them more permeable to the dye.

7.3 Simple staining

- Uses a single basic dye (methylene blue, crystal violet or carbol fuchsin).
- Shows the shape, size and arrangement of cells - coccus, bacillus, spirillum, and single, diplo, strepto or staphylo arrangements.
- Does not distinguish between different types of bacteria.
- Procedure: prepare smear, heat-fix, flood with dye for 30 seconds to 1 minute, wash, blot, examine under oil immersion.

7.4 Negative staining

- The cell is not stained; instead the **background** is stained dark, and the unstained cell appears as a bright, clear object.
- Uses acidic dyes such as nigrosin or India ink, which are repelled by the negatively charged cell surface and so do not enter the cell.
- Does not require heat-fixing, so the cells remain alive and their true shape and capsule are preserved.
- Uses: observing capsules, cell shape and size, and spirochaetes that are difficult to stain.

7.5 Differential staining

Differential staining uses more than one dye and separates bacteria into groups according to their staining behaviour. The most important example is Gram staining.

7.5.1 Gram staining

Developed by Hans Christian Gram in 1884. It divides bacteria into Gram-positive and Gram-negative.

– Apply **crystal violet** (primary stain) for 30 seconds - all cells become purple.

– Apply **Gram's iodine** (mordant) for 1 minute - forms a crystal violet-iodine complex inside the cells.

– Apply **alcohol or acetone-alcohol** (decolouriser) for a few seconds - Gram-positive cells retain the purple complex; Gram-negative cells lose the colour and become colourless.

– Apply **safranin** (counterstain) for 30 seconds - Gram-negative cells take up the pink/red colour; Gram-positive cells remain purple.

- Reason for the difference: Gram-positive cells have a thick peptidoglycan wall that traps the crystal violet-iodine complex when alcohol is applied. Gram-negative cells have a thin peptidoglycan layer and an outer membrane; the alcohol dissolves the outer membrane and washes the complex out.
- Uses: classification of bacteria, identification of pathogens, and choice of antibiotic.

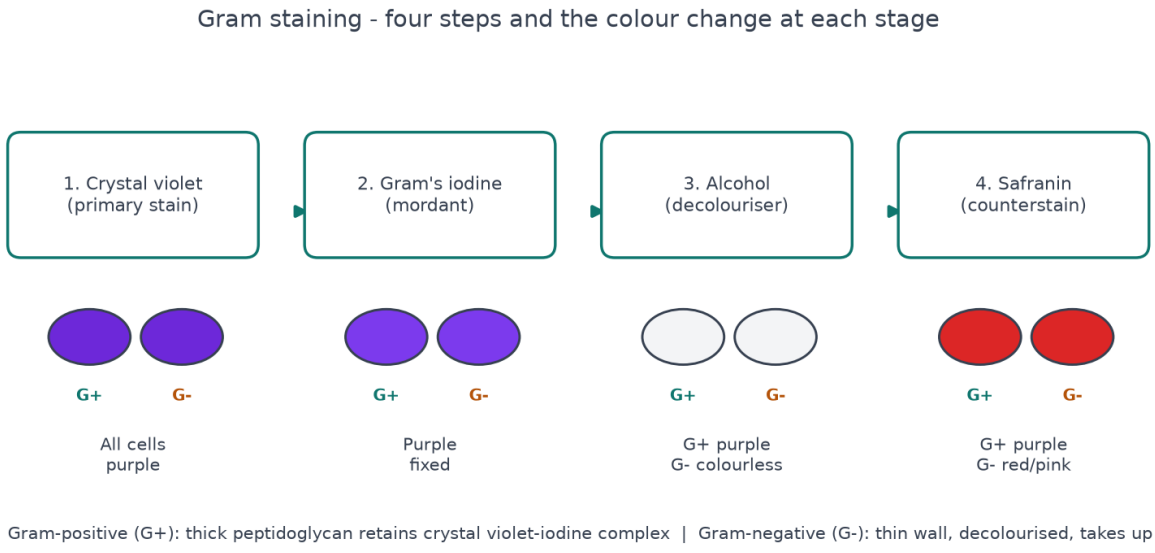


Figure 7.1 - Gram staining: the four steps and the colour of Gram-positive and Gram-negative cells at each stage.

7.5.2 Acid-fast staining (Ziehl-Neelsen)

- Used to detect mycobacteria (*Mycobacterium tuberculosis*, *M. leprae*) whose walls contain a waxy lipid, mycolic acid.
- Cells are stained with **carbol fuchsin** and heated to drive the dye through the waxy wall; they are then decolourised with acid-alcohol.
- Acid-fast cells retain the red colour; non-acid-fast cells are decolourised and counterstained blue with methylene blue.

7.6 Structural staining

Structural staining demonstrates a particular structure of the cell, such as the capsule, spore or flagella.

7.6.1 Capsule staining

- The capsule is a non-ionic, slimy outer layer that repels dyes and does not take up stains easily.
- Methods: negative staining with India ink or nigrosin, in which the capsule appears as a clear halo around the cell; or special stains such as Anthony's or Hiss's method.
- Capsules resist phagocytosis and are important in virulence.

7.6.2 Endospore staining

- Endospores are highly resistant structures and are impermeable to ordinary dyes.
- **Schaeffer-Fulton method**: a smear is flooded with **malachite green** and steamed for several minutes to drive the dye into the spore; it is then washed and counterstained with **safranin**.
- Spores appear green; the vegetative cells appear red.
- Important for identifying *Bacillus* and *Clostridium* and for checking the efficiency of sterilisation.

7.6.3 Flagella staining

- Flagella are too thin to be seen directly.
- A mordant (tannic acid and potassium alum) is used to thicken the flagella and deposit dye on them, and the stain is then applied; the flagella appear as dark, wavy filaments.
- Used to study the number and arrangement of flagella (monotrichous, lophotrichous, amphitrichous, peritrichous).

Table 7.1 - Types of staining methods

Type	Dyes used	Purpose	Example method
Simple	One basic dye	Shape, size, arrangement	Methylene blue stain
Negative	Acidic dye (nigrosin, India ink)	Capsule, cell outline, living cells	Nigrosin negative stain
Differential - Gram	Crystal violet, iodine, safranin	Group bacteria as G+ or G-	Gram stain
Differential - acid-fast	Carbol fuchsin, methylene blue	Detect mycobacteria	Ziehl-Neelsen
Structural - capsule	India ink, special stains	Demonstrate capsule	Anthony's method
Structural - spore	Malachite green, safranin	Demonstrate endospores	Schaeffer-Fulton
Structural - flagella	Mordant + dye	Demonstrate flagella	Leifson's method

7.7 Agricultural relevance

- Gram staining is a first step in identifying bacterial plant pathogens such as *Xanthomonas* (Gram-negative) and *Clavibacter* (Gram-positive).
- Endospore staining helps in testing the efficiency of sterilisation of media and in studying *Bacillus* biofertilizers such as *Bacillus subtilis*.
- Capsule staining helps in studying the virulence of animal pathogens such as *Pasteurella*.

Exam focus

Key points to remember

- Stains give colour and contrast to transparent cells; basic dyes bind negatively charged cell material.
- Simple staining uses one dye; negative staining stains the background, not the cell.
- Gram staining is a differential stain; four steps - crystal violet, iodine, alcohol, safranin.
- Gram-positive cells stay purple; Gram-negative cells become pink because of the wall structure.
- Structural stains demonstrate capsule (negative stain), spore (Schaeffer-Fulton) and flagella.
- A mordant fixes the dye; a decolouriser removes it; a counterstain colours the decolourised cells.

Short-answer questions with model answers

1. Give the principle of Gram staining.

Gram-positive cells have a thick peptidoglycan wall that retains the crystal violet-iodine complex during alcohol decolourisation and so remain purple. Gram-negative cells have a thin peptidoglycan layer and an outer membrane that is dissolved by alcohol, so the complex is washed out and they take up the safranin counterstain, appearing pink.

2. What is a mordant? Give an example.

A mordant is a substance that fixes the dye in the cell and increases the intensity of the stain. Example: Gram's iodine in Gram staining, and phenol in carbol fuchsin.

3. Why is negative staining useful?

The cell is not stained, so it stays alive and its true shape and capsule are preserved. It is used to demonstrate capsules and to observe organisms such as spirochaetes that are difficult to stain.

Long-answer questions with model answer outlines

1. Describe the different types of staining methods with examples.

Outline: (i) Need for staining. (ii) Principle - basic and acidic dyes, mordant, decolouriser, counterstain. (iii) Simple staining. (iv) Negative staining. (v) Differential staining - Gram and acid-fast. (vi) Structural staining - capsule, spore, flagella. (vii) Comparison table and applications.

2. Explain the Gram staining procedure and the reason for the difference in colour.

Outline: (i) Introduction and use. (ii) Step 1 crystal violet. (iii) Step 2 iodine. (iv) Step 3 alcohol decolourisation. (v) Step 4 safranin. (vi) Structural reason for retention or loss. (vii) Diagram and agricultural importance.

Lecture 8
Sterilization - Principle - Physical Agents and Chemical Methods

Sterilization - Principle - Physical Agents and Chemical Methods

8.1 Definition and principle

Sterilisation is the process of destroying all forms of microbial life, including bacteria, fungi, viruses and their resistant spores, in or on an object or medium. **Disinfection** means destroying or removing pathogens but not necessarily all microorganisms, and usually refers to inanimate objects. **Antisepsis** is disinfection applied to living tissue. **Sanitisation** is the reduction of the microbial population to a level considered safe for public health.

- The principle of sterilisation is to kill or remove microbes by damaging their essential structures - proteins, nucleic acids, cell walls or membranes - or by physically removing them from the material.
- A **sterilising agent** is one that kills all microbes, including spores; a **disinfectant** kills pathogens on inanimate objects; an **antiseptic** is used on skin.
- **Bactericidal** agents kill bacteria; **bacteriostatic** agents only inhibit their growth, which resumes when the agent is removed.
- **Microbicidal** kills microbes; **microbistatic** inhibits growth.

8.2 Physical agents of sterilisation

8.2.1 Heat - moist heat

Moist heat kills by coagulating and denaturing proteins. It is more effective than dry heat at the same temperature because steam transfers heat better and water helps denature proteins.

- **Boiling** (100 C, 10-30 minutes) kills vegetative cells but not most spores; it is disinfection rather than sterilisation.
- **Autoclave** (steam under pressure) - the standard method. At 15 psi (pounds per square inch) above atmospheric pressure, steam reaches 121 C; at 121 C, 15 minutes kills all cells and spores. Used for media, glassware, rubber, surgical instruments and waste.
- **Tyndallisation** - intermittent steaming at 100 C for 20-30 minutes on three successive days. Spores that survive germinate between treatments and are killed on the next heating. Used for media that would be spoiled by autoclaving (for example, media containing sugars, milk or gelatin).
- **Pasteurisation** - heating to 63 C for 30 minutes (holder method) or 72 C for 15 seconds (flash or HTST method), followed by rapid cooling. It kills pathogens and reduces spoilage organisms but does not sterilise. Used for milk, fruit juices and wine.

8.2.2 Heat - dry heat

Dry heat kills by oxidation of cell constituents and by burning.

- **Red heat / flaming** - holding an inoculating loop or needle in a flame until red hot; used for loops, needles and the mouths of culture tubes.

- **Hot-air oven** - 160 C for 2 hours or 170-180 C for 1 hour. Used for glassware such as petri dishes, pipettes, test tubes, and for metal instruments and oils that are damaged by steam.

8.2.3 Filtration

- Removes microbes physically rather than killing them. Used for heat-labile liquids such as sera, sugar solutions, antibiotics and vitamins.
- **Membrane filters** of pore size 0.22 um or 0.45 um are used for sterilising liquids; the liquid is forced through and the microbes are trapped on the membrane.
- **HEPA filters** (high-efficiency particulate air) remove microbes from air and are used in laminar-flow cabinets and clean rooms.

8.2.4 Radiation

- **Ultraviolet (UV) light** - non-ionising; wavelength about 260 nm is most effective; it damages DNA by forming thymine dimers. It has poor penetrating power, so it is used only for surface and air sterilisation (for example, in laminar-flow hoods and operation theatres).
- **Ionising radiation** - gamma rays (from cobalt-60) and X-rays. They have high penetrating power and are used for sterilising plastic syringes, catheters, surgical gloves and some foods. This is called "cold sterilisation" because it does not raise the temperature.

8.2.5 Other physical methods

- **Ultrasonication** - high-frequency sound waves disrupt cells; used for cleaning instruments and for breaking cells.
- **High pressure** - very high hydrostatic pressure is used in food processing to kill microbes.
- **Desiccation** - drying inhibits growth but does not sterilise; many spores survive.

8.3 Chemical methods

- **Alcohols** - 70% ethanol or isopropanol; denature proteins and dissolve lipids; used for skin and surfaces. Pure alcohol is less effective than 70% because it dehydrates the cell too fast.
- **Aldehydes** - formaldehyde (as formalin) and glutaraldehyde; alkylate proteins and nucleic acids; used for fumigation and for sterilising instruments and heat-sensitive equipment.
- **Halogens** - chlorine (as bleaching powder, sodium hypochlorite) and iodine (as tincture of iodine or povidone-iodine); oxidise cell material; used for water treatment, dairy equipment and skin.
- **Phenols and cresols** - denature proteins and disrupt membranes; used as disinfectants (Lysol, cresol).
- **Gases** - ethylene oxide and propylene oxide; alkylate proteins; used for sterilising plastics, catheters and heat-sensitive items in a closed chamber.
- **Dyes** - such as crystal violet, malachite green, acriflavine; interfere with cell metabolism; used in some topical preparations.
- **Detergents** - quaternary ammonium compounds (cationic detergents) disrupt membranes; used for cleaning and for skin antisepsis.
- **Heavy metals** - salts of mercury, silver and copper; combine with cell proteins; used in some antiseptics and in agriculture (for example, Bordeaux mixture for plant disease control).

- **Hydrogen peroxide** - produces free radicals; used for surfaces and wounds.

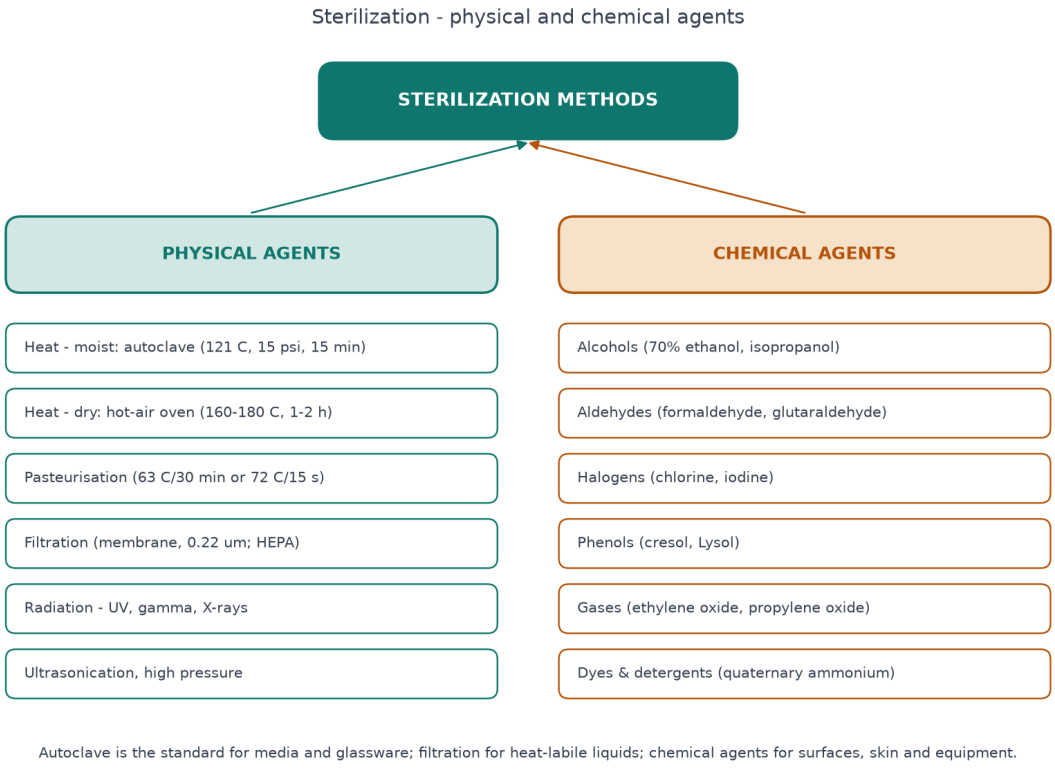


Figure 8.1 - Chart of physical and chemical agents used for sterilisation.

8.4 Factors affecting the efficiency of sterilisation

- Type and number of microorganisms present - spores are far more resistant than vegetative cells.
- Concentration and nature of the sterilising agent.
- Temperature and time of exposure - higher temperature shortens the required time.
- Presence of organic matter, which can protect microbes.
- pH of the medium.
- Accessibility of the agent to the microbes (penetration).

Worked example 8.1 - Sterilisation time and the decimal reduction value

Problem: A culture contains 1,000,000 (10^6) spores. The decimal reduction time D (the time to kill 90% of the spores at a given temperature) is 2 minutes. How long is needed to reduce the count to 1 spore?

Step 1: Find how many 90% reductions (log reductions) are needed. From 10^6 to 10^0 is 6 log reductions.

Step 2: Each log reduction takes D = 2 minutes.

Step 3: Time = number of log reductions x D = 6 x 2 = 12 minutes.

Answer: 12 minutes at the given temperature. This illustrates why a safety margin (usually 15 minutes at 121 C) is used in autoclaving to allow for the most resistant spores.

Table 8.1 - Sterilisation methods with examples

Method	Agent and conditions	Kills spores?	Typical use
Autoclave	Moist heat, 121 C, 15 psi, 15 min	Yes	Media, glassware, instruments
Boiling	Moist heat, 100 C, 10-30 min	No (mostly)	Disinfection of instruments
Tyndallisation	Steam 100 C, 20-30 min x 3 days	Yes	Sugar and milk media
Hot-air oven	Dry heat, 160-170 C, 1-2 h	Yes	Glassware, metal, oils
Pasteurisation	63 C/30 min or 72 C/15 s	No	Milk, juices, wine
Filtration	Membrane 0.22 um	Yes (removes)	Sera, antibiotics, vitamins
UV light	260 nm	Poor	Surfaces, air, hoods
Gamma radiation	Cobalt-60	Yes	Plastics, gloves, some foods
Ethylene oxide	Gas	Yes	Heat-sensitive plastics
Alcohol	70% ethanol	No	Skin, surfaces, benches

8.5 Agricultural and dairy applications

- Autoclaving is used to prepare sterile culture media for isolating *Rhizobium*, *Azotobacter* and *Trichoderma*.
- Pasteurisation is applied to milk in dairies; the HTST method (72 C for 15 seconds) is common in Indian milk plants.
- Chlorine compounds are used to sterilise dairy equipment, poultry sheds and drinking water.
- Gamma radiation is used to sterilise packed spices and some horticultural produce.
- Hot-air ovens and flaming are routine in every microbiology laboratory.

Exam focus

Key points to remember

- Sterilisation destroys all life including spores; disinfection destroys pathogens; antisepsis is on living tissue.
- Autoclave: 121 C, 15 psi, 15 minutes - the standard method.
- Moist heat is more effective than dry heat; dry heat needs 160-170 C for 1-2 hours.
- Filtration is used for heat-labile liquids; UV for surfaces; gamma rays for plastics and food.
- Pasteurisation kills pathogens but does not sterilise.
- Chemical agents include alcohols, aldehydes, halogens, phenols, gases and detergents.

Short-answer questions with model answers

1. Define sterilisation, disinfection and antisepsis.

Sterilisation is the complete destruction of all microbial life including spores. Disinfection is the destruction or removal of pathogens from inanimate objects, not necessarily all microbes. Antisepsis is

disinfection applied to living tissue.

2. Why is moist heat more effective than dry heat?

Moist heat transfers heat better and water helps to coagulate and denature proteins, so it kills microbes at lower temperature and in shorter time than dry heat. Spores are killed at 121 C by moist heat but need 160-170 C with dry heat.

3. Give the standard autoclave conditions and state one use.

121 C at 15 psi for 15 minutes. It is used to sterilise culture media, glassware and surgical instruments.

Long-answer questions with model answer outlines

1. Describe the physical methods of sterilisation.

Outline: (i) Definition and principle. (ii) Moist heat - boiling, autoclave, tyndallisation, pasteurisation. (iii) Dry heat - flaming, hot-air oven. (iv) Filtration - membrane, HEPA. (v) Radiation - UV, gamma. (vi) Other - ultrasonication, high pressure. (vii) Comparison table and applications.

2. Explain the chemical methods of sterilisation with examples.

Outline: (i) Principle of chemical disinfection. (ii) Alcohols. (iii) Aldehydes. (iv) Halogens. (v) Phenols. (vi) Gases. (vii) Dyes, detergents and heavy metals. (viii) Factors affecting efficiency and agricultural uses.

Lecture 9
Isolation and Enrichment Culture Techniques; Preservation Techniques

Isolation and Enrichment Culture Techniques; Preservation Techniques

9.1 Introduction

Microbes occur in nature in mixed populations. To study a microorganism, it must first be separated from all others and grown as a **pure culture** - a population derived from a single cell. The methods used to separate and grow microbes are called isolation techniques; enrichment is used when the desired microbe is present in very small numbers.

9.2 Culture media

A **culture medium** is a nutrient preparation used to grow microorganisms. It must supply carbon, nitrogen, minerals, water and growth factors, and must be at a suitable pH, temperature and oxygen level.

- **Synthetic (defined) medium** - the exact chemical composition is known (for example, glucose-mineral salts medium).
- **Complex (undefined) medium** - contains substances of unknown composition such as peptone, beef extract, yeast extract. Nutrient agar and nutrient broth are examples.
- **Solid medium** - contains agar (1.5-2%), a polysaccharide from seaweed, which solidifies the medium but is not digested by most microbes. Agar melts at about 95 C and solidifies at about 42 C.
- **Semisolid medium** - 0.5% agar; used for motility tests.
- **Liquid (broth) medium** - no agar; used for growing large numbers of cells.
- **Selective medium** - contains substances that inhibit unwanted microbes and allow the desired ones to grow (for example, MacConkey agar for Gram-negative bacteria).
- **Differential medium** - shows a visible difference between colonies (for example, blood agar, MacConkey agar).
- **Enriched medium** - contains extra nutrients for fastidious organisms (for example, blood agar).

9.3 Isolation techniques

9.3.1 Streak plate method

- A sterilised loop is dipped in the mixed sample and streaked across the surface of a solid medium in a series of successive streaks, flaming the loop between streaks.
- The number of cells deposited decreases with each streak, so that single cells become separated and grow into separate, well-isolated colonies.
- Each colony, being derived from one cell, is a pure culture.
- It is the most common and most useful isolation method.

9.3.2 Pour plate method

- A measured volume of diluted sample is placed in a sterile petri dish and molten cooled agar (about 45 C) is poured over it and mixed.
- After incubation, colonies develop both on the surface and within the agar.
- It is also used for counting viable cells (standard plate count).

9.3.3 Spread plate method

- A measured volume of diluted sample is spread evenly over the surface of a prepared solid medium with a sterile bent glass rod (spreader).
- Colonies develop on the surface, which makes counting easier.

9.3.4 Single-cell isolation

- Micromanipulator or a micro-pipette is used to pick up a single cell under the microscope and transfer it to a fresh medium. It gives a guaranteed pure culture but is tedious.

9.4 Enrichment culture technique

- Enrichment is used to increase the proportion of a desired microbe in a mixed population before isolation.
- The medium and conditions are designed to favour the desired microbe and suppress others.
- Examples: growing *Rhizobium* on a medium containing mannitol and yeast extract; enriching nitrogen-fixing bacteria in a nitrogen-free medium, where only nitrogen fixers can grow; enriching sulphate reducers in a medium with sulphate; using a high-salt medium to select halophiles.
- After one or more enrichment cycles, the sample is plated and colonies are isolated.

9.5 Maintenance and preservation of pure cultures

Once a pure culture is obtained, it must be maintained without contamination and without change in its characteristics.

- **Refrigeration** - cultures on slants or stabs are stored at 4-5 C and subcultured every 2-3 months. Simple but risks mutation and contamination.
- **Mineral oil overlay** - a layer of sterile mineral oil over a slant reduces oxygen and drying; cultures last longer (about 1 year).
- **Lyophilisation (freeze-drying)** - the culture is frozen and dried under vacuum in the presence of a protective agent (skim milk or serum). The ampoule is sealed. Cultures remain viable for many years and this is the standard method for culture collections.
- **Cryopreservation** - cultures are frozen at very low temperature, usually in liquid nitrogen at -196 C or in a deep freezer at -70 to -80 C, with a cryoprotectant such as glycerol or DMSO. This is widely used for long-term storage.
- **Subculturing / periodic transfer** - the culture is transferred to fresh medium at regular intervals. Simple but labour-intensive and prone to contamination and genetic change.
- **Drying** - spores and some fungi are preserved by drying on soil, sand or silica gel.

9.6 Culture collections

Well-known culture collections maintain type cultures and supply them to researchers: American Type Culture Collection (ATCC, USA), National Collection of Type Cultures (NCTC, UK), National Collection of Industrial Microorganisms (NCIM, India) and the Microbial Type Culture Collection (MTCC, India, at IMTECH Chandigarh).

Table 9.1 - Isolation, enrichment and preservation methods

Method	Principle	Main use
Streak plate	Progressive dilution on solid medium	Isolating pure cultures
Pour plate	Sample mixed with molten agar	Isolation and viable count
Spread plate	Sample spread on surface	Isolation and viable count
Enrichment culture	Selective medium favours one microbe	Isolating microbes present in low numbers
Refrigeration	Low temperature slows metabolism	Short-term storage
Mineral oil	Reduces oxygen and drying	Medium-term storage
Lyophilisation	Freeze-drying under vacuum	Long-term storage in collections
Cryopreservation	Very low temperature (-70 to -196 C)	Long-term storage

9.7 Agricultural relevance

- Enrichment and isolation are used to obtain efficient *Rhizobium* strains from root nodules for biofertilizer production.
- Pure cultures of *Azotobacter*, *Azospirillum*, phosphate-solubilising bacteria and *Trichoderma* are maintained and supplied to farmers as inoculants.
- Lyophilisation and cryopreservation are used by culture collections and biofertilizer laboratories to maintain mother cultures.

Exam focus

Key points to remember

- A pure culture is derived from a single cell; it is essential for all microbiological study.
- Agar (1.5-2%) is the common solidifying agent; it is not digested by most microbes.
- Streak plate is the most common isolation method; pour and spread plates are used for counts.
- Enrichment uses a selective medium to favour the desired microbe.
- Lyophilisation and cryopreservation are the best long-term preservation methods.
- Subculturing every 2-3 months with refrigeration is the simplest short-term method.

Short-answer questions with model answers

1. What is a pure culture and why is it important?

A pure culture is a population of cells derived from a single cell and containing only one type of microorganism. It is important because a mixed population cannot be used to study the characteristics, physiology or pathogenicity of a single microbe.

2. Describe the streak plate method.

A loopful of the sample is streaked over the surface of a solid medium in successive streaks, flaming the loop between streaks. The cells are progressively diluted so that single cells become separated and grow into isolated colonies, each of which is a pure culture.

3. Explain lyophilisation.

Lyophilisation is freeze-drying: the culture is frozen and then dried under vacuum with a protective substance such as skim milk. The dried culture is sealed in an ampoule and remains viable for many years. It is the standard method of preserving cultures in collections.

Long-answer questions with model answer outlines

1. Describe the methods of isolation of pure cultures.

Outline: (i) Need for pure cultures. (ii) Culture media types. (iii) Streak plate. (iv) Pour plate. (v) Spread plate. (vi) Single-cell isolation. (vii) Enrichment technique and examples.

2. Explain the techniques used for the preservation of microorganisms.

Outline: (i) Need for preservation. (ii) Refrigeration and periodic subculturing. (iii) Mineral oil overlay. (iv) Lyophilisation. (v) Cryopreservation. (vi) Drying on soil or silica gel. (vii) Culture collections and agricultural importance.

Lecture 10
Evolutionary Relationship - Position of Microbes in the Living World; C...

Evolutionary Relationship - Position of Microbes in the Living World; Concepts and Developments in Classification of Microorganisms

10.1 The position of microbes in the living world

Early classification divided all living things into two kingdoms - plants and animals. Microorganisms did not fit this scheme well: some are motile like animals, others are photosynthetic like plants, and bacteria are structurally unlike either. As knowledge grew, new systems were proposed.

- **Two-kingdom system** (Linnaeus) - Plantae and Animalia. Microorganisms were placed awkwardly in one or the other.
- **Three-kingdom system** (Haeckel, 1866) - Protista, Plantae and Animalia. Protista contained all unicellular organisms.
- **Four-kingdom system** (Copeland, 1938) - Monera (prokaryotes), Protista (eukaryotic unicellular), Plantae and Animalia.
- **Five-kingdom system** (R. H. Whittaker, 1969) - Monera, Protista, Fungi, Plantae and Animalia. This system is based on cell type (prokaryote or eukaryote), level of organisation, mode of nutrition and reproduction. It is widely taught.
- **Six-kingdom system** (Carl Woese, 1977) - based on molecular phylogeny (16S ribosomal RNA sequences). The prokaryotes are divided into two very different groups, Bacteria (Eubacteria) and Archaea, giving the kingdoms Eubacteria, Archaebacteria, Protista, Fungi, Plantae and Animalia.
- **Three-domain system** (Woese) - all life is placed in three domains: Bacteria, Archaea and Eukarya. This is the system most accepted today. It shows that Archaea are as different from Bacteria as they are from eukaryotes.

10.2 Evolutionary relationship of microbes

- All living organisms share a common ancestry. Comparisons of the base sequence of ribosomal RNA, especially 16S rRNA in prokaryotes and 18S rRNA in eukaryotes, are used to construct evolutionary trees.
- Woese's work showed that life on earth split very early into three lines - Bacteria, Archaea and Eukarya.
- The endosymbiotic theory (Lynn Margulis) explains that mitochondria and chloroplasts of eukaryotic cells arose from bacteria that were taken up by a larger host cell and became permanent partners. Evidence: mitochondria and chloroplasts have their own circular DNA and their own ribosomes, similar to bacteria.
- Microbes are the oldest known life forms; fossil evidence of bacteria-like cells goes back about 3.5 billion years.

10.3 Concepts and developments in classification of microorganisms

Classification (taxonomy) is the science of grouping organisms. It has three parts:

- **Nomenclature** - naming organisms according to the rules, using the binomial system of Carolus Linnaeus: a genus name and a species name, both italicised, genus with a capital letter and species with a small letter (for example, *Escherichia coli*, *Rhizobium leguminosarum*).
- **Identification** - determining the group to which an unknown organism belongs.
- **Classification** - arranging organisms into groups (taxa) on the basis of shared characteristics.

10.3.1 Basis of classification

- **Morphological** - cell shape, size, arrangement, presence of flagella, spores, capsules.
- **Staining** - Gram reaction, acid-fastness.
- **Cultural** - appearance of colonies, growth on different media.
- **Biochemical** - ability to ferment sugars, produce enzymes, utilise different carbon and nitrogen sources.
- **Serological** - reaction with specific antibodies.
- **Genetic and molecular** - DNA base composition (G+C content), DNA-DNA hybridisation, ribosomal RNA sequences. These are the most reliable and are the basis of modern phylogeny.

10.3.2 The taxonomic hierarchy

The ranks, from the broadest to the narrowest, are: Domain, Kingdom, Phylum (or Division), Class, Order, Family, Genus, Species. A species in bacteriology is a group of strains that share many characteristics; a strain is a population derived from a single isolation.

10.4 Agricultural relevance

- Correct classification is essential to identify plant pathogens accurately - for example, distinguishing *Xanthomonas* from *Pseudomonas* - and so to choose the right control measures.
- Classification of *Rhizobium* species according to the host legume (for example, *R. leguminosarum* biovar *viciae* for pea and *trifolieae* for clover) guides the selection of the correct inoculant.
- Molecular classification helps in selecting efficient biofertilizer and biocontrol strains.

Table 10.1 - Systems of classification of living organisms

System	Author and year	Main groups
Two kingdoms	Linnaeus	Plantae, Animalia
Three kingdoms	Haeckel, 1866	Protista, Plantae, Animalia
Four kingdoms	Copeland, 1938	Monera, Protista, Plantae, Animalia
Five kingdoms	Whittaker, 1969	Monera, Protista, Fungi, Plantae, Animalia
Six kingdoms	Woese, 1977	Eubacteria, Archaebacteria, Protista, Fungi, Plantae, Animalia
Three domains	Woese	Bacteria, Archaea, Eukarya

Exam focus

Key points to remember

- Whittaker's five-kingdom system (1969) is widely used; Woese's three-domain system is the modern one.
- Woese used 16S ribosomal RNA sequences to show Bacteria, Archaea and Eukarya are the three main lines.
- The endosymbiotic theory explains the origin of mitochondria and chloroplasts from bacteria.
- Taxonomy has three parts: nomenclature, identification and classification.
- The binomial system was given by Linnaeus; genus and species names are italicised.
- Modern classification uses molecular methods, especially rRNA sequencing and DNA base composition.

Short-answer questions with model answers

1. Give Whittaker's five-kingdom system.

Monera (prokaryotes), Protista (eukaryotic unicellular organisms), Fungi, Plantae and Animalia. The system is based on cell type, level of organisation, mode of nutrition and reproduction.

2. What is the three-domain system and who proposed it?

The three-domain system was proposed by Carl Woese on the basis of 16S ribosomal RNA comparisons. It places all life in three domains: Bacteria, Archaea and Eukarya. It shows that Archaea are distinct from Bacteria.

3. State the endosymbiotic theory.

The endosymbiotic theory (Lynn Margulis) states that mitochondria and chloroplasts of eukaryotic cells originated from bacteria that were engulfed by a larger host cell and became permanent partners. Their own circular DNA and ribosomes support this idea.

Long-answer questions with model answer outlines

1. Describe the position of microorganisms in the living world and the systems of classification.

Outline: (i) Early two-kingdom system and its limitations. (ii) Three-, four-, five- and six-kingdom systems. (iii) Woese's three-domain system and its basis. (iv) Evolutionary relationships and rRNA. (v) Endosymbiotic theory. (vi) Summary table.

2. Explain the concepts and methods used in the classification of microorganisms.

Outline: (i) Meaning of taxonomy and its three parts. (ii) Binomial nomenclature. (iii) Morphological, staining, cultural and biochemical criteria. (iv) Serological and molecular criteria. (v) Taxonomic hierarchy. (vi) Agricultural importance of accurate classification.

Lecture 11
Groups of Microorganisms - Prokaryotes and Eukaryotes; Archaea and Thei...

Groups of Microorganisms - Prokaryotes and Eukaryotes; Archaea and Their Ecology; Differences among Archaea, Eubacteria and Eukaryotes; Rickettsia and Chlamydia

11.1 Prokaryotes and eukaryotes

The fundamental division of cellular organisms is into prokaryotes and eukaryotes. Prokaryotes (Bacteria and Archaea) have no membrane-bound nucleus and no membrane-bound organelles. Eukaryotes (fungi, algae, protozoa, plants and animals) have a true nucleus and membrane-bound organelles.

Table 11.1 - Prokaryotes vs eukaryotes

Feature	Prokaryotes	Eukaryotes
Nucleus	Absent; nucleoid without membrane	True nucleus with membrane
DNA	Circular, not associated with histones (except archaea)	Linear, associated with histones
Ribosomes	70S (50S + 30S)	80S (60S + 40S)
Membrane-bound organelles	Absent	Present (mitochondria, ER, Golgi)
Cell wall	Peptidoglycan (bacteria)	Cellulose (plants), chitin (fungi)
Size	0.5-5 um	10-100 um
Cell division	Binary fission	Mitosis and meiosis
Flagella	Simple, no 9+2 structure	Complex, 9+2 microtubule structure

11.2 Archaea and their ecology

The Archaea (formerly archaebacteria) form a separate domain. They resemble bacteria in size and shape and in being prokaryotic, but differ greatly in biochemistry and genetics. Many live in extreme environments and are therefore called **extremophiles**.

- **Methanogens** - anaerobic archaea that produce methane. Found in marshes, paddy soils, the rumen of cattle, and sewage digesters. Example: *Methanobacterium*, *Methanosarcina*. They are important in the global carbon cycle and in biogas production.
- **Halophiles** - live in very salty environments such as salt pans and the Dead Sea. Example: *Halobacterium*. Their proteins need high salt to function.
- **Thermoacidophiles (thermophiles)** - live at high temperature and low pH, such as in hot springs and volcanic vents. Example: *Sulfolobus*, *Thermoplasma*.
- **Psychrophiles** - cold-loving archaea found in polar and deep-sea environments.
- Cell walls of archaea lack peptidoglycan; they contain pseudopeptidoglycan or proteins or glycoproteins.

- Membrane lipids of archaea contain branched-chain hydrocarbons (isoprenoid ethers) instead of the straight-chain fatty acids joined by ester bonds found in bacteria and eukaryotes.

11.3 Differences among archaea, eubacteria and eukaryotes

Table 11.2 - Archaea vs eubacteria vs eukaryotes

Feature	Archaea	Eubacteria	Eukaryotes
Cell type	Prokaryotic	Prokaryotic	Eukaryotic
Cell wall	No peptidoglycan; pseudopeptidoglycan/protein	Peptidoglycan (murein)	Cellulose/chitin/absent
Membrane lipids	Branched isoprenoid ethers	Straight-chain fatty acid esters	Straight-chain fatty acid esters
Ribosome	70S	70S	80S
tRNA initiator	Methionine	Formylmethionine	Methionine
RNA polymerase	Complex, many subunits	Simple, few subunits	Complex, many subunits
Introns	Present in some genes	Rare or absent	Common
Habitat	Often extreme environments	Wide range, all habitats	Wide range
Examples	*Methanobacterium*, *Halobacterium*	*E. coli*, *Rhizobium*	Fungi, algae, protozoa

11.4 Rickettsia and Chlamydia

Both are small, obligate intracellular parasites; that is, they can multiply only inside a living host cell. Because of this they were once thought to be viruses, but they are true bacteria: they have both DNA and RNA, ribosomes, a cell wall with peptidoglycan (in *Rickettsia*) and can be cultured in living tissue or in embryonated eggs, not in ordinary media.

11.4.1 Rickettsia

- Small, Gram-negative, rod-shaped bacteria; they do not form spores.
- Obligate intracellular parasites; they cannot grow outside a living cell.
- Transmitted by arthropod vectors - ticks, lice, fleas and mites.
- Cause diseases: epidemic typhus (*Rickettsia prowazekii*), Rocky Mountain spotted fever (*R. rickettsii*), scrub typhus (*Orientia tsutsugamushi*).
- They have a cell wall containing peptidoglycan and are sensitive to tetracycline.
- They can be grown in embryonated eggs or tissue culture.

11.4.2 Chlamydia

- Small, Gram-negative, obligate intracellular bacteria.
- Have a distinctive life cycle with two forms - the **elementary body** (infectious, extracellular, resistant) and the **reticulate body** (reproductive, intracellular).
- Their cell wall lacks typical peptidoglycan (or has very little); they depend on the host cell for energy (ATP).

- Cause diseases: trachoma, inclusion conjunctivitis, lymphogranuloma venereum (*Chlamydia trachomatis*), psittacosis (*C. psittaci*), and pneumonia (*C. pneumoniae*).
- Respond to tetracycline and azithromycin.

Table 11.3 - Rickettsia vs Chlamydia

Feature	Rickettsia	Chlamydia
Shape	Rod (bacillus)	Spherical to oval
Cell wall peptidoglycan	Present	Absent or very little
Energy (ATP)	Produces its own ATP	Depends on host cell for ATP
Life cycle	Simple binary fission	Two forms: elementary and reticulate body
Transmission	Arthropod vectors (ticks, lice, fleas, mites)	Direct contact, aerosols, sexual contact
Diseases	Typhus, spotted fevers	Trachoma, psittacosis, pneumonia
Culture	Embryonated eggs, tissue culture	Cell culture, embryonated eggs

11.5 Agricultural relevance

- Methanogens in paddy fields and cattle rumen are central to methane production and to the carbon cycle; rumen archaea help in the digestion of feed.
- Chlamydia causes diseases in animals - enzootic abortion in sheep (*C. abortus*) and psittacosis in poultry - which are of veterinary importance.
- Rickettsia-like organisms (*Pochonia*, rickettsia-like bacteria in plants) can affect crops and insect vectors, and are studied for insect control.

Exam focus

Key points to remember

- Prokaryotes have no true nucleus or membrane-bound organelles; eukaryotes have both.
- Prokaryotic ribosomes are 70S; eukaryotic ribosomes are 80S.
- Archaea form a separate domain; their walls lack peptidoglycan and their membrane lipids are branched ethers.
- Archaea include methanogens, halophiles, thermoacidophiles and psychrophiles.
- Rickettsia and Chlamydia are obligate intracellular bacteria, not viruses.
- Rickettsia are transmitted by arthropods; Chlamydia have elementary and reticulate body stages and depend on the host for ATP.

Short-answer questions with model answers

1. Give four differences between prokaryotes and eukaryotes.

(i) Prokaryotes have no true nucleus; eukaryotes have a membrane-bound nucleus. (ii) Prokaryotic ribosomes are 70S; eukaryotic are 80S. (iii) Prokaryotes lack membrane-bound organelles; eukaryotes have mitochondria, ER and Golgi. (iv) Prokaryotic DNA is circular and not histone-bound; eukaryotic DNA is linear and histone-bound.

2. How do archaea differ from eubacteria in cell wall and membrane lipids?

Archaeal cell walls lack peptidoglycan and contain pseudopeptidoglycan, proteins or glycoproteins, whereas eubacterial walls contain peptidoglycan (murein). Archaeal membrane lipids are branched isoprenoid ethers, while eubacterial membrane lipids are straight-chain fatty acid esters.

3. Why are Rickettsia and Chlamydia not classified as viruses?

They are true bacteria because they have both DNA and RNA, ribosomes and a cell wall (peptidoglycan in Rickettsia), and they reproduce by binary fission. They are obligate intracellular parasites but are cellular, unlike viruses.

Long-answer questions with model answer outlines

1. Compare archaea, eubacteria and eukaryotes.

Outline: (i) Three domains of life. (ii) Cell type. (iii) Cell wall and membrane lipids. (iv) Ribosomes and RNA polymerase. (v) Introns and tRNA initiator. (vi) Habitats and examples. (vii) Comparison table.

2. Describe Rickettsia and Chlamydia and bring out their similarities and differences.

Outline: (i) Both are obligate intracellular bacteria. (ii) Rickettsia - shape, vector transmission, diseases. (iii) Chlamydia - elementary and reticulate bodies, energy dependence, diseases. (iv) Comparison table. (v) Veterinary and agricultural importance.

Lecture 12
Bergey's Manual of Systematic Bacteriology (Outline Only); Economic Imp...

Bergey's Manual of Systematic Bacteriology (Outline Only); Economic Importance of Bacteria

12.1 Bergey's Manual - an outline

Bergey's Manual of Determinative Bacteriology, first published in 1923 and edited by David Hendricks Bergey, is the standard reference for identifying and classifying bacteria. It was later expanded into Bergey's Manual of Systematic Bacteriology, which gives detailed descriptions of bacterial groups. The most recent edition divides the bacteria into a number of major groups (phyla), based mainly on 16S rRNA sequence data rather than only on morphology.

- Early editions used Gram reaction, shape, oxygen requirement, motility and biochemical characters.
- Modern editions (the second edition, in five volumes) use molecular phylogeny and recognise major phyla.
- Volume 1 covers the Archaea and the deeply branching and phototrophic Bacteria.
- Volume 2 covers the Proteobacteria - the largest and most diverse group, including *Escherichia*, *Pseudomonas*, *Rhizobium*, *Agrobacterium* and *Xanthomonas*.
- Volume 3 covers the low G+C Gram-positive bacteria (*Bacillus*, *Clostridium*, *Lactobacillus*, *Staphylococcus*, *Streptococcus*, *Mycoplasma*).
- Volume 4 covers the high G+C Gram-positive bacteria (*Actinomyces*, *Streptomyces*, *Mycobacterium*, *Corynebacterium*).
- Volume 5 covers the Planctomycetes, Spirochaetes, Fibrobacteres, Bacteroidetes and Fusobacteria.
- It is used mainly for identification (determinative function) and for classification (systematic function).

12.2 Economic importance of bacteria

Bacteria are of enormous importance, both beneficial and harmful.

12.2.1 Beneficial roles

- **Soil fertility and nitrogen fixation** - *Rhizobium* fixes nitrogen in symbiosis with legumes; free-living *Azotobacter*, *Azospirillum* and *Clostridium* also fix nitrogen; nitrifying bacteria (*Nitrosomonas*, *Nitrobacter*) convert ammonia to nitrate; *Bacillus* and *Pseudomonas* solubilise phosphate.
- **Decomposition and nutrient cycling** - bacteria decompose organic matter and release nutrients; they are essential in the carbon, nitrogen, sulphur and phosphorus cycles.
- **Biofertilizers and biocontrol** - *Rhizobium*, *Azotobacter*, *Azospirillum*, phosphate-solubilising bacteria and *Bacillus subtilis* and *Pseudomonas fluorescens* as biocontrol agents.
- **Dairy industry** - *Lactobacillus* and *Streptococcus* convert milk to curd, cheese, butter and yoghurt; they give characteristic flavour and prevent spoilage.
- **Fermented foods** - idli, dosa, dhokla, sauerkraut, soy sauce and vinegar.

- **Industrial products** - acetic acid, lactic acid, citric acid, butanol, acetone, ethanol, enzymes (proteases, amylases), vitamins and amino acids.
- **Antibiotics and pharmaceuticals** - streptomycin from *Streptomyces griseus*, chloramphenicol from *Streptomyces venezuelae*, and many others.
- **Sewage treatment and bioremediation** - bacteria break down organic wastes in sewage; they degrade oil spills and pesticide residues.
- **Biogas production** - methanogenic bacteria convert organic waste to methane.
- **Genetic engineering** - *E. coli* is the most widely used host for producing human insulin, growth hormone and other products.
- **Retting of fibres** - bacteria help in the retting of jute and coir.
- **Vaccines and research** - bacteria are used to produce vaccines and as model organisms in genetics.

12.2.2 Harmful roles

- **Plant diseases** - bacterial blight of rice (*Xanthomonas oryzae*), citrus canker (*Xanthomonas citri*), crown gall (*Agrobacterium tumefaciens*), bacterial wilt (*Ralstonia solanacearum*), soft rot (*Erwinia*), fire blight of apple and pear (*Erwinia amylovora*).
- **Human diseases** - tuberculosis, cholera, typhoid, tetanus, diphtheria, plague, leprosy, anthrax, food poisoning.
- **Animal diseases** - anthrax, brucellosis, black quarter, mastitis, tuberculosis in cattle.
- **Food spoilage** - bacteria cause souring of milk, putrefaction of meat and spoilage of canned foods, sometimes producing dangerous toxins (*Clostridium botulinum*).
- **Denitrification** - converts nitrate to nitrogen gas, causing loss of soil nitrogen.
- **Corrosion and biofouling** - sulphur bacteria cause corrosion of pipes and metal structures.

Table 12.1 - Economic importance of bacteria

Beneficial	Harmful
Nitrogen fixation (*Rhizobium*)	Plant diseases (blight, canker, wilt)
Decomposition and nutrient cycling	Human and animal diseases
Dairy products (curd, cheese)	Food spoilage and food poisoning
Antibiotics and industrial chemicals	Denitrification causing nitrogen loss
Sewage treatment and bioremediation	Corrosion of pipes and metal
Genetic engineering and vaccines	Weeds? No - but crop losses through disease

12.3 Agricultural relevance

- Biofertilizer production in India depends on *Rhizobium*, *Azotobacter*, *Azospirillum* and phosphate-solubilising bacteria.
- Biogas plants on farms use methanogenic bacteria to convert dung to fuel and enriched slurry.

- Bacterial biocontrol agents such as *Bacillus subtilis* and *Pseudomonas fluorescens* are widely used to control soil-borne plant pathogens.

Exam focus

Key points to remember

- Bergey's Manual is the standard reference for bacterial identification and classification.
- Modern editions use 16S rRNA sequence data; the Proteobacteria form the largest group.
- Bacteria are beneficial in nitrogen fixation, decomposition, dairy, industry, sewage treatment and genetic engineering.
- Bacteria are harmful as plant and animal pathogens, as food spoilers and in denitrification.
- *Rhizobium* is the most important nitrogen-fixing bacterium in agriculture.

Short-answer questions with model answers

1. What is Bergey's Manual?

Bergey's Manual of Determinative Bacteriology and its successor, Bergey's Manual of Systematic Bacteriology, is the standard reference for the identification and classification of bacteria. Modern editions classify bacteria mainly on the basis of 16S rRNA sequences.

2. Give four beneficial roles of bacteria in agriculture.

(i) Nitrogen fixation by *Rhizobium* and *Azotobacter*. (ii) Decomposition of organic matter and release of nutrients. (iii) Phosphate solubilisation. (iv) Biocontrol of plant pathogens by *Bacillus subtilis* and *Pseudomonas fluorescens*. They also help in biogas production and composting.

3. Name four bacterial plant diseases with their causal organisms.

Bacterial blight of rice (*Xanthomonas oryzae*), citrus canker (*Xanthomonas citri*), crown gall (*Agrobacterium tumefaciens*) and bacterial wilt (*Ralstonia solanacearum*).

Long-answer questions with model answer outlines

1. Write an outline of Bergey's Manual of Systematic Bacteriology.

Outline: (i) History and purpose. (ii) Shift from morphological to molecular basis. (iii) The five volumes and the groups they cover. (iv) Proteobacteria as the largest group. (v) Uses in identification and classification.

2. Discuss the economic importance of bacteria.

Outline: (i) Introduction. (ii) Beneficial - soil fertility, nitrogen fixation, decomposition, dairy, fermented foods, industry, antibiotics, sewage treatment, biogas, genetic engineering. (iii) Harmful - plant diseases, human and animal diseases, food spoilage, denitrification, corrosion. (iv) Conclusion on their dual role.

PART III - THE BACTERIAL CELL, FUNGI, ALGAE AND VIRUSES

Mounish Raj · MouRa · Agri AI

Lecture 13
Bacteria - Size, Shape, Structure and Arrangement of Cells

Bacteria - Size, Shape, Structure and Arrangement of Cells

13.1 Size of bacteria

Bacteria are the smallest cellular organisms. Their size is measured in micrometres (um); 1 um = 1/1000 of a millimetre = 1000 nanometres.

- Most bacteria range from about 0.5 to 5 um in length and about 0.2 to 2 um in width.
- The smallest bacteria are about 0.2 um across (for example, *Mycoplasma*, which also lacks a cell wall) and the largest may reach 0.5 mm in some giant bacteria such as *Thiomargarita*, but these are exceptions.
- Because of their small size, bacteria have a very large surface-to-volume ratio, which allows rapid uptake of nutrients and rapid growth.
- Size is usually measured with an ocular micrometer or by electron microscopy. Some bacteria are just at the limit of resolution of the light microscope.

13.2 Shape of bacteria

13.2.1 Coccus (plural cocci)

Spherical or nearly spherical cells, about 0.5-1 um in diameter. Examples: *Streptococcus*, *Staphylococcus*, *Micrococcus*.

13.2.2 Bacillus (plural bacilli)

Rod-shaped or cylindrical cells; length usually greater than width. Examples: *Bacillus*, *Escherichia coli*, *Rhizobium*, *Lactobacillus*.

13.2.3 Spirillum (plural spirilla)

Rigid, spiral or helical cells with a corkscrew shape. Examples: *Spirillum*, *Aquaspirillum*.

13.2.4 Vibrio

Comma-shaped, curved rods; less than one complete turn of the spiral. Example: *Vibrio cholerae*.

13.2.5 Spirochaete

Flexible, thin, spiral cells with axial filaments; they move by flexing. Examples: *Treponema*, *Leptospira*, *Borrelia*.

13.2.6 Other shapes

- **Pleomorphic** - variable shape (for example, *Corynebacterium*).
- **Filamentous** - grow as branching filaments (for example, *Streptomyces*, actinomycetes).
- **Square, star-shaped or stalked** - found in some aquatic bacteria.

13.3 Arrangement of cells

The arrangement of cells results from the plane in which the cells divide and whether they remain attached after division.

- **Coccus arrangements**
- *Diplococcus* - cells in pairs (*Streptococcus pneumoniae*).
- *Streptococcus* - cells in chains (division in one plane).
- *Staphylococcus* - irregular clusters like a bunch of grapes (division in several planes).
- *Tetrad* - groups of four (division in two planes).
- *Sarcina* - cubes of eight (division in three planes).
- **Bacillus arrangements**
- *Diplobacillus* - in pairs.
- *Streptobacillus* - in chains.
- *Palisade* - cells lying side by side like a fence (for example, *Corynebacterium*).

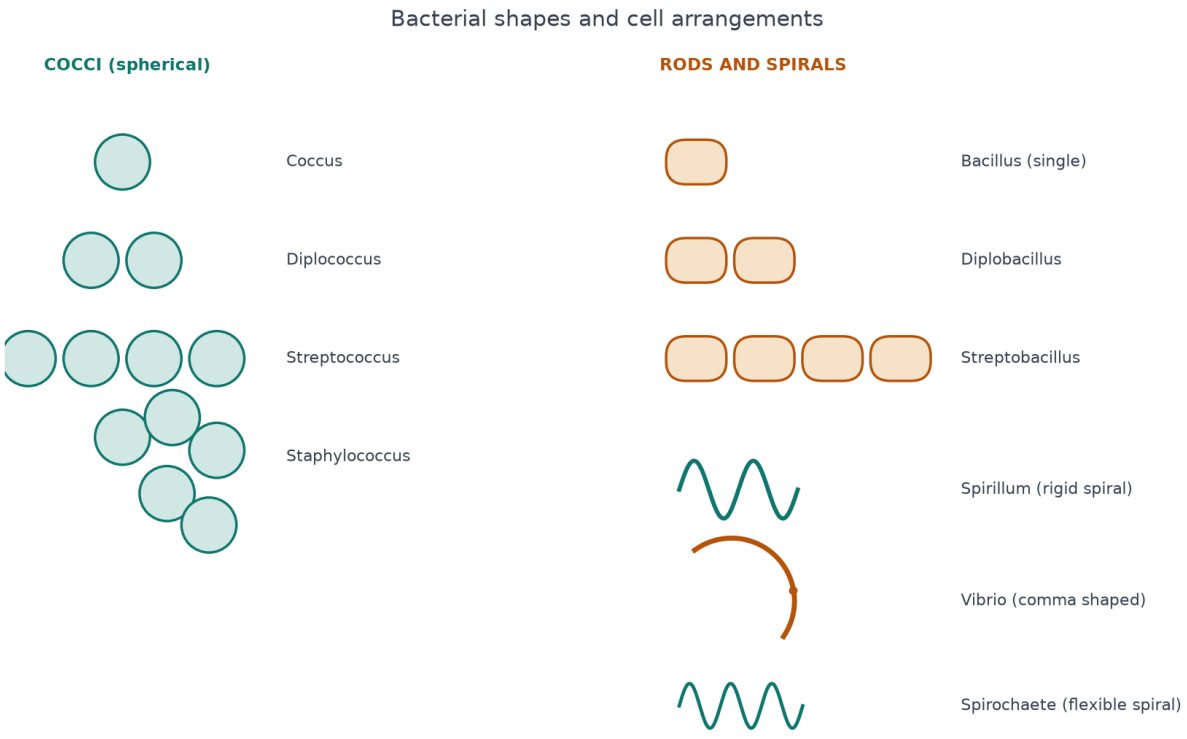


Figure 13.1 - Bacterial shapes (coccus, bacillus, spirillum, vibrio, spirochaete) and the common cell arrangements (diplo, strepto, staphylo).

13.4 Structure of the bacterial cell - an overview

A bacterial cell has a fairly simple structure compared with a eukaryotic cell. Its main parts are the capsule (when present), cell wall, plasma membrane, cytoplasm, nucleoid, ribosomes, plasmids, and surface appendages such as flagella and pili. These are described in detail in Lecture 14.

13.5 Agricultural relevance

- Shape and arrangement are used in the first stages of identification of plant pathogens - for example, *Xanthomonas* (rods) from *Streptococcus* (cocci).
- *Rhizobium* in root nodules appears as pleomorphic, bacteroid forms; recognising them helps confirm nodulation.
- *Streptomyces*, a filamentous bacterium, is important as a source of antibiotics and as a biocontrol agent in soil.

Table 13.1 - Shapes of bacteria with examples and approximate size

Shape	Description	Approx. size	Example
Coccus	Spherical	0.5-1 um diameter	*Staphylococcus aureus*
Bacillus	Rod	0.5-1 x 1-5 um	*Escherichia coli*
Spirillum	Rigid spiral	0.5 x 5-20 um	*Spirillum volutans*
Vibrio	Comma shaped	0.5 x 1-3 um	*Vibrio cholerae*
Spirochaete	Flexible spiral	0.1-0.5 x 5-250 um	*Treponema pallidum*

Exam focus

Key points to remember

- Most bacteria are 0.5-5 um in size; the unit is the micrometre.
- Basic shapes: coccus, bacillus, spirillum, vibrio and spirochaete.
- Arrangement depends on the plane of division and whether cells stay attached.
- Diplo = pairs; strepto = chains; staphlo = clusters; sarcina = cubes of eight.
- *Mycoplasma* is the smallest and lacks a cell wall.

Short-answer questions with model answers

1. Draw and label the common shapes of bacteria.

Coccus (spherical, e.g. *Streptococcus*), bacillus (rod, e.g. *E. coli*), spirillum (rigid spiral), vibrio (comma shaped, e.g. *Vibrio cholerae*) and spirochaete (flexible spiral, e.g. *Treponema*). Refer to Figure 13.1.

2. How are cocci arranged? Give examples.

Diplococci in pairs (*Streptococcus pneumoniae*), streptococci in chains, staphylococci in irregular clusters (*Staphylococcus aureus*), tetrads in fours and sarcinae in cubes of eight. The arrangement depends on the plane of cell division.

3. What is the size range of bacteria?

Most bacteria are about 0.5-5 um in length and 0.2-2 um in width. *Mycoplasma* is the smallest at about 0.2 um and lacks a cell wall.

Long-answer questions with model answer outlines

1. Describe the size, shape and arrangement of bacterial cells.

Outline: (i) Size range and units. (ii) Shapes - coccus, bacillus, spirillum, vibrio, spirochaete, other shapes. (iii) Arrangements of cocci and bacilli. (iv) Relation of arrangement to plane of division. (v) Diagram. (vi) Importance in identification.

2. Explain how the shape and arrangement of bacteria are used in their identification, with examples.

Outline: (i) Microscopic examination as the first step. (ii) Shape characters. (iii) Arrangement characters. (iv) Examples from plant pathology. (v) *Rhizobium* bacteroids. (vi) Limitations and need for biochemical tests.

Lecture 14
Bacteria - External and Internal Structures and Their Functionality

Bacteria - External and Internal Structures and Their Functionality

14.1 Introduction

The bacterial cell is studied in terms of its external structures (those outside the plasma membrane) and internal structures (those within the cytoplasm). Each structure has a specific function, and many of them are important in disease, in survival and in agriculture.

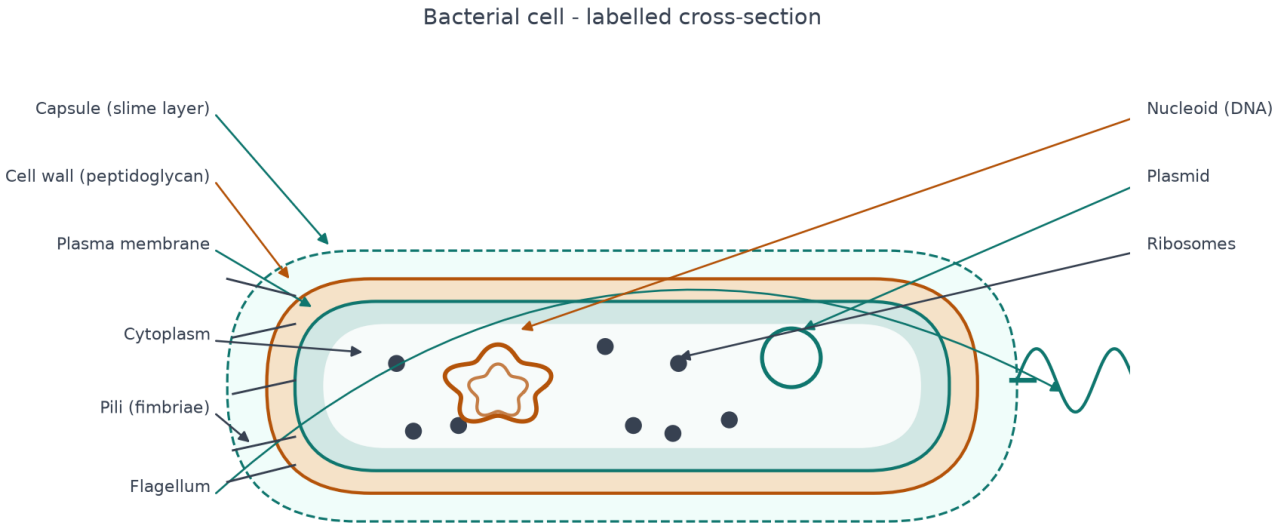


Figure 14.1 - Labelled cross-section of a bacterial cell showing capsule, cell wall, plasma membrane, cytoplasm, nucleoid, ribosomes, plasmid, flagellum and pili.

14.2 External structures

14.2.1 Capsule and slime layer

- The **capsule** is a well-defined, gelatinous layer outside the cell wall; a **slime layer** is a loose, unorganised layer.
- Made mostly of polysaccharides (sometimes polypeptides, for example poly-D-glutamic acid in *Bacillus anthracis*).
- Functions: protects against drying, protects against phagocytosis and so increases virulence, helps adhesion to surfaces and to host cells, and stores food.
- Demonstrated by negative staining or capsule stains.

14.2.2 Flagella

- Long, thin, whip-like appendages used for **motility** (movement).
- Made of the protein **flagellin**; the structure has three parts - the filament, the hook and the basal body.
- About 20 nm thick and up to 15-20 um long; too thin to be seen in a light microscope without special staining.
- Arrangement: **monotrichous** (single flagellum at one end), **amphitrichous** (one at each end), **lophotrichous** (a tuft at one end), **peritrichous** (all over the cell).
- They rotate like a propeller; rotation is driven by a proton gradient across the membrane.
- Functions: movement towards nutrients (chemotaxis) and away from harmful substances, and a role in disease and in colonising plant roots.

14.2.3 Pili (fimbriae)

- Short, thin, hair-like appendages, made of protein (pilin), shorter and straighter than flagella.
- Do not function in movement.
- Functions: attachment to surfaces and to host cells; **sex pili** (F pili) are longer and are used in conjugation to transfer DNA from one cell to another.
- Pili are important in the colonisation of plant roots by *Rhizobium* and in the attachment of pathogens.

14.2.4 Cell wall

- A rigid layer outside the plasma membrane that gives shape, strength and protection against osmotic lysis.
- Made of **peptidoglycan (murein)**, a polymer of N-acetylglucosamine (NAG) and N-acetylmuramic acid (NAM) cross-linked by short peptide chains.
- Gram-positive walls have a thick peptidoglycan layer (up to 90% of the wall) with teichoic acids.
- Gram-negative walls have a thin peptidoglycan layer, an outer membrane containing lipopolysaccharide (LPS, the endotoxin), and lipoproteins.
- Penicillin acts by inhibiting peptidoglycan synthesis; lysozyme breaks the bond between NAG and NAM.
- *Mycoplasma* has no cell wall; archaea have no peptidoglycan.

14.2.5 Plasma membrane

- A thin, flexible layer just inside the cell wall, made of a phospholipid bilayer with proteins (the fluid mosaic model).
- It is **selectively permeable** and controls the entry and exit of substances.
- Functions: transport of nutrients, excretion of waste, respiration (it carries the electron transport chain in bacteria), and synthesis of cell wall and capsule components.
- It forms inward folds called **mesosomes**, which increase surface area.

14.3 Internal structures

14.3.1 Cytoplasm

- The semi-fluid, granular material inside the plasma membrane.
- Contains water, dissolved nutrients, enzymes, ribosomes and the genetic material.
- It is the site of most metabolic reactions.

14.3.2 Nucleoid

- The bacterial chromosome - a single, circular, double-stranded DNA molecule, not enclosed in a nuclear membrane.
- It carries all the essential genes of the cell and is attached to the plasma membrane at one point.
- It is haploid; there is normally only one copy per cell.

14.3.3 Plasmids

- Small, circular, extrachromosomal DNA molecules that replicate independently of the chromosome.
- Not essential for normal growth but carry useful genes, for example:
- *F factor* - transfer of DNA during conjugation.
- *R factor* - resistance to antibiotics.
- *Ti plasmid* of *Agrobacterium tumefaciens* - causes crown gall disease; widely used in plant genetic engineering.
- Plasmids are important tools in genetic engineering.

14.3.4 Ribosomes

- Small granular structures, the site of protein synthesis.
- Bacterial ribosomes are **70S**, made of a 50S and a 30S subunit.
- They differ from the 80S eukaryotic ribosomes, which is why many antibiotics (streptomycin, tetracycline, erythromycin) act on bacterial ribosomes without harming human cells.

14.3.5 Inclusion bodies and storage granules

- Store reserve materials such as glycogen, poly-beta-hydroxybutyrate (PHB), polyphosphate (volutin or metachromatic granules) and sulphur.
- Help the cell survive when nutrients are scarce.

14.3.6 Endospores

- Highly resistant, dormant structures formed by some Gram-positive bacteria (*Bacillus*, *Clostridium*) when nutrients are exhausted.
- Contain a core with DNA and dipicolinic acid, surrounded by a cortex and a spore coat.
- Resistant to heat, drying, radiation and chemicals; they can survive for years.
- Not a means of reproduction - one cell forms one spore.
- Because spores resist heat, autoclaving at 121 C is required for sterilisation.

Table 14.1 - Bacterial structures and their functions

Structure	Composition	Function
Capsule	Polysaccharide/protein	Protection, virulence, adhesion
Flagellum	Flagellin protein	Motility
Pilus	Pilin protein	Attachment; sex pilus for conjugation
Cell wall	Peptidoglycan	Shape, rigidity, protection
Plasma membrane	Phospholipid + protein	Selective permeability, respiration
Nucleoid	Circular DNA	Genetic information
Plasmid	Small circular DNA	Extra genes - F, R, Ti factors
Ribosome	RNA + protein (70S)	Protein synthesis
Inclusion body	Glycogen, PHB, polyphosphate	Storage of reserve food
Endospore	DNA + dipicolinic acid + coats	Survival under adverse conditions

14.4 Agricultural relevance

- The Ti plasmid of *Agrobacterium tumefaciens* is used to transfer useful genes into plants.
- Flagella and pili help *Rhizobium* and *Azospirillum* move towards and attach to plant roots.
- Capsule formation helps plant pathogens resist plant defence and survive on seed and in soil.
- R-plasmids are responsible for the spread of antibiotic resistance among bacteria, including animal pathogens.

Exam focus

Key points to remember

- External structures: capsule, cell wall, plasma membrane, flagella, pili.
- Internal structures: cytoplasm, nucleoid, plasmids, ribosomes, inclusion bodies, endospores.
- Peptidoglycan is the main component of the bacterial cell wall.
- The bacterial chromosome is a single circular DNA; plasmids are extrachromosomal and carry extra genes.
- Bacterial ribosomes are 70S; this is the basis of the selective action of many antibiotics.
- Endospores are resistant dormant structures, not reproductive bodies.

Short-answer questions with model answers

1. Give the functions of the capsule.

The capsule protects the cell from drying and from phagocytosis, increases virulence, helps the cell adhere to surfaces and host tissues, and stores food.

2. Distinguish between flagella and pili.

Flagella are long, whip-like appendages of flagellin used for motility; they are 20 nm thick and up to 20 um long. Pili are short, hair-like appendages of pilin used for attachment and, in the case of sex pili, for

the transfer of DNA during conjugation; they do not move the cell.

3. What is a plasmid? Give two examples and their functions.

A plasmid is a small, circular, extrachromosomal DNA molecule that replicates independently. The F factor carries genes for conjugation, the R factor carries antibiotic-resistance genes, and the Ti plasmid of *Agrobacterium tumefaciens* causes crown gall and is used in plant genetic engineering.

Long-answer questions with model answer outlines

1. Describe the external and internal structures of a bacterial cell and their functions.

Outline: (i) Introduction and diagram. (ii) External - capsule, cell wall, plasma membrane, flagella, pili. (iii) Internal - cytoplasm, nucleoid, plasmid, ribosomes, inclusion bodies, endospores. (iv) Functions of each. (v) Comparison table. (vi) Agricultural and medical importance.

2. Explain the structure and function of the bacterial cell wall and plasma membrane.

Outline: (i) Location and composition of the cell wall. (ii) Peptidoglycan structure. (iii) Gram-positive and Gram-negative wall differences. (iv) Action of penicillin and lysozyme. (v) Plasma membrane - fluid mosaic model, selective permeability, respiration. (vi) Mesosomes. (vii) Importance.

Lecture 15
Morphology and Classification of Fungi and Their Economic Importance

Morphology and Classification of Fungi and Their Economic Importance

15.1 Introduction

Fungi are eukaryotic, heterotrophic organisms that obtain food by absorption. They have no chlorophyll and cannot photosynthesise. They may be unicellular (yeasts) or multicellular (moulds and mushrooms). They are found in soil, water, air and on plants and animals.

15.2 Morphology of fungi

15.2.1 Thallus and mycelium

- The body of a fungus is called the **thallus**; it may be a single cell (yeast) or a filamentous mass (mould).
- The filaments are called **hyphae**; a mass of hyphae forms the **mycelium**.
- **Septate hyphae** have cross walls (septa) that divide the hypha into cells (*Aspergillus*, *Penicillium*).
- **Aseptate (coenocytic) hyphae** have no cross walls and contain many nuclei in a continuous cytoplasm (*Rhizopus*, *Mucor*).

15.2.2 Cell structure

- Fungal cells are eukaryotic, with a true nucleus, mitochondria, endoplasmic reticulum and vacuoles.
- The cell wall is made mainly of **chitin**, with some cellulose and glucans; chitin also occurs in insects.
- The cytoplasm contains stored food as glycogen and oil.

15.2.3 Yeasts

- Unicellular, oval or spherical fungi; they reproduce mainly by **budding**.
- Example: *Saccharomyces cerevisiae*, used in baking, brewing and wine making.

15.2.4 Reproduction

- **Vegetative** - fragmentation of mycelium, budding (in yeasts), fission and formation of spores.
- **Asexual** - spores such as conidia, sporangiospores, arthrospores, chlamydospores; produced in large numbers.
- **Sexual** - by the fusion of compatible nuclei, producing ascospores (in Ascomycetes), basidiospores (in Basidiomycetes) and zygospores (in Zygomycetes).
- *Rhizopus* (bread mould) is a good example for study: it has stolons, rhizoids and sporangiophores bearing sporangia.

15.3 Classification of fungi

The older system divided fungi into four classes - Phycomycetes, Ascomycetes, Basidiomycetes and Deuteromycetes (Fungi Imperfecti). A modern outline is given below.

- **Division Zygomycota (Phycomycetes)** - coenocytic hyphae; sexual spore is a zygospore; asexual spores are sporangiospores. Example: *Rhizopus*, *Mucor*.
- **Division Ascomycota (Ascomycetes)** - septate hyphae; sexual spores are ascospores formed in a sac called an ascus; asexual spores are conidia. Includes yeasts and many moulds and the morels. Examples: *Saccharomyces*, *Aspergillus*, *Penicillium*, *Neurospora*, *Claviceps*.
- **Division Basidiomycota (Basidiomycetes)** - septate hyphae; sexual spores are basidiospores formed on a club-shaped basidium. Includes mushrooms, puffballs, rusts and smuts. Examples: *Agaricus*, *Puccinia* (wheat rust), *Ustilago* (smut).
- **Division Deuteromycota (Fungi Imperfecti)** - no known sexual stage; reproduce only asexually. Examples: *Alternaria*, *Fusarium*, *Colletotrichum*, *Trichoderma*, *Cercospora*.
- **Oomycetes (water moulds)** - now placed with the stramenopiles, not true fungi; cell wall contains cellulose; example: *Phytophthora infestans* (late blight of potato), *Pythium*.

15.4 Economic importance of fungi

15.4.1 Beneficial roles

- **Decomposition** - fungi break down complex organic matter such as cellulose and lignin and release nutrients into the soil.
- **Mycorrhiza** - a symbiotic association of fungi with plant roots; helps in the absorption of water and phosphorus. Ectomycorrhiza (e.g. *Boletus*) and endomycorrhiza or VAM (*Glomus*) are important in agriculture and forestry.
- **Antibiotics** - penicillin from *Penicillium notatum* and *P. chrysogenum*; griseofulvin from *Penicillium*; cephalosporins from *Cephalosporium*.
- **Food and beverages** - yeasts in bread, beer, wine; moulds in cheese (Roquefort, Camembert); soy sauce, tempeh and idli.
- **Organic acids and enzymes** - citric acid from *Aspergillus niger*; gluconic acid; amylases, proteases, pectinases.
- **Vitamins and steroids** - riboflavin (vitamin B2) from *Ashbya*; gibberellins from *Gibberella fujikuroi*; cortisone and other steroids.
- **Mushroom cultivation** - *Agaricus bisporus* (button mushroom), *Pleurotus* (oyster mushroom), *Volvariella* (paddy straw mushroom) - a source of protein and income for farmers.
- **Biocontrol** - *Trichoderma* and *Gliocladium* control soil-borne pathogens.
- **Single-cell protein** - yeast biomass as a protein source for feed.

15.4.2 Harmful roles

- **Plant diseases** - wheat rust (*Puccinia graminis*), smut of wheat (*Ustilago*), late blight of potato (*Phytophthora infestans*), blast of rice (*Magnaporthe oryzae*), wilt diseases (*Fusarium*), anthracnose (*Colletotrichum*), powdery and downy mildews, damping off of seedlings (*Pythium*, *Rhizoctonia*).

- **Human diseases (mycoses)** - ringworm (*Trichophyton*), athlete's foot, candidiasis (*Candida albicans*), aspergillosis.
- **Animal diseases** - ringworm in cattle; mycotic abortion.
- **Food spoilage and mycotoxins** - moulds spoil stored grain and produce mycotoxins such as aflatoxin (*Aspergillus flavus*), which is carcinogenic and a serious problem in groundnut and maize.
- **Wood decay** - dry rot and wet rot of timber.

Table 15.1 - Classification of fungi with examples and importance

Group	Hyphae	Sexual spore	Example	Importance
Zygomycota	Coenocytic	Zygospore	*Rhizopus*	Bread mould; spoilage
Ascomycota	Septate	Ascospore	*Aspergillus*, *Saccharomyces*	Citric acid, antibiotics, yeast
Basidiomycota	Septate	Basidiospore	*Agaricus*, *Puccinia*	Mushrooms; wheat rust
Deuteromycota	Septate	None known	*Trichoderma*, *Fusarium*	Biocontrol; wilt disease
Oomycetes	Coenocytic	Oospore	*Phytophthora*	Late blight of potato

15.5 Agricultural relevance

- Mycorrhizal fungi improve phosphorus uptake and drought tolerance of crops.
- *Trichoderma* is used as a biocontrol agent against *Rhizoctonia* and *Pythium*.
- Aflatoxin contamination of groundnut and maize must be prevented by proper drying and storage.
- Mushroom cultivation gives additional income and uses agricultural waste.

Exam focus

Key points to remember

- Fungi are eukaryotic, heterotrophic, absorptive; cell wall contains chitin.
- Hyphae may be septate or coenocytic; a mass of hyphae is the mycelium.
- Yeasts are unicellular and reproduce by budding.
- Main groups: Zygomycota, Ascomycota, Basidiomycota, Deuteromycota (and Oomycetes).
- Beneficial - decomposition, mycorrhiza, antibiotics, food, organic acids, mushrooms, biocontrol.
- Harmful - plant diseases, mycoses, food spoilage and mycotoxins.

Short-answer questions with model answers

1. Draw the structure of a mould and label its parts.

A mould such as *Rhizopus* has aseptate hyphae forming a mycelium; stolons run along the surface, rhizoids anchor the fungus, and sporangiophores bear sporangia containing sporangiospores.

2. Distinguish septate from coenocytic hyphae.

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Septate hyphae have cross walls (septa) that divide the filament into cells with one or more nuclei each (*Aspergillus*, *Penicillium*). Coenocytic hyphae have no cross walls and contain many nuclei in a continuous cytoplasm (*Rhizopus*, *Mucor*).

3. Give four beneficial and two harmful roles of fungi.

Beneficial: decomposition and nutrient cycling; mycorrhiza; production of antibiotics (penicillin) and organic acids (citric acid); food and beverages (bread, cheese, mushroom). Harmful: plant diseases such as wheat rust and late blight; food spoilage and mycotoxins such as aflatoxin.

Long-answer questions with model answer outlines

1. Describe the morphology and classification of fungi.

Outline: (i) General features - eukaryotic, heterotrophic, absorptive. (ii) Thallus, hyphae, mycelium, septate and coenocytic. (iii) Cell structure and chitin wall. (iv) Yeasts. (v) Reproduction - vegetative, asexual, sexual. (vi) Classification - Zygomycota, Ascomycota, Basidiomycota, Deuteromycota, Oomycetes. (vii) Table.

2. Discuss the economic importance of fungi in agriculture.

Outline: (i) Introduction. (ii) Beneficial - decomposition, mycorrhiza, antibiotics, food, organic acids, vitamins, mushroom cultivation, biocontrol. (iii) Harmful - plant diseases with examples, mycotoxins, food spoilage, wood decay. (iv) Management of fungal diseases. (v) Conclusion.

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Lecture 16
Morphology and Classification of Algae and Their Economic Importance

Morphology and Classification of Algae and Their Economic Importance

16.1 Introduction

Algae are eukaryotic, chlorophyll-bearing organisms that carry out oxygenic photosynthesis. They are mostly aquatic but also occur in soil, on rocks, on tree bark and in association with fungi (lichens). Algae range from unicellular forms to large multicellular seaweeds. They have no true roots, stems or leaves.

16.2 Morphology of algae

16.2.1 Thallus

- The plant body is called a **thallus** - it is not differentiated into root, stem and leaf.
- Forms: unicellular (*Chlamydomonas*), colonial (*Volvox*), filamentous (*Spirogyra*, *Oscillatoria*), membranous, and giant multicellular forms (*Ulva*, *Sargassum*).

16.2.2 Cell structure

- Eukaryotic with a true nucleus; chloroplasts contain chlorophyll **a** and, in different groups, chlorophyll b, c, d or e.
- Cell wall is usually made of **cellulose** and pectin; in diatoms it contains silica; in some it is absent.
- Reserve food: starch (green algae), laminarin and mannitol (brown algae), floridean starch (red algae), oils (diatoms).
- Accessory pigments give characteristic colours - green (chlorophyll), brown (fucoxanthin), red (phycoerythrin), blue-green (phycocyanin).
- Flagella, when present, are of the 9+2 type.

16.2.3 Reproduction

- **Vegetative** - fragmentation, cell division, budding.
- **Asexual** - by spores such as zoospores (motile), aplanospores, akinetes.
- **Sexual** - by fusion of gametes: isogamy (similar gametes), anisogamy (dissimilar), and oogamy (large non-motile egg with small motile sperm).

16.3 Classification of algae

Algae are classified mainly on the basis of pigmentation, reserve food and cell wall composition.

- **Chlorophyceae (green algae)** - chlorophyll a and b; reserve food starch; cellulose wall. Examples: *Chlamydomonas*, *Volvox*, *Spirogyra*, *Chlorella*, *Ulva*.

- **Cyanophyceae (blue-green algae or cyanobacteria)** - chlorophyll a and phycocyanin; reserve food cyanophycean starch; prokaryotic cells; cell wall of peptidoglycan. Examples: *Nostoc*, *Anabaena*, *Oscillatoria*. These are now classified as bacteria (cyanobacteria) but are traditionally studied with algae.
- **Bacillariophyceae (diatoms)** - chlorophyll a and c, fucoxanthin; cell wall of silica (frustule) in two overlapping halves; reserve food oil. Examples: *Navicula*, *Pinnularia*.
- **Phaeophyceae (brown algae)** - chlorophyll a and c, fucoxanthin; reserve food laminarin and mannitol; cell wall of cellulose and alginic acid. Examples: *Laminaria*, *Sargassum*, *Fucus*.
- **Rhodophyceae (red algae)** - chlorophyll a and phycoerythrin; reserve food floridean starch; cell wall of cellulose and agar. Examples: *Polysiphonia*, *Gelidium*, *Gracilaria*.
- **Euglenophyceae** - chlorophyll a and b; motile with flagella; reserve food paramylon. Example: *Euglena*.

16.4 Economic importance of algae

16.4.1 Beneficial roles

- **Nitrogen fixation** - blue-green algae such as *Anabaena*, *Nostoc* and *Tolypothrix* fix atmospheric nitrogen. *Azolla* (a water fern) with *Anabaena azollae* is used as a biofertilizer in paddy fields.
- **Biofertilizer** - blue-green algae are applied to flooded rice fields, where they fix nitrogen and add organic matter; *Anabaena*, *Nostoc* and *Aulosira* are used in India.
- **Soil reclamation** - cyanobacteria help in reclaiming saline and alkaline soils (usar land) by adding organic matter and nitrogen.
- **Food** - *Chlorella*, *Spirulina* (a cyanobacterium) are rich in protein and are used as health foods and feed supplements. *Porphyra*, *Laminaria* and *Ulva* are eaten as sea vegetables.
- **Agar and agarose** - from red algae (*Gelidium*, *Gracilaria*); used as a solidifying agent in culture media and in electrophoresis.
- **Alginic acid and carrageenan** - from brown and red algae; used as thickeners and stabilisers in food and industry.
- **Diatomaceous earth** - fossilised diatom shells used as a filter aid, as an abrasive and as an insecticide.
- **Sewage treatment and bioremediation** - algae in oxidation ponds help remove nutrients and organic matter; they also absorb heavy metals.
- **Source of oxygen** - algae produce a large part of the oxygen on earth through photosynthesis.
- **Single-cell protein and biofuel** - microalgae are grown for protein and for lipids for biodiesel.
- **As laboratory material** - *Chlorella*, *Chlamydomonas* and diatoms are used in research.

16.4.2 Harmful roles

- **Water blooms and eutrophication** - heavy growth of algae (water bloom) in nutrient-rich water depletes oxygen and kills fish; some cyanobacteria release toxins.
- **Red tides** - caused by dinoflagellates; kill fish and poison shellfish.
- **Fouling** - algae grow on ships, pipes and nets and cause biofouling and corrosion.

- **Some are parasites or cause disease** - *Prototheca* can cause disease in animals.

Table 16.1 - Classification of algae

Class	Main pigments	Reserve food	Cell wall	Example
Chlorophyceae	Chlorophyll a, b	Starch	Cellulose	*Spirogyra*
Cyanophyceae	Chlorophyll a, phycocyanin	Cyanophycean starch	Peptidoglycan	*Anabaena*
Bacillariophyceae	Chlorophyll a, c, fucoxanthin	Oil	Silica	*Navicula*
Phaeophyceae	Chlorophyll a, c, fucoxanthin	Laminarin, mannitol	Cellulose, algin	*Sargassum*
Rhodophyceae	Chlorophyll a, phycoerythrin	Floridean starch	Cellulose, agar	*Gracilaria*

16.5 Agricultural relevance

- *Azolla-Anabaena* is a cheap biofertilizer for paddy; it adds about 20-30 kg nitrogen per hectare.
- Blue-green algae are applied to rice fields and to alkaline soils.
- *Spirulina* is grown as a protein-rich feed supplement.
- Diatomaceous earth is used as a natural insecticide against stored-grain pests.

Exam focus

Key points to remember

- Algae are eukaryotic, photosynthetic thallophytes with chlorophyll a.
- Classification is based on pigments, reserve food and cell wall.
- Five main classes: Chlorophyceae, Cyanophyceae, Bacillariophyceae, Phaeophyceae, Rhodophyceae.
- Cyanobacteria (blue-green algae) fix nitrogen and are used as biofertilizers.
- Agar comes from red algae; alginic acid from brown algae.
- Harmful effects include water blooms, eutrophication and red tides.

Short-answer questions with model answers

1. What are algae? Give the basis of their classification.

Algae are eukaryotic, chlorophyll-bearing, photosynthetic thallophytes that lack true roots, stems and leaves. They are classified on the basis of pigmentation, nature of reserve food and cell wall composition.

2. Name the major classes of algae with one example each.

Chlorophyceae (*Spirogyra*), Cyanophyceae (*Anabaena*), Bacillariophyceae (*Navicula*), Phaeophyceae (*Sargassum*) and Rhodophyceae (*Gracilaria*).

3. How are blue-green algae useful in agriculture?

Blue-green algae such as *Anabaena*, *Nostoc* and *Aulosira* fix atmospheric nitrogen and are used as biofertilizers in paddy fields. *Azolla* with *Anabaena* is a valuable green manure for rice. They also help

reclaim alkaline soils.

Long-answer questions with model answer outlines

1. Describe the morphology and classification of algae.

Outline: (i) General features. (ii) Thallus types. (iii) Cell structure and pigments. (iv) Reserve food. (v) Reproduction - vegetative, asexual, sexual. (vi) Classification into five classes with characters. (vii) Table.

2. Discuss the economic importance of algae.

Outline: (i) Introduction. (ii) Beneficial - nitrogen fixation, biofertilizer, soil reclamation, food, agar, algin, diatomaceous earth, sewage treatment, oxygen, biofuel. (iii) Harmful - water blooms, eutrophication, red tides, fouling. (iv) Conclusion on their role in agriculture and industry.

Lecture 17
Viruses and Their Properties; Bacteriophages - Lytic and Lysogenic Cycl...

Viruses and Their Properties; Bacteriophages - Lytic and Lysogenic Cycles; Virulent and Temperate Phages

17.1 Introduction

Viruses are the smallest known infectious agents. They are acellular, that is, they are not cells; they consist of a nucleic acid core surrounded by a protein coat. They have no independent metabolism and can multiply only inside a living host cell. Because of this they are called **obligate intracellular parasites**. The study of viruses is called **virology**.

17.2 Properties of viruses

- They are much smaller than bacteria - about 20 to 300 nm; they can be seen only with an electron microscope.
- They contain only **one** type of nucleic acid - either DNA or RNA, never both.
- They have no cytoplasm, no ribosomes, no enzymes for energy production and no cell wall.
- They cannot grow or divide by themselves; they are reproduced by the host cell machinery.
- They are **host specific** - a virus infects only certain hosts (plant, animal or bacterial).
- They are resistant to antibiotics; they can be inactivated by heat, formaldehyde, chlorine and UV light.
- They can be crystallised and stored like chemical substances, yet they show the property of reproduction inside a host - hence they are regarded as a link between the living and the non-living.

17.3 Structure of viruses

- **Nucleic acid core** - the genetic material, either DNA or RNA; may be single-stranded or double-stranded, linear or circular.
- **Capsid** - the protein coat made of many small subunits called **capsomeres**. It protects the nucleic acid and helps in attachment to the host.
- **Envelope** - an additional outer layer of lipid and protein, present in some viruses (for example, influenza virus, HIV). It is derived from the host cell membrane.
- **Spikes** - glycoprotein projections on the envelope that help the virus attach to the host cell.
- **Symmetry** - viruses show helical, icosahedral (cubical) or complex symmetry. Bacteriophages have a complex symmetry (a head and a tail).

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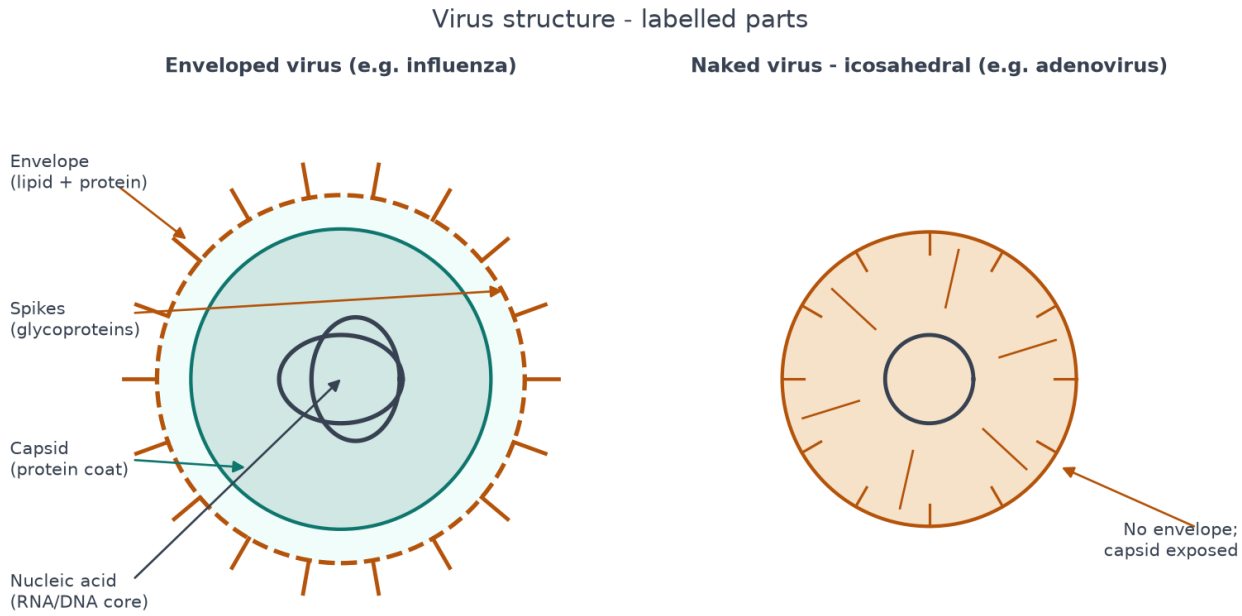


Figure 17.1 - Structure of an enveloped virus and a naked virus, labelled with capsid, nucleic acid, envelope and spikes.

17.4 Bacteriophages

Bacteriophages ("phages") are viruses that infect bacteria. They were discovered by Twort (1915) and d'Herelle (1917). The typical phage (T4) has a **head** (icosahedral, containing the DNA), a **collar**, a **sheath**, a **core**, a **base plate** and **tail fibres**.

- The phage attaches to the bacterial surface by its tail fibres.
- The DNA is injected into the cell through the tail; the empty protein coat (ghost) remains outside.

17.5 Lytic cycle (virulent phages)

A virulent phage always destroys the host cell. The steps are:

– **Attachment (adsorption)** - the phage tail fibres bind to specific receptors on the bacterial cell wall.

– **Penetration (injection)** - the phage DNA is injected into the host cell.

– **Biosynthesis (replication)** - the host DNA is broken down, and the phage DNA takes over the host machinery to make many copies of phage DNA and phage proteins.

– **Maturation (assembly)** - new phage particles are assembled from the DNA and protein subunits.

– **Lysis and release** - the host cell bursts (lyses) and releases many new phages (50-200), which then infect other cells. The cycle takes about 20-30 minutes.

17.6 Lysogenic cycle (temperate phages)

A temperate phage can either lyse the cell or, more often, establish a stable relationship with it.

- After penetration, the phage DNA **integrates into the host chromosome** and becomes a **prophage**.

- The prophage is copied along with the host chromosome at every cell division, so the daughter cells also carry it.
- The host cell is not destroyed; it is called a **lysogen**.
- Under certain conditions (UV light, chemicals, stress), the prophage is **induced** and enters the lytic cycle, producing new phages and lysing the cell.
- Temperate phages may also carry useful genes; in transduction they carry bacterial genes from one cell to another.

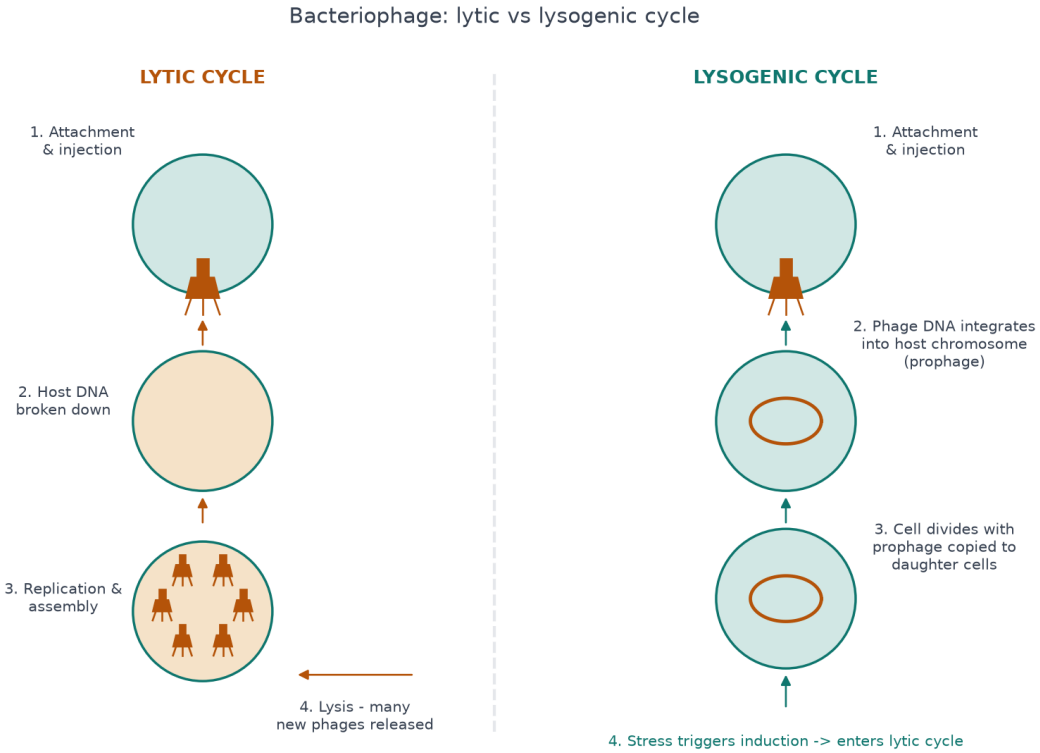


Figure 17.2 - Bacteriophage life cycles: the lytic cycle (left) and the lysogenic cycle (right), both starting from phage attachment.

17.7 Virulent and temperate phages

- **Virulent phages** - always follow the lytic cycle and kill the host (for example, T4, T7).
- **Temperate phages** - can follow either the lytic or the lysogenic cycle (for example, lambda phage, P1).

Table 17.1 - Comparison of viruses and bacteria

Feature	Virus	Bacterium
Size	20-300 nm	500-5000 nm
Structure	Nucleic acid + protein coat	Complete cell
Nucleic acid	DNA or RNA (one only)	DNA and RNA
Ribosomes	Absent	Present
Metabolism	None of its own	Independent
Multiplication	Only inside a host cell	Binary fission
Antibiotic sensitivity	Resistant	Sensitive

17.8 Agricultural relevance

- Plant viruses cause serious crop losses: tobacco mosaic virus, cucumber mosaic virus, tomato leaf curl virus, banana bunchy top virus, sugarcane mosaic virus, rice tungro virus and potato virus Y.
- Most plant viruses are transmitted by insect vectors such as aphids, whiteflies and leafhoppers; control therefore includes vector management.
- Bacteriophages are used in phage typing to identify bacteria and are being studied as biocontrol agents against bacterial plant pathogens such as *Xanthomonas* and *Ralstonia*.

Exam focus

Key points to remember

- Viruses are acellular obligate intracellular parasites; they contain DNA or RNA, never both.
- They consist of a nucleic acid core and a protein coat (capsid); some have an envelope and spikes.
- They are visible only under the electron microscope (20-300 nm).
- Bacteriophages infect bacteria; T4 is a virulent phage; lambda is a temperate phage.
- Lytic cycle: attachment, penetration, biosynthesis, maturation, lysis.
- Lysogenic cycle: the phage DNA integrates as a prophage and is copied with the host chromosome.

Short-answer questions with model answers

1. List any four properties of viruses.

(i) They are much smaller than bacteria (20-300 nm) and visible only under an electron microscope. (ii) They contain only one type of nucleic acid, DNA or RNA. (iii) They have no independent metabolism and multiply only inside a living host cell. (iv) They are resistant to antibiotics and can be crystallised.

2. Distinguish between the lytic and lysogenic cycles.

In the lytic cycle the phage DNA takes over the host machinery, new phages are formed and the host cell is lysed and destroyed. In the lysogenic cycle the phage DNA integrates into the host chromosome as a prophage, is copied at each cell division without harming the host, and enters the lytic cycle only when induced.

3. Draw and label the structure of a bacteriophage.

A T4 phage has an icosahedral head containing DNA, a collar, a sheath and core, a base plate and tail fibres used for attachment.

Long-answer questions with model answer outlines

1. Describe the properties and structure of viruses.

Outline: (i) Introduction and definition. (ii) Properties - size, nucleic acid, no metabolism, host specificity, resistance. (iii) Structure - nucleic acid, capsid, capsomeres, envelope, spikes, symmetry. (iv) Diagram. (v) Viruses as a link between living and non-living. (vi) Plant viruses and their transmission.

2. Explain the lytic and lysogenic cycles of bacteriophages.

Outline: (i) Introduction to bacteriophages. (ii) Structure of a phage. (iii) Lytic cycle steps with diagram. (iv) Lysogenic cycle and prophage. (v) Induction. (vi) Virulent vs temperate phages. (vii) Applications - phage typing and biocontrol.

Lecture 18
Reproduction in Bacteria - Population Growth and Growth Phases - Genera...

Reproduction in Bacteria - Population Growth and Growth Phases - Generation Time and Specific Growth Rate

18.1 Reproduction in bacteria

Bacteria reproduce mainly by **binary fission** (also called transverse fission), an asexual process. The steps are:

– The cell grows and the DNA replicates.

– The two DNA molecules attach to the plasma membrane and move apart.

– A septum (cross wall) forms in the middle of the cell.

– The cell divides into two identical daughter cells.

- Under favourable conditions, *E. coli* can divide every 20 minutes.
- Other methods of reproduction (not true reproduction but genetic transfer): budding (some bacteria), fragmentation (actinomycetes), conidia and spores (actinomycetes and some fungi-like bacteria), and endospores (a survival, not a reproductive, structure).
- Genetic transfer by transformation, conjugation and transduction is described in later lectures and is not reproduction in the strict sense.

18.2 Population growth

When bacteria are grown in a closed (batch) culture, the population does not increase indefinitely. The number of cells increases in a characteristic pattern called the **growth curve**, which has four phases.

18.3 The growth curve and its phases

18.3.1 Lag phase

- The cells are adapting to the new medium; they synthesise enzymes and other cell components but do not divide.
- The number of cells does not increase, though the cells increase in size and metabolic activity.
- The length of the lag phase depends on the age of the inoculum and the difference between the old and new media.

18.3.2 Log phase (exponential phase)

- The cells divide at a constant, maximum rate; the number doubles at regular intervals.
- When the number is plotted on a logarithmic scale, the curve is a straight line.
- Growth is balanced; the cells are most uniform and are used for biochemical and genetic studies.
- Nutrients are plentiful and waste products are low.

18.3.3 Stationary phase

- The growth rate equals the death rate, so the total number of viable cells remains constant.
- Caused by exhaustion of an essential nutrient, accumulation of toxic waste products, and crowding.
- The cells become smaller and more resistant; many secondary metabolites (such as antibiotics) are produced in this phase.

18.3.4 Death (decline) phase

- The death rate exceeds the growth rate and the number of viable cells falls.
- Cells die due to starvation and toxic products; some may lyse.
- A few cells may survive for long periods.

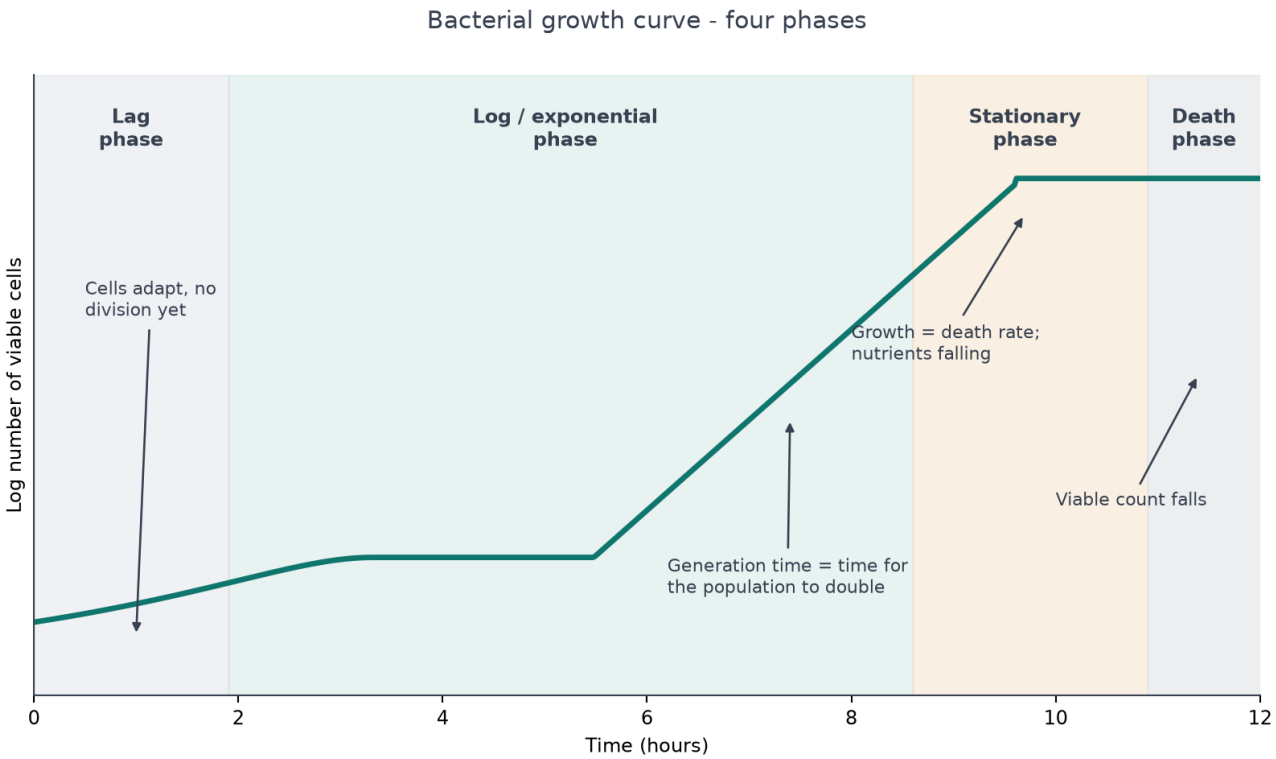


Figure 18.1 - The bacterial growth curve: lag, log (exponential), stationary and death phases, with time and log number of viable cells as axes.

18.4 Generation time (doubling time)

The **generation time** is the time taken for the population to double in number during the log phase. It is also called the doubling time.

- Generation time varies with the organism and conditions: *E. coli* about 20 minutes; *Mycobacterium tuberculosis* about 18-24 hours; *Nitrosomonas* about 10-12 hours.
- The number of cells after n generations is given by: N = N0 x 2^n, where N0 is the initial number and N the final number.
- Taking logarithms: n = (log N - log N0) / log 2 = 3.3 (log N - log N0).
- Generation time g = t / n, where t is the time interval.

18.5 Specific growth rate

The **specific growth rate** (mu) expresses the rate of growth per unit of biomass and is a measure of how fast the population is growing.

- During the log phase, growth is exponential: N = N0 x e^(mu t), so mu = (ln N - ln N0) / t.
- It is usually expressed in reciprocal hours (per hour, h^-1).
- The relation between specific growth rate and generation time is: mu = ln 2 / g = 0.693 / g.

Worked example 18.1 - Generation time and specific growth rate

Problem: A culture of *E. coli* has 1,000 cells per mL at the start of the log phase. After 3 hours the count is 64,000 cells per mL. Calculate the number of generations, the generation time and the specific growth rate.

Step 1: Write the formula N = N0 x 2^n. Here N0 = 1000, N = 64000, t = 3 h.

Step 2: Find n. 2^n = N / N0 = 64000 / 1000 = 64. Since 2^6 = 64, n = 6 generations.

Step 3: Generation time g = t / n = 3 hours / 6 = 0.5 hour = 30 minutes.

Step 4: Specific growth rate mu = 0.693 / g = 0.693 / 0.5 = 1.386 per hour.

Answer: 6 generations; generation time 30 minutes; specific growth rate about 1.39 h^-1.

Worked example 18.2 - Growth after a given time

Problem: A culture starts with 100 cells and has a generation time of 20 minutes. How many cells are present after 2 hours (120 minutes)?

Step 1: Number of generations n = total time / generation time = 120 / 20 = 6.

Step 2: N = N0 x 2^n = 100 x 2^6 = 100 x 64 = 6400 cells.

Answer: 6,400 cells after 2 hours.

Table 18.1 - The four phases of the growth curve

Phase	Cell number	Growth rate	Cause	Use
Lag	No change	Zero	Adaptation, enzyme synthesis	-
Log	Increases rapidly	Maximum, constant	Plenty of nutrients	Biochemical and genetic studies
Stationary	Constant	Growth = death	Nutrient exhaustion, waste	Secondary metabolite production
Death	Falls	Death > growth	Starvation, toxins	-

18.6 Methods of measuring growth (brief)

- Direct microscopic count, viable plate count (standard plate count), turbidity (optical density) measurement, dry weight and metabolic activity (for example, CO2 production).
- The viable plate count and turbidity methods are described in Lecture 20.

18.7 Agricultural relevance

- The generation time of *Rhizobium* determines how quickly a biofertilizer culture can be built up in the laboratory.
- In milk, the growth of lactic acid bacteria causes curdling; their growth rate is controlled by refrigeration.
- In food spoilage, a short generation time of 20 minutes means a single cell can become millions in a few hours - hence the need for rapid refrigeration.

Exam focus

Key points to remember

- Bacteria reproduce mainly by binary fission.
- The growth curve has four phases - lag, log, stationary and death.
- In the log phase the cells divide at the maximum, constant rate.
- Generation time is the time for the population to double: N = N0 x 2^n and g = t / n.
- Specific growth rate mu = 0.693 / g and is expressed per hour.
- *E. coli* has a generation time of about 20 minutes under ideal conditions.

Short-answer questions with model answers

1. Draw and explain the bacterial growth curve.

The curve has four phases: lag (adaptation, no division), log or exponential (rapid, constant division), stationary (growth equals death, nutrients exhausted) and death (viable count falls). Refer to Figure 18.1.

2. Define generation time and give the formula.

Generation time (doubling time) is the time taken for a population to double during the log phase. It is given by g = t / n, where n = 3.3 (log N - log N0) and t is the time interval.

3. What is specific growth rate?

The specific growth rate (mu) is the rate of growth per unit biomass, given by mu = (ln N - ln N0) / t during the log phase, and related to generation time by mu = 0.693 / g. It is expressed in per hour.

Long-answer questions with model answer outlines

1. Describe the growth curve of bacteria and the factors that produce each phase.

Outline: (i) Binary fission and population growth. (ii) Batch culture. (iii) Lag phase - adaptation. (iv) Log phase - balanced growth. (v) Stationary phase - nutrient exhaustion and waste. (vi) Death phase. (vii) Diagram and uses of each phase.

2. Explain generation time and specific growth rate with worked examples.

Outline: (i) Definition of generation time. (ii) Formula N = N0 x 2^n and g = t / n. (iii) Worked example. (iv) Definition of specific growth rate. (v) Formula mu = (ln N - ln N0)/t and mu = 0.693/g. (vi) Factors affecting both. (vii) Agricultural significance.

PART IV - GROWTH, NUTRITION AND METABOLISM

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Lecture 19
Batch and Continuous Culture - Chemostat and Turbidostat; Synchronous C...

Batch and Continuous Culture - Chemostat and Turbidostat; Synchronous Culture; Diauxic Growth Curve

19.1 Batch culture

A **batch culture** is a closed system in which a fixed amount of medium is inoculated and the organisms grow until a nutrient is exhausted or a toxic product accumulates. The population passes through all four phases of the growth curve (lag, log, stationary, death). It is the usual method in the laboratory for growing cultures and for producing metabolites such as antibiotics.

- Advantages: simple, cheap, easy to set up.
- Disadvantages: nutrient concentration and pH change, waste products accumulate, growth rate is not constant, and the cells are not in a steady physiological state.

19.2 Continuous culture

A **continuous culture** is an open system in which fresh medium is added and spent medium (with cells) is removed at the same rate, so that the volume and the nutrient concentration remain constant. The cells can be kept growing at a constant rate for long periods. Two common types are the chemostat and the turbidostat.

19.2.1 Chemostat

- Fresh medium is supplied at a constant rate and culture is removed at the same rate, keeping the volume constant.
- Growth rate is controlled by the rate of addition of a **limiting nutrient** (usually the carbon source or nitrogen source).
- The **dilution rate** (D) is the flow rate divided by the volume of the culture. At steady state, the specific growth rate of the cells equals the dilution rate (mu = D).
- By changing the dilution rate, the growth rate of the culture can be controlled precisely.
- Uses: study of growth physiology, continuous production of cells, enzymes and metabolites.

19.2.2 Turbidostat

- Fresh medium is added automatically whenever the cell density (turbidity) rises above a set level, and the excess culture is removed.
- The cell density is kept constant by a photocell that measures the turbidity.
- Unlike the chemostat, no nutrient is deliberately made limiting; the cells grow at their maximum rate for the given conditions.
- Uses: maintaining a constant, high cell density for studies of growth at the maximum rate.

19.3 Synchronous culture

A **synchronous culture** is one in which all the cells divide at the same time, so that the population passes through the cell cycle together. This is useful for studying events of the cell cycle such as DNA replication and cell division.

- Methods of obtaining synchronous cultures:
- **Selection** - cells of a particular size or age are separated physically (for example, by filtration or by centrifugation in a gradient) and then allowed to grow.
- **Induction** - all the cells are brought to the same physiological state by a treatment such as a temperature change or starvation, then released together. For example, a "baby machine" collects newly divided cells.
- Synchrony is usually lost after a few generations because of small differences in generation time.

19.4 Diauxic growth

Diauxic growth (biphasic growth) occurs when a bacterium is grown in a medium containing two sugars, such as glucose and lactose. The organism uses the sugar it prefers (glucose) first; when this is exhausted, growth stops briefly (a second lag) while the enzymes for the second sugar (lactose) are synthesised; growth then resumes using the second sugar.

- The resulting curve shows two log phases separated by a lag - hence the name "diauxic".
- This is an example of **catabolite repression**: glucose represses the synthesis of the enzymes for lactose.
- It demonstrates that bacteria regulate their enzyme synthesis according to the nutrients available, and that they use the most easily metabolised substrate first.

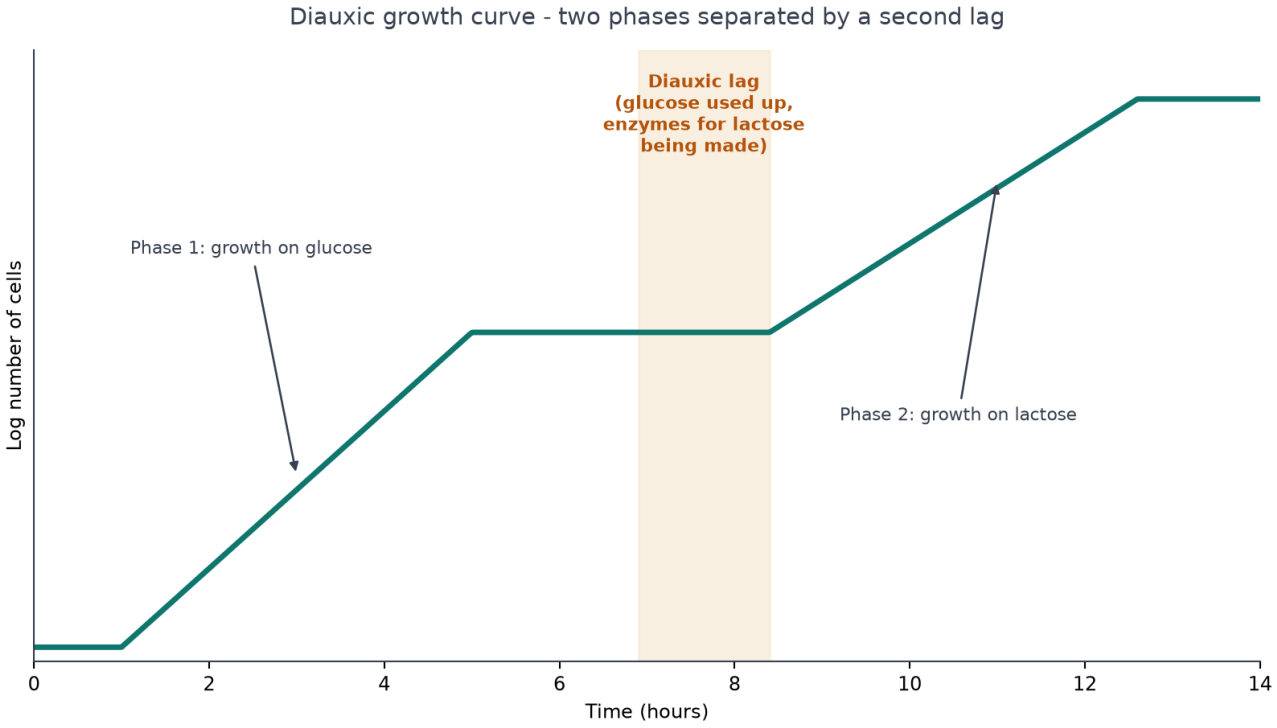


Figure 19.1 - Diauxic growth curve: growth on glucose, a lag while lactose enzymes are formed, then a second growth phase on lactose.

Table 19.1 - Batch and continuous culture compared

Feature	Batch culture	Continuous culture
System	Closed	Open
Medium	Fixed, not renewed	Continuously renewed
Growth phases	All four phases occur	Cells held in log phase
Growth rate	Changes with time	Constant (controlled)
Steady state	Not achieved	Achieved
Main use	Small-scale culture, metabolite production	Growth studies, continuous production

19.5 Agricultural and industrial relevance

- Continuous culture (chemostat) is used in industry for the continuous production of single-cell protein, enzymes and organic acids.
- In the rumen of cattle, the microbial population behaves like a continuous culture, with feed coming in and digesta moving out.
- In a biogas plant, the digester works as a continuous or semi-continuous system.
- In dairy fermentations, batch cultures of starter organisms are used for curd and cheese.

Exam focus

Key points to remember

- Batch culture is closed; the population passes through all four growth phases.
- Continuous culture is open; fresh medium is added and spent medium removed at the same rate.
- In a chemostat the growth rate is controlled by a limiting nutrient and mu = D.
- In a turbidostat the cell density is kept constant by controlling the flow with a turbidity sensor.
- A synchronous culture has all cells dividing together; it is used to study the cell cycle.
- Diauxic growth shows two log phases separated by a lag, due to catabolite repression.

Short-answer questions with model answers

1. Distinguish between batch and continuous culture.

In a batch culture a fixed amount of medium is inoculated and the cells pass through lag, log, stationary and death phases; nutrients fall and waste accumulates. In a continuous culture fresh medium is added and spent medium removed at the same rate, so the cells remain in the log phase in a steady state.

2. What is a chemostat?

A chemostat is a continuous culture device in which fresh medium is added at a constant rate and culture is removed at the same rate, keeping the volume constant. Growth rate is controlled by the rate of supply of a limiting nutrient, and at steady state the specific growth rate equals the dilution rate.

3. Explain diauxic growth.

When a bacterium is grown on two sugars such as glucose and lactose, it uses glucose first; after glucose is exhausted there is a lag while lactose-degrading enzymes are synthesised, after which growth resumes on lactose. The curve therefore shows two growth phases separated by a lag, a result of

catabolite repression.

Long-answer questions with model answer outlines

1. Describe continuous culture systems and their applications.

Outline: (i) Limitation of batch culture. (ii) Principle of continuous culture. (iii) Chemostat - dilution rate, limiting nutrient, mu = D. (iv) Turbidostat - turbidity control. (v) Comparison. (vi) Applications in industry and research. (vii) Rumen and biogas examples.

2. Explain synchronous culture and the diauxic growth curve.

Outline: (i) Meaning of synchronous culture. (ii) Selection and induction methods. (iii) Uses in cell-cycle studies. (iv) Diauxic growth - definition. (v) Glucose and lactose example. (vi) Catabolite repression. (vii) Diagram and significance.

Lecture 20
Conditions for Growth - Temperature Requirements - Aerobes and Anaerobe...

Conditions for Growth - Temperature Requirements - Aerobes and Anaerobes - Other Factors Influencing Growth; Methods of Assessment of Growth

20.1 Introduction

Microbial growth means an increase in the number of cells, not in the size of a cell. Growth depends on many environmental factors; the most important are temperature, oxygen, pH, water activity, nutrients and osmotic pressure. Each organism has a range of tolerance and an optimum for each factor.

20.2 Temperature requirements

Temperature affects the rate of all enzyme reactions. Each microbe has a **minimum**, an **optimum** and a **maximum** temperature for growth, called the **cardinal temperatures**.

- **Psychrophiles** - grow at low temperature, optimum about 10-15 C, maximum about 20 C; found in polar regions, deep sea and refrigerated food. Example: *Pseudomonas* species that spoil refrigerated food.
- **Psychrotrophs (psychrotolerant)** - grow at low temperature but have an optimum above 20 C; important in food spoilage in refrigerators. Example: *Listeria monocytogenes*.
- **Mesophiles** - optimum about 25-40 C (usually 37 C); most bacteria, including all pathogens of humans and animals and most soil bacteria. Example: *E. coli*, *Rhizobium*.
- **Thermophiles** - optimum about 45-60 C; found in hot springs, compost and solar-heated soils. Example: *Bacillus stearothermophilus*.
- **Hyperthermophiles (extreme thermophiles)** - optimum above 80 C; usually archaea from hot vents. Example: *Thermus aquaticus* (source of Taq polymerase), *Pyrococcus*.
- At high temperature, proteins denature and membranes become too fluid; at low temperature, enzyme activity is too slow and membranes become rigid.

20.3 Oxygen requirements

- **Obligate aerobes** - must have oxygen; they use it as the final electron acceptor. Example: *Pseudomonas*, *Mycobacterium*.
- **Obligate (strict) anaerobes** - cannot grow in the presence of oxygen; oxygen is toxic to them. Example: *Clostridium*, *Bacteroides*, methanogens.
- **Facultative anaerobes** - grow better with oxygen but can also grow without it by fermentation. Example: *E. coli*, *Staphylococcus*, yeasts.
- **Microaerophiles** - need a low concentration of oxygen (about 2-10%). Example: *Campylobacter*, *Helicobacter*.
- **Aerotolerant anaerobes** - do not use oxygen but are not harmed by it. Example: *Lactobacillus*, *Streptococcus*.

- **Capnophiles** - need a high concentration of carbon dioxide. Example: *Neisseria*.
- Oxygen is toxic to anaerobes because it forms toxic products such as superoxide radicals and hydrogen peroxide, which anaerobes lack the enzymes to destroy.

20.4 Other factors influencing growth

- **pH** - most bacteria grow best at pH 6.5-7.5 (neutrophiles); acidophiles at low pH (for example, *Lactobacillus*, pH 4-5; *Thiobacillus* at pH 2); alkaliphiles at high pH. Fungi prefer slightly acidic pH (5-6). Moulds tolerate a wider pH range than bacteria, which is why they spoil acidic foods.
- **Water activity (a w)** - the availability of water. Most bacteria need a high water activity; halophiles tolerate high salt; xerophiles tolerate drying. Adding salt or sugar (as in pickles and jams) reduces water activity and prevents spoilage.
- **Osmotic pressure** - high salt or sugar causes plasmolysis; halotolerant organisms survive. *Halobacterium* requires high salt.
- **Nutrients** - carbon, nitrogen, minerals, growth factors and vitamins must be available.
- **Osmotic and hydrostatic pressure**, **radiation** and **hydrostatic pressure** also influence growth.
- **Biotic factors** - competition, antagonism, and products of other microbes.

20.5 Methods of assessment of growth

20.5.1 Direct methods

- **Direct microscopic count** - a known volume of culture is placed in a counting chamber (haemocytometer or Petroff-Hausser chamber) and the cells are counted under the microscope. It is quick but cannot distinguish living from dead cells.
- **Viable plate count (standard plate count)** - the sample is diluted and plated; after incubation, each colony is assumed to have arisen from one cell. The result is expressed as colony-forming units (CFU) per mL. It counts only living cells. It is accurate but takes time (24-48 hours).
- **Membrane filter count** - the sample is passed through a membrane filter, which is placed on a medium; colonies grow on the filter. Useful for water and dilute samples.

20.5.2 Indirect methods

- **Turbidity (optical density)** - the culture is placed in a colorimeter or spectrophotometer and the light absorbed or scattered is measured; the optical density is proportional to the cell number. It is rapid but cannot distinguish living from dead cells.
- **Dry weight** - the cells are washed, dried and weighed. Useful for filamentous organisms and for large cell masses.
- **Metabolic activity** - measurement of oxygen uptake, carbon dioxide production, or acid production; used for estimating the number of metabolically active cells.
- **Most probable number (MPN)** - a statistical method in which the sample is diluted in a series of tubes and the number of positive tubes is used to estimate the number of cells (used for coliforms in water).

Worked example 20.1 - Colony forming unit calculation from a dilution plate

Problem: One mL of a sample was diluted serially. From the 10^-5 dilution, 0.1 mL was spread on a plate. After incubation, 65 colonies appeared. Calculate the number of CFU per mL of the original sample.

Step 1: Write the formula. CFU per mL = number of colonies / (volume plated x dilution).

Step 2: Substitute values. Number of colonies = 65; volume plated = 0.1 mL; dilution = 10^-5.

Step 3: Compute. CFU per mL = 65 / (0.1 x 10^-5) = 65 / 10^-6 = 65 x 10^6 = 6.5 x 10^7.

Answer: The original sample contained about 6.5 x 10^7 colony forming units per mL.

Worked example 20.2 - Dilution series for a plate count

Problem: A sample is expected to contain about 10^7 cells per mL. Design a dilution series to give 30-300 colonies per plate when 1 mL is plated.

Step 1: Decide the target. If about 100 colonies are wanted on the plate, the dilution should give about 100 cells per mL.

Step 2: Find the dilution needed. Original count is 10^7 per mL; to get 10^2 per mL, a dilution of 10^5 is needed.

Step 3: Prepare the series. Transfer 1 mL of sample into 9 mL of diluent (this is 10^-1), mix; repeat to 10^-2, 10^-3, 10^-4 and 10^-5. Each step dilutes ten times.

Step 4: Plate 1 mL from the 10^-5 dilution. Expect about 100 colonies, which lies in the acceptable 30-300 range.

Answer: Use a ten-fold dilution series up to 10^-5 and plate 1 mL from the 10^-5 (and, as a check, the 10^-4) dilution.

Table 20.1 - Classification of bacteria by temperature and oxygen requirement

Group	Temperature range (optimum)	Example
Psychrophile	10-15 C	*Pseudomonas* (polar)
Psychrotroph	20-30 C (grows at 5 C)	*Listeria monocytogenes*
Mesophile	25-40 C	*E. coli*, *Rhizobium*
Thermophile	45-60 C	*Bacillus stearothermophilus*
Hyperthermophile	above 80 C	*Thermus aquaticus*
Oxygen group	Requirement	Example
Obligate aerobe	Needs O2	*Pseudomonas*
Obligate anaerobe	O2 is toxic	*Clostridium*
Facultative anaerobe	Better with O2, grows without	*E. coli*
Microaerophile	Low O2 (2-10%)	*Campylobacter*
Aerotolerant anaerobe	Does not use O2, not harmed	*Lactobacillus*

20.6 Agricultural relevance

- Refrigeration of milk, meat and vegetables slows the growth of mesophiles but allows psychrotrophs to grow - hence the importance of the cold chain.
- Composting reaches 55-65 C, favouring thermophiles that destroy pathogens and weed seeds.
- Water activity control by drying, salting and sugaring is used in the preservation of farm produce.
- Viable plate counts are used to assess the quality of biofertilizer inoculants (the number of viable *Rhizobium* cells per gram of carrier).

Exam focus

Key points to remember

- Cardinal temperatures are the minimum, optimum and maximum for growth.
- Mesophiles (optimum 25-40 C) include most bacteria and all pathogens of warm-blooded animals.
- Obligate aerobes need oxygen; obligate anaerobes are killed by it.
- pH, water activity, osmotic pressure and nutrients also control growth.
- Viable plate count measures living cells as CFU per mL; turbidity measures total cell mass rapidly.
- CFU per mL = colonies / (volume plated x dilution).

Short-answer questions with model answers

1. Classify bacteria according to their temperature requirements.

Psychrophiles (optimum 10-15 C), psychrotrophs (grow at low temperature, optimum above 20 C), mesophiles (25-40 C), thermophiles (45-60 C) and hyperthermophiles (above 80 C).

2. Why is oxygen toxic to obligate anaerobes?

Oxygen forms toxic products such as superoxide radicals and hydrogen peroxide. Obligate anaerobes lack the enzymes (superoxide dismutase, catalase) needed to destroy these products, so oxygen kills them.

3. Give the formula for the viable plate count.

CFU per mL = number of colonies / (volume plated in mL x dilution factor).

Long-answer questions with model answer outlines

1. Describe the factors that influence the growth of microorganisms.

Outline: (i) Meaning of growth. (ii) Temperature - cardinal temperatures and groups. (iii) Oxygen - aerobes, anaerobes, facultative, microaerophiles, aerotolerant. (iv) pH. (v) Water activity and osmotic pressure. (vi) Nutrients. (vii) Biotic factors. (viii) Agricultural significance.

2. Explain the methods used to assess microbial growth.

Outline: (i) Direct microscopic count. (ii) Viable plate count with formula. (iii) Membrane filter count. (iv) Turbidity method. (v) Dry weight. (vi) Metabolic activity. (vii) MPN. (viii) Worked examples and choice of method.

Lecture 21
Nutritional Types of Bacteria; Metabolic Diversity in Microbes

Nutritional Types of Bacteria; Metabolic Diversity in Microbes

21.1 Nutritional requirements

All living cells need a source of carbon, a source of energy, a source of nitrogen, minerals and water, and often growth factors. Microorganisms differ greatly in the sources they can use, and this gives rise to different nutritional types.

- **Carbon source** - autotrophs use carbon dioxide; heterotrophs use organic carbon such as sugars, organic acids and amino acids.
- **Energy source** - phototrophs use light; chemotrophs use chemical compounds.
- **Electron (hydrogen) donor** - lithotrophs use inorganic compounds; organotrophs use organic compounds.
- **Nitrogen source** - from nitrogen gas (nitrogen fixers), nitrate, ammonia or organic nitrogen.
- **Growth factors** - vitamins, amino acids and bases that the organism cannot synthesise; required in small amounts.

21.2 Nutritional types of bacteria

The nutritional type is named by combining the source of energy and the source of carbon (and, in a fuller scheme, the electron donor).

- **Photoautotrophs** - energy from light, carbon from carbon dioxide. Examples: cyanobacteria, green sulphur bacteria.
- **Photoheterotrophs** - energy from light, carbon from organic compounds. Examples: purple non-sulphur bacteria.
- **Chemoautotrophs (chemolithoautotrophs)** - energy from the oxidation of inorganic compounds, carbon from carbon dioxide. Examples: nitrifying bacteria (*Nitrosomonas*, *Nitrobacter*), sulphur bacteria (*Thiobacillus*), iron bacteria.
- **Chemoheterotrophs (chemoorganoheterotrophs)** - energy and carbon both from organic compounds. This is the largest group and includes most bacteria, fungi and protozoa. Example: *E. coli*, *Rhizobium*, *Bacillus*.
- **Saprophytes** - chemoheterotrophs that live on dead organic matter; important decomposers.
- **Parasites** - chemoheterotrophs that live on or in a living host and obtain nutrients from it; many are pathogens.

21.3 Metabolic diversity in microbes

Microbes show a remarkable variety of metabolic pathways, which allows them to live in almost every environment on earth.

- **Ways of obtaining energy** - photosynthesis (light), respiration (organic or inorganic electron donors with an external electron acceptor), fermentation (organic compounds as both donor and acceptor),

and chemolithotrophy (oxidation of inorganic compounds).

- **Ways of obtaining carbon** - autotrophy (CO2 fixation, for example by the Calvin cycle) and heterotrophy.
- **Oxygen relationships** - aerobic, anaerobic, facultative, microaerophilic, aerotolerant.
- **Nitrogen metabolism** - nitrogen fixation (conversion of N2 to ammonia by nitrogenase), nitrification (ammonia to nitrite to nitrate), denitrification (nitrate to nitrogen gas), ammonification (organic N to ammonia).
- **Special metabolisms** - methanogenesis (production of methane by archaea), sulphate reduction, iron oxidation and reduction, anoxygenic and oxygenic photosynthesis, and hydrogen metabolism.
- This metabolic diversity is the basis of the role of microbes in the biogeochemical cycles of carbon, nitrogen, sulphur, phosphorus and iron.

21.4 The nitrogen cycle and its microbes

- **Nitrogen fixation** - *Rhizobium* (symbiotic, in legume nodules), *Azotobacter* and *Azospirillum* (free-living/associative), cyanobacteria (*Anabaena*, *Nostoc*). Converts N2 to NH3.
- **Nitrification** - *Nitrosomonas* converts ammonia to nitrite; *Nitrobacter* converts nitrite to nitrate. This makes nitrogen available to plants.
- **Denitrification** - *Pseudomonas*, *Thiobacillus denitrificans* convert nitrate to N2 gas; causes loss of soil nitrogen.
- **Ammonification** - decomposition of organic nitrogen to ammonia by many bacteria and fungi.

Table 21.1 - Nutritional types of bacteria

Nutritional type	Energy source	Carbon source	Example
Photoautotroph	Light	CO2	Cyanobacteria, green sulphur bacteria
Photoheterotroph	Light	Organic compounds	Purple non-sulphur bacteria
Chemoautotroph	Inorganic compounds	CO2	*Nitrosomonas*, *Thiobacillus*
Chemoheterotroph	Organic compounds	Organic compounds	*E. coli*, *Rhizobium*, fungi

21.5 Agricultural relevance

- Chemoautotrophic nitrifying bacteria make nitrogen available to crops; denitrifying bacteria cause loss of nitrogen from waterlogged soils.
- Symbiotic and free-living nitrogen fixers reduce the need for chemical nitrogen fertilizer.
- Sulphur-oxidising bacteria (*Thiobacillus*) are used to reclaim alkaline soils and to produce sulphuric acid for soil amendment.

Exam focus

Key points to remember

- Nutritional types are named by the source of energy and carbon.
- Photoautotrophs use light and CO2; chemoautotrophs use inorganic compounds and CO2.
- Chemoheterotrophs use organic compounds for both energy and carbon and form the largest group.
- Saprophytes live on dead matter; parasites live on living hosts.
- Metabolic diversity underlies the biogeochemical cycles.
- Nitrification, denitrification, nitrogen fixation and ammonification are the key steps of the nitrogen cycle.

Short-answer questions with model answers

1. Classify bacteria according to their nutritional requirements.

Photoautotrophs (light energy, CO2 carbon), photoheterotrophs (light, organic carbon), chemoautotrophs (inorganic energy source, CO2 carbon) and chemoheterotrophs (organic energy and carbon source).

2. What are growth factors? Give an example.

Growth factors are organic compounds such as vitamins, amino acids and bases that an organism cannot synthesise and must obtain ready-made in small amounts. Example: biotin and thiamine required by some *Lactobacillus* species.

3. Name the bacteria involved in nitrification and denitrification.

Nitrification: *Nitrosomonas* (ammonia to nitrite) and *Nitrobacter* (nitrite to nitrate). Denitrification: *Pseudomonas* and *Thiobacillus denitrificans* (nitrate to nitrogen gas).

Long-answer questions with model answer outlines

1. Explain the nutritional types of bacteria with examples.

Outline: (i) Nutritional requirements. (ii) Basis of classification - energy, carbon, electron donor. (iii) Photoautotrophs. (iv) Photoheterotrophs. (v) Chemoautotrophs. (vi) Chemoheterotrophs, saprophytes and parasites. (vii) Table and agricultural importance.

2. Describe metabolic diversity in microorganisms and its significance.

Outline: (i) Meaning of metabolic diversity. (ii) Energy-yielding mechanisms. (iii) Carbon assimilation. (iv) Oxygen relationships. (v) Nitrogen metabolism and the nitrogen cycle. (vi) Special metabolisms - methanogenesis, sulphate reduction, iron metabolism. (vii) Role in biogeochemical cycles and agriculture.

Lecture 22
Aerobic Respiration and Anaerobic Respiration

Aerobic Respiration and Anaerobic Respiration

22.1 Introduction

Respiration is the process by which cells obtain energy by the stepwise oxidation of a substrate (usually glucose), with the electrons being passed to a final electron acceptor. If the final acceptor is oxygen, the process is **aerobic respiration**; if it is some other inorganic compound, the process is **anaerobic respiration**. In both cases energy is conserved as ATP.

22.2 Aerobic respiration

Aerobic respiration is the complete oxidation of glucose to carbon dioxide and water using oxygen as the final electron acceptor. It gives the maximum energy.

22.2.1 Glycolysis (Embden-Meyerhof pathway)

- Glucose is converted to two molecules of pyruvate in a series of ten enzyme reactions in the cytoplasm.
- Two molecules of ATP are used and four are produced, giving a net gain of 2 ATP, along with 2 NADH.

22.2.2 Pyruvate oxidation and the Krebs cycle (TCA cycle)

- Pyruvate is converted to acetyl-CoA with the release of one CO2 and the formation of one NADH per pyruvate.
- Acetyl-CoA enters the Krebs cycle, where it is oxidised to two CO2 with the production of 3 NADH, 1 FADH2 and 1 ATP (GTP) per turn of the cycle.
- The Krebs cycle occurs in the cytoplasm in bacteria (in the mitochondrial matrix in eukaryotes).

22.2.3 Electron transport chain and oxidative phosphorylation

- The NADH and FADH2 produced pass their electrons to a chain of carriers in the membrane - flavoproteins, cytochromes and quinones.
- As electrons move down the chain, protons are pumped out of the cell, creating a proton gradient (the proton motive force).
- Oxygen is the final electron acceptor and combines with electrons and protons to form water.
- The proton gradient drives ATP synthesis by the enzyme ATP synthase - this is **oxidative phosphorylation**.
- In bacteria, the electron transport chain is located in the plasma membrane.

22.2.4 Energy yield

- The complete oxidation of one molecule of glucose yields about 36-38 ATP (some textbooks give 38, others 36 or 30 depending on the assumptions about the transport of NADH). The standard range

taught is 36-38 ATP.

- Most of the ATP (about 34) comes from oxidative phosphorylation, with only 2 from glycolysis and 2 from the Krebs cycle.

22.3 Anaerobic respiration

In anaerobic respiration the final electron acceptor is an inorganic compound other than oxygen. The electron transport chain is similar to that of aerobic respiration, but the terminal acceptor is different and the energy yield is lower.

- Common final electron acceptors and their products:
- **Nitrate** reduced to nitrite or nitrogen gas (nitrate respiration; *Pseudomonas*, *Bacillus*).
- **Sulphate** reduced to hydrogen sulphide (*Desulfovibrio*).
- **Carbon dioxide** reduced to methane (methanogens).
- **Ferric iron** reduced to ferrous iron.
- Energy yield is less than aerobic respiration (fewer ATP per glucose) but more than fermentation.
- Anaerobic respiration is important in waterlogged soils, sediments, the rumen and sewage digesters.

Table 22.1 - Aerobic vs anaerobic respiration

Feature	Aerobic respiration	Anaerobic respiration
Final electron acceptor	Oxygen	Inorganic compounds (nitrate, sulphate, CO2, Fe3+)
Products	CO2 and water	Nitrite/N2, H2S, methane, Fe2+
Electron transport chain	Present	Present
ATP yield per glucose	About 36-38	Less than aerobic (about 2-36)
Organisms	Most aerobes	*Pseudomonas*, *Desulfovibrio*, methanogens
Environment	Well-aerated	Waterlogged, sediments, rumen

22.4 Agricultural relevance

- In waterlogged paddy soils, anaerobic respiration dominates; sulphate reduction produces hydrogen sulphide, which can damage roots and cause "akiochi" disease in rice.
- Denitrification in waterlogged soils causes loss of nitrogen fertilizer as N2 gas.
- In the rumen, anaerobic respiration and fermentation by microbes digest cellulose and produce volatile fatty acids and methane.

Exam focus

Key points to remember

- Aerobic respiration oxidises glucose completely to CO2 and water, using oxygen as the final acceptor.
- Glycolysis gives 2 ATP and 2 NADH; the Krebs cycle gives 3 NADH, 1 FADH2 and 1 ATP per turn.

- The electron transport chain builds a proton gradient that drives ATP synthesis (oxidative phosphorylation).
- Aerobic respiration yields about 36-38 ATP per glucose.
- Anaerobic respiration uses nitrate, sulphate, CO2 or Fe3+ as the final electron acceptor and yields less energy.
- In bacteria the electron transport chain is in the plasma membrane.

Short-answer questions with model answers

1. Define aerobic and anaerobic respiration.

Aerobic respiration is the complete oxidation of a substrate with oxygen as the final electron acceptor, producing CO2 and water. Anaerobic respiration uses an inorganic compound other than oxygen (nitrate, sulphate, CO2 or Fe3+) as the final electron acceptor, producing nitrite or nitrogen, hydrogen sulphide, methane or ferrous iron.

2. Where does the electron transport chain occur in bacteria and what is its role?

In bacteria the electron transport chain is located in the plasma membrane. It transfers electrons from NADH and FADH2 to the final acceptor, pumping protons to create a proton gradient that drives ATP synthesis.

3. How much ATP is produced by aerobic respiration of one glucose molecule?

About 36-38 ATP in the standard textbook range, most of it from oxidative phosphorylation.

Long-answer questions with model answer outlines

1. Describe aerobic respiration with its stages and energy yield.

Outline: (i) Definition. (ii) Glycolysis. (iii) Pyruvate oxidation and Krebs cycle. (iv) Electron transport chain and oxidative phosphorylation. (v) Chemiosmotic theory and proton motive force. (vi) Energy yield. (vii) Location in bacteria and significance.

2. Explain anaerobic respiration and compare it with aerobic respiration.

Outline: (i) Definition and principle. (ii) Final electron acceptors and products. (iii) Organisms involved. (iv) Energy yield. (v) Comparison table. (vi) Importance in waterlogged soils, rumen and sewage. (vii) Agricultural significance.

Lecture 23
Fermentative Mode of Respiration

Fermentative Mode of Respiration

23.1 Introduction

Fermentation is a mode of energy generation in which an organic compound is both the electron donor and the final electron acceptor. There is no electron transport chain and no external electron acceptor such as oxygen. The substrate is only partially oxidised, so the energy yield is low. Fermentation occurs in the cytoplasm and does not require oxygen.

- The term "fermentation" is used in two senses: in the strict biochemical sense, an anaerobic process in which an organic compound is the electron acceptor; in the industrial sense, any large-scale microbial process, aerobic or anaerobic (for example, citric acid "fermentation").
- Fermentation is carried out by **facultative anaerobes** and **obligate anaerobes**, and also by yeasts.
- The purpose of fermentation is to regenerate the oxidised coenzyme NAD+ from NADH so that glycolysis can continue and ATP can be produced.

23.2 Glycolysis in fermentation

Fermentation begins with glycolysis, which converts one glucose to two pyruvate with a net gain of 2 ATP and 2 NADH. The pyruvate is then converted to various end products, and in the process NADH is oxidised back to NAD+.

23.3 Types of fermentation

23.3.1 Lactic acid fermentation

- Pyruvate is reduced to lactic acid by the enzyme lactate dehydrogenase, using NADH.
- **Homolactic fermentation** - lactic acid is the only product; carried out by *Lactobacillus*, *Streptococcus*, *Leuconostoc*. Used in curd, cheese, yoghurt and sauerkraut.
- **Heterolactic fermentation** - produces lactic acid, ethanol and CO2; carried out by *Leuconostoc mesenteroides*.
- In muscle cells, lactic acid fermentation occurs during vigorous exercise when oxygen is limiting.

23.3.2 Alcoholic fermentation

- Pyruvate is decarboxylated to acetaldehyde with the release of CO2; acetaldehyde is then reduced to ethanol by alcohol dehydrogenase using NADH.
- Carried out by yeasts such as *Saccharomyces cerevisiae*.
- Used in the production of beer, wine, bread (CO2 makes the dough rise) and industrial alcohol.

23.3.3 Other fermentations

- **Mixed acid fermentation** - *E. coli* produces lactic, acetic, succinic and formic acids, ethanol and CO2; detected by the methyl red test.

- **Butanediol fermentation** - *Enterobacter* produces butanediol and ethanol; detected by the Voges-Proskauer test.
- **Butyric acid fermentation** - *Clostridium butyricum* produces butyric acid, CO2 and hydrogen.
- **Propionic acid fermentation** - *Propionibacterium* produces propionic acid and CO2; used in Swiss cheese.
- **2,3-butanediol, acetone-butanol and butanol-isopropanol fermentations** - industrial importance.
- **Stickland reaction** - fermentation of pairs of amino acids by *Clostridium*.

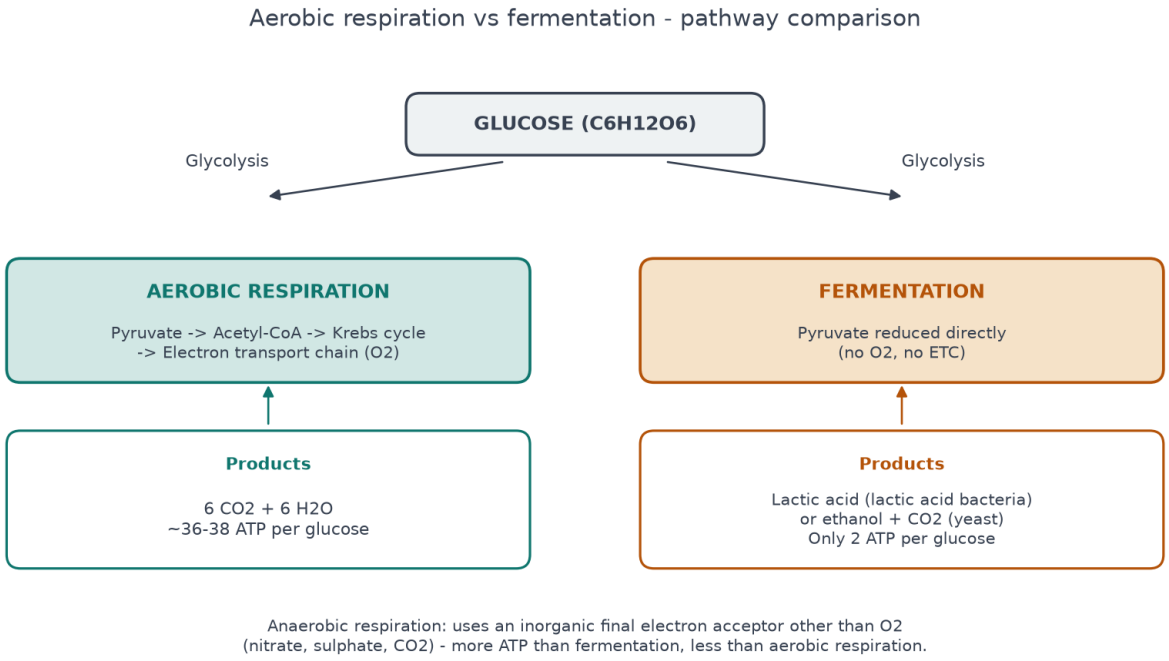


Figure 23.1 - Comparison of aerobic respiration and fermentation starting from glucose, showing the different products and ATP yields.

Table 23.1 - Aerobic respiration vs anaerobic respiration vs fermentation

Feature	Aerobic respiration	Anaerobic respiration	Fermentation
Final electron acceptor	O2	Inorganic (NO3-, SO4 2-, CO2)	Organic compound (pyruvate)
Electron transport chain	Present	Present	Absent
Oxygen needed	Yes	No	No
Substrate oxidation	Complete	Complete	Partial
End products	CO2 + H2O	Nitrite/N2, H2S, CH4	Organic acids, alcohols, CO2
ATP per glucose	About 36-38	Intermediate	Only 2
Organisms	Aerobes	Some anaerobes	Facultative and obligate anaerobes, yeasts

23.4 Agricultural and industrial relevance

- Curd, cheese, butter, ghee and yoghurt depend on lactic acid fermentation by *Lactobacillus* and *Streptococcus*.
- Silage (preserved fodder) is produced by natural lactic acid fermentation, which lowers the pH and preserves the fodder.
- Bread, idli, dosa and dhokla depend on alcoholic and lactic acid fermentation by yeasts and bacteria.
- Vinegar is produced by the oxidation of ethanol to acetic acid by *Acetobacter*.
- Biogas and butanol-acetone production use fermentative bacteria.
- In waterlogged soils, fermentative bacteria produce organic acids and alcohols, which can harm plant roots.

Exam focus

Key points to remember

- In fermentation an organic compound is both the electron donor and the final electron acceptor.
- There is no electron transport chain and no oxygen requirement.
- Fermentation yields only 2 ATP per glucose, because the substrate is only partly oxidised.
- Lactic acid fermentation (curd, silage) and alcoholic fermentation (bread, wine) are the main types.
- Fermentation regenerates NAD+ so that glycolysis can continue.
- Fermentations are used to identify bacteria in the laboratory (methyl red, Voges-Proskauer tests).

Short-answer questions with model answers

1. Define fermentation and give two examples.

Fermentation is a mode of energy generation in which an organic compound acts as both the electron donor and the final electron acceptor, without an electron transport chain or oxygen. Examples: lactic acid fermentation by *Lactobacillus* (curd) and alcoholic fermentation by *Saccharomyces cerevisiae* (bread, wine).

2. Why is the energy yield of fermentation low?

Because the substrate is only partially oxidised and the electrons are transferred to an organic compound instead of to oxygen through an electron transport chain; hence only 2 ATP per glucose are produced.

3. Give the equation for alcoholic fermentation.

Glucose -> 2 ethanol + 2 CO2 + 2 ATP (via glycolysis and the reduction of acetaldehyde to ethanol).

Long-answer questions with model answer outlines

1. Explain the fermentative mode of energy generation with types and examples.

Outline: (i) Definition and principle. (ii) Role of glycolysis and NAD+ regeneration. (iii) Lactic acid fermentation - homo- and heterolactic. (iv) Alcoholic fermentation. (v) Other fermentations - mixed acid, butanediol, butyric, propionic. (vi) Industrial and agricultural uses. (vii) Comparison with respiration.

2. Compare aerobic respiration, anaerobic respiration and fermentation.

Outline: (i) Definition of each. (ii) Electron acceptor. (iii) Role of the electron transport chain. (iv) Oxidation of substrate. (v) End products. (vi) ATP yield. (vii) Organisms and environments. (viii) Table and significance.

Mounish Raj · MouRa · Agri AI

Lecture 24
Oxygenic and Anoxygenic Modes of Photosynthesis

Oxygenic and Anoxygenic Modes of Photosynthesis

24.1 Introduction

Photosynthesis is the process by which light energy is captured and converted into chemical energy, with the fixation of carbon dioxide into organic matter. It is carried out by plants, algae and several groups of bacteria. In photosynthesis, light energy is used to generate ATP and reducing power, which are then used to fix CO2.

24.2 Photosynthetic pigments and machinery

- **Chlorophylls** - chlorophyll a is the main pigment in all photosynthetic organisms; chlorophyll b, c, d and e occur in different groups. Bacteriochlorophylls occur in photosynthetic bacteria.
- **Accessory pigments (carotenoids and phycobilins)** - absorb light of other wavelengths and pass the energy to chlorophyll; they also protect against photo-oxidation.
- The pigments are organised in **photosystems** (antenna complexes and reaction centres) embedded in membranes - the thylakoid membranes of chloroplasts in plants and algae, and the plasma membrane or chromatophores in photosynthetic bacteria.
- Photosynthetic bacteria have bacteriochlorophyll and often use specialised membrane structures called chromatophores.

24.3 Oxygenic photosynthesis

Oxygenic photosynthesis is carried out by **cyanobacteria**, algae and green plants. It uses water as the electron donor and releases **oxygen** as a by-product.

- Overall equation: 6 CO2 + 6 H2O + light -> C6H12O6 + 6 O2.
- Two photosystems are involved - **photosystem I (PS I)** and **photosystem II (PS II)**.
- In PS II, light energy splits water (photolysis) into oxygen, protons and electrons; the electrons pass along an electron transport chain to PS I.
- As in respiration, the electron transport chain pumps protons and generates ATP by **photophosphorylation** (cyclic and non-cyclic).
- In PS I, electrons are raised to a higher energy level and used to reduce NADP+ to NADPH.
- The ATP and NADPH produced are used to fix CO2 into carbohydrate by the **Calvin cycle** (dark reaction).

24.4 Anoxygenic photosynthesis

Anoxygenic photosynthesis is carried out by several groups of bacteria - green sulphur bacteria, purple sulphur bacteria and purple non-sulphur bacteria. It does **not** produce oxygen, because water is not used as the electron donor.

- Electron donors are inorganic or organic compounds such as hydrogen sulphide (H2S), hydrogen, thiosulphate or organic compounds.
- Example: green sulphur bacteria use H2S as the electron donor and deposit sulphur globules; purple sulphur bacteria also use H2S.
- Equation (example with H2S): CO2 + 2 H2S + light -> CH2O + 2 S + H2O.
- Only one photosystem (similar to PS I) is present; hence no oxygen is evolved.
- Photosynthetic bacteria are mostly anaerobic and often live in sulphur springs, mud and stagnant water where light and H2S are available.

24.5 Comparison of oxygenic and anoxygenic photosynthesis

Table 24.1 - Oxygenic vs anoxygenic photosynthesis

Feature	Oxygenic photosynthesis	Anoxygenic photosynthesis
Organisms	Cyanobacteria, algae, green plants	Green/purple sulphur and non-sulphur bacteria
Pigments	Chlorophyll a and b	Bacteriochlorophylls
Electron donor	Water	H2S, H2, thiosulphate, organic compounds
Oxygen evolved	Yes	No
Photosystems	Two (PS I and PS II)	One
Habitat	Aerobic, illuminated	Anaerobic, illuminated, H2S-rich

- Photosynthesis in bacteria is important in nature because it adds organic matter in environments where plants cannot grow, and some photosynthetic bacteria are important in the sulphur cycle and in wastewater treatment.

24.6 Agricultural relevance

- Cyanobacteria in paddy fields carry out oxygenic photosynthesis, add organic matter, fix nitrogen and contribute to soil fertility.
- Algal photosynthesis in ponds and rice fields produces oxygen, which supports aerobic decomposition and aquatic life.
- Photosynthetic bacteria are used in wastewater treatment and in the production of single-cell protein and hydrogen.

Exam focus

Key points to remember

- Photosynthesis converts light energy into chemical energy and fixes CO2.
- Oxygenic photosynthesis uses water as the electron donor, involves two photosystems and releases oxygen.
- Anoxygenic photosynthesis uses H2S, H2 or organic compounds, has one photosystem and releases no oxygen.

- Cyanobacteria, algae and plants carry out oxygenic photosynthesis; green and purple bacteria carry out anoxygenic photosynthesis.
- ATP is produced by photophosphorylation; CO2 is fixed by the Calvin cycle.

Short-answer questions with model answers

1. Distinguish oxygenic from anoxygenic photosynthesis.

Oxygenic photosynthesis (cyanobacteria, algae, plants) uses water as the electron donor, has two photosystems and releases oxygen. Anoxygenic photosynthesis (green and purple sulphur and non-sulphur bacteria) uses H2S, H2 or organic compounds as the electron donor, has one photosystem and does not release oxygen.

2. Write the overall equation for oxygenic photosynthesis.

6 CO2 + 6 H2O + light energy -> C6H12O6 + 6 O2.

3. What are bacteriochlorophylls and where are they found?

Bacteriochlorophylls are photosynthetic pigments found in photosynthetic bacteria (green and purple sulphur and non-sulphur bacteria) in place of chlorophyll; they absorb light in the near-infrared region.

Long-answer questions with model answer outlines

1. Describe oxygenic photosynthesis with its light and dark reactions.

Outline: (i) Definition. (ii) Pigments and photosystems. (iii) PS II and photolysis of water. (iv) Electron transport and photophosphorylation. (v) PS I and NADPH formation. (vi) Calvin cycle (dark reaction). (vii) Overall equation and significance.

2. Explain anoxygenic photosynthesis and compare it with oxygenic photosynthesis.

Outline: (i) Definition and organisms. (ii) Electron donors. (iii) Pigments and single photosystem. (iv) Products and absence of oxygen. (v) Habitats. (vi) Comparison table. (vii) Role in the sulphur cycle and agriculture.

PART V - BIOENERGETICS, GENETICS AND GENETIC ENGINEERING

Mounish Raj · MouRa · Agri AI

Lecture 25
Energy Generation by Substrate Level Phosphorylation, Oxidative and Pho...

Energy Generation by Substrate Level Phosphorylation, Oxidative and Photophosphorylation

25.1 Introduction

All cells need energy, and the universal energy currency of the cell is **adenosine triphosphate (ATP)**. ATP is formed from ADP and inorganic phosphate (Pi) with the input of energy. The formation of ATP is called **phosphorylation**, and there are three main ways in which it occurs in microorganisms: substrate level phosphorylation, oxidative phosphorylation and photophosphorylation.

25.2 Structure and role of ATP

- ATP consists of adenine, ribose and three phosphate groups; the bonds between the phosphate groups are high-energy bonds.
- Hydrolysis of ATP to ADP + Pi releases about 7.3 kcal (about 30.5 kJ) per mole, which drives endergonic reactions.
- ATP is the energy currency; NADH and FADH2 are the electron carriers; NADPH is the reducing power for biosynthesis.

25.3 Substrate level phosphorylation

- In this process, ATP is formed directly by the transfer of a phosphate group from a high-energy organic compound to ADP, without the involvement of an electron transport chain.
- It occurs in the cytoplasm during glycolysis and the Krebs cycle.
- Examples:
- In glycolysis, the conversion of 1,3-bisphosphoglycerate to 3-phosphoglycerate and of phosphoenolpyruvate to pyruvate yields ATP.
- In the Krebs cycle, the conversion of succinyl-CoA to succinate yields one ATP (GTP) per turn.
- Net yield from glycolysis is 2 ATP per glucose; from the Krebs cycle, 2 ATP per glucose (one per turn, two turns).
- It is the only way in which ATP is produced during fermentation.

25.4 Oxidative phosphorylation

- ATP is formed as a result of the transfer of electrons through an electron transport chain to a final electron acceptor (oxygen in aerobic respiration).
- Electrons from NADH and FADH2 pass through a series of carriers (flavoproteins, quinones, cytochromes).
- As electrons pass, protons are pumped across the membrane, creating a **proton gradient** (proton motive force).

- The protons flow back through the enzyme **ATP synthase**, and this flow drives the synthesis of ATP from ADP and Pi.
- This explanation is the **chemiosmotic theory**, proposed by Peter Mitchell (Nobel Prize, 1978).
- Oxidative phosphorylation accounts for most of the ATP produced in aerobic respiration (about 34 of the 36-38 ATP per glucose).
- In bacteria the electron transport chain and ATP synthase are located in the plasma membrane.

25.5 Photophosphorylation

- ATP is formed using light energy in photosynthetic organisms.
- Light energy excites the chlorophyll in the reaction centre; the excited electron passes through an electron transport chain, and the resulting proton gradient drives ATP synthesis by ATP synthase - again following the chemiosmotic principle.
- Two types:
- **Cyclic photophosphorylation** - the electron returns to the original chlorophyll molecule; only ATP is produced, not NADPH. Occurs mainly in anoxygenic photosynthetic bacteria with one photosystem.
- **Non-cyclic photophosphorylation** - the electron does not return; both ATP and NADPH are produced. Occurs in oxygenic photosynthesis with two photosystems (PS I and PS II).

25.6 Comparison of the three types

Table 25.1 - Types of phosphorylation

Feature	Substrate level	Oxidative	Photo-
Source of energy	High-energy organic compound	Oxidation of NADH/FADH2	Light
Electron transport chain	Absent	Present	Present
Membrane required	No	Yes	Yes
Final electron acceptor	Organic compound	O2 or other inorganic acceptor	Cyclic or none (non-cyclic: NADP+)
Products	ATP only	ATP (and water)	ATP (cyclic); ATP + NADPH (non-cyclic)
Location	Cytoplasm	Plasma membrane (bacteria)	Thylakoid/plasma membrane
Occurrence	Glycolysis, Krebs cycle, fermentation	Respiration	Photosynthesis

Worked example 25.1 - Accounting for the ATP of aerobic respiration

Problem: From one molecule of glucose, state how many ATP come from substrate level phosphorylation and how many from oxidative phosphorylation, giving the total.

Step 1: Substrate level phosphorylation. Glycolysis gives 2 ATP (net) and the Krebs cycle gives 2 ATP. Total = 4 ATP.

Step 2: Oxidative phosphorylation. Glycolysis gives 2 NADH; pyruvate oxidation gives 2 NADH; the Krebs cycle gives 6 NADH and 2 FADH2. Total 10 NADH and 2 FADH2.

Step 3: Using the standard textbook conversion (about 3 ATP per NADH and 2 ATP per FADH2), oxidative phosphorylation gives about 10 x 3 + 2 x 2 = 34 ATP.

Step 4: Total = 4 + 34 = 38 ATP; textbooks give a range of about 36-38 ATP because of losses in transport.

Answer: about 4 ATP from substrate level phosphorylation and about 34 from oxidative phosphorylation, giving a total of about 36-38 ATP per glucose.

25.7 Agricultural relevance

- Energy generation governs the efficiency with which microbes decompose organic matter and release nutrients in the soil.
- In the rumen and in biogas plants, the balance between fermentation, anaerobic respiration and methanogenesis determines the products formed.
- Cyanobacteria in paddy fields use photophosphorylation to fix carbon and nitrogen, adding to soil fertility.

Exam focus

Key points to remember

- ATP is the energy currency of the cell.
- Substrate level phosphorylation transfers a phosphate from an organic compound to ADP; no electron transport chain is involved.
- Oxidative phosphorylation uses an electron transport chain and a proton gradient (chemiosmotic theory).
- Photophosphorylation uses light energy; cyclic gives only ATP, non-cyclic gives ATP and NADPH.
- ATP synthase is the enzyme that makes ATP in oxidative and photophosphorylation.

Short-answer questions with model answers

1. Define substrate level phosphorylation with an example.

Substrate level phosphorylation is the direct formation of ATP by the transfer of a phosphate group from a high-energy organic compound to ADP, without an electron transport chain. Example: conversion of phosphoenolpyruvate to pyruvate in glycolysis.

2. State the chemiosmotic theory.

Proposed by Peter Mitchell, the chemiosmotic theory states that the energy from electron transport is used to pump protons across a membrane, creating a proton gradient (proton motive force). The protons flow back through ATP synthase, and this flow drives the synthesis of ATP.

3. Distinguish cyclic from non-cyclic photophosphorylation.

In cyclic photophosphorylation the excited electron returns to the original chlorophyll and only ATP is produced; it occurs in anoxygenic bacteria with one photosystem. In non-cyclic photophosphorylation the electron does not return and both ATP and NADPH are produced; it occurs in oxygenic photosynthesis with two photosystems.

Long-answer questions with model answer outlines

1. Describe the three mechanisms of ATP generation in microorganisms.

Outline: (i) ATP as energy currency. (ii) Substrate level phosphorylation - mechanism, examples, yield. (iii) Oxidative phosphorylation - electron transport chain, proton motive force, chemiosmotic theory, ATP synthase. (iv) Photophosphorylation - cyclic and non-cyclic. (v) Comparison table. (vi) Worked example and significance.

2. Explain oxidative phosphorylation and the chemiosmotic theory.

Outline: (i) Location in bacteria. (ii) Electron carriers of the chain. (iii) Proton pumping and proton motive force. (iv) ATP synthase and its role. (v) Chemiosmotic theory of Peter Mitchell. (vi) ATP yield. (vii) Comparison with substrate level phosphorylation.

Lecture 26
Central Dogma of Life; Genetic Elements in Bacteria - Structure and Fun...

Central Dogma of Life; Genetic Elements in Bacteria - Structure and Functions of Bacterial Chromosome, Plasmid and Transposons

26.1 The central dogma of molecular biology

The central dogma, proposed by Francis Crick, describes the flow of genetic information in a cell:

- **DNA -> RNA -> Protein**.
- **Replication** - DNA is copied to make more DNA; this passes information to daughter cells.
- **Transcription** - the information in DNA is copied into messenger RNA (mRNA).
- **Translation** - the mRNA is read on ribosomes and the information is used to build a protein from amino acids.
- A special case is **reverse transcription**, in which RNA is copied back into DNA (in retroviruses such as HIV), which is an exception to the general flow.
- The sequence of bases in DNA determines the sequence of amino acids in a protein, and hence the structure and function of the cell.

26.2 The bacterial chromosome

- The bacterial chromosome is a single, circular, double-stranded DNA molecule.
- It is not enclosed in a nuclear membrane; it lies in the cytoplasm in a region called the **nucleoid**.
- It is **haploid** - only one copy per cell is normally present.
- It is not associated with histone proteins (unlike eukaryotic DNA), though it is folded and supercoiled and attached to the plasma membrane.
- It carries all the essential genes for growth and reproduction.
- *E. coli* has a chromosome of about 4.6 million base pairs carrying about 4,300 genes.
- It replicates by a semi-conservative process, starting at a fixed point (origin of replication) and proceeding in both directions.

26.3 Plasmids

- Plasmids are small, circular, double-stranded DNA molecules that occur in addition to the chromosome and replicate independently.
- They are not essential for normal growth but carry genes that give the cell special properties.
- They are much smaller than the chromosome (a few thousand to a few hundred thousand base pairs).
- Types of plasmids:

- **F (fertility) factor** - carries genes for conjugation and the sex pilus; the F+ cell acts as a donor.
- **R (resistance) factor** - carries genes for resistance to antibiotics; a serious problem in medicine and agriculture.
- **Col plasmids** - carry genes for the production of bacteriocins (proteins that kill related bacteria).
- **Virulence plasmids** - carry genes for toxins or other virulence factors (for example, the Ti plasmid of *Agrobacterium tumefaciens*).
- **Degradative plasmids** - carry genes for the breakdown of unusual substances such as pesticides and hydrocarbons; useful in bioremediation.
- Plasmids are widely used as **vectors** (carriers) in genetic engineering.

26.4 Transposons (transposable genetic elements)

- Transposons are segments of DNA that can move from one place to another within the genome - a process called **transposition**.
- They were discovered by Barbara McClintock in maize (Nobel Prize, 1983).
- **Insertion sequences (IS elements)** are the simplest transposons; they carry only the genes needed for their own movement.
- A **composite transposon** consists of an IS element at each end with a central region carrying other genes (for example, antibiotic-resistance genes).
- Transposons can move genes, including antibiotic-resistance genes, from a plasmid to the chromosome or from one plasmid to another, and thus contribute to the spread of resistance.
- They can also cause mutations when they insert into a gene.

Table 26.1 - Genetic elements in bacteria

Element	Nature	Size	Function
Chromosome	Circular dsDNA	Large (about 4.6 Mb in *E. coli*)	Carries all essential genes
Plasmid	Small circular dsDNA	A few kb to 100+ kb	Extra genes - F, R, Ti, Col, degradative
Transposon	DNA segment that can move	About 1-20 kb	Transposition; spread of genes, mutations
Insertion sequence	Simplest transposon	About 0.7-2 kb	Carries only transposition genes

26.5 Agricultural relevance

- The Ti plasmid of *Agrobacterium tumefaciens* is used to transfer genes of interest into crop plants, including genes for pest resistance and drought tolerance.
- R-plasmids spread antibiotic resistance among animal pathogens and among bacteria in manure and soil.
- Degradative plasmids are used in bioremediation of pesticide-contaminated soil.

Exam focus

Key points to remember

- The central dogma is DNA -> RNA -> Protein; reverse transcription is an exception.
- The bacterial chromosome is a single, circular, double-stranded DNA in the nucleoid, haploid and not histone-bound.
- Plasmids are small, circular, extrachromosomal DNA molecules that replicate independently.
- F factor, R factor, Col plasmid, virulence plasmid and degradative plasmid are the main types.
- Transposons are DNA segments that can move within the genome and can carry other genes.
- Insertion sequences are the simplest transposons.

Short-answer questions with model answers

1. State the central dogma of molecular biology.

The central dogma, proposed by Crick, states that genetic information flows from DNA to RNA to protein: DNA is replicated, transcribed into mRNA, and the mRNA is translated into protein. Reverse transcription (RNA to DNA) is a special exception.

2. What is a plasmid? Give three types with their functions.

A plasmid is a small, circular, extrachromosomal DNA molecule that replicates independently of the chromosome. The F factor carries conjugation genes, the R factor carries antibiotic-resistance genes, and the Ti plasmid of *Agrobacterium tumefaciens* causes crown gall and is used in plant genetic engineering.

3. What are transposons?

Transposons are segments of DNA that can move from one position to another in the genome. They may carry other genes such as antibiotic-resistance genes and can cause mutations when they insert into a gene.

Long-answer questions with model answer outlines

1. Explain the central dogma and describe the genetic elements of bacteria.

Outline: (i) Central dogma - replication, transcription, translation. (ii) Reverse transcription. (iii) Bacterial chromosome - structure and function. (iv) Plasmids - types and functions. (v) Transposons and insertion sequences. (vi) Table. (vii) Applications in genetic engineering and antibiotic resistance.

2. Describe the structure and functions of the bacterial chromosome and plasmids.

Outline: (i) Bacterial chromosome - circular, dsDNA, nucleoid, haploid, supercoiled. (ii) Replication - semi-conservative, origin, bidirectional. (iii) Plasmids - definition and properties. (iv) Types - F, R, Col, virulence, degradative. (v) Use as vectors. (vi) Agricultural and medical importance.

Lecture 27
Mutation in Bacteria - Principles and Types; Mutagens - Physical, Chemi...

Mutation in Bacteria - Principles and Types; Mutagens - Physical, Chemical and Biological

27.1 What is a mutation

A **mutation** is a permanent, heritable change in the base sequence of DNA. It may alter a single base pair (point mutation) or a larger segment of DNA. Mutations are the ultimate source of genetic variation and hence of evolution, and they are also important in producing improved strains of microorganisms.

27.2 Principles of mutation

- Mutations occur **spontaneously** at a low rate (about 1 in 10^6 to 10^8 cells per generation) or can be **induced** by agents called mutagens.
- Spontaneous mutations arise from errors in DNA replication, from tautomeric shifts in bases, or from the action of natural mutagens.
- A mutation may be **silent** (no change in the protein), **missense** (one amino acid changed), **nonsense** (a stop codon created, giving a shortened protein) or **frameshift** (insertion or deletion of bases, changing the reading frame).
- A mutation may be **lethal**, **harmful** or, rarely, **beneficial**.
- Mutations can be **reverted** (back mutation) or suppressed by a second mutation.
- Mutation is different from **adaptation**: a mutation is a random, heritable change, while adaptation is a physiological response to the environment.

27.3 Types of mutation

- **Point (substitution) mutation** - a single base is replaced by another. It may be a transition (purine to purine, or pyrimidine to pyrimidine) or a transversion (purine to pyrimidine).
- **Frameshift mutation** - insertion or deletion of one or a few bases, which shifts the reading frame and changes all the amino acids after the point.
- **Deletion** - loss of a segment of DNA.
- **Insertion** - addition of a segment of DNA.
- **Duplication** - a segment is repeated.
- **Inversion** - a segment is reversed in orientation.
- **Nonsense mutation** - a codon is changed to a stop codon.
- **Missense mutation** - a codon is changed to one coding for a different amino acid.
- **Silent mutation** - the codon changes but the amino acid does not (because of the degeneracy of the genetic code).

27.4 Mutagens

A mutagen is an agent that increases the frequency of mutation above the spontaneous rate.

27.4.1 Physical mutagens

- **Ultraviolet (UV) light** - absorbed by DNA, forming pyrimidine dimers (thymine dimers) that block replication; the cell may repair them, or a mutation may result.
- **Ionising radiation** (X-rays, gamma rays) - cause single- and double-strand breaks in DNA and free radical damage.
- **Heat** - can cause depurination and other damage.

27.4.2 Chemical mutagens

- **Base analogues** - substances similar to normal bases that are incorporated into DNA and cause mispairing (for example, 5-bromouracil, 2-aminopurine).
- **Deaminating agents** - remove amino groups from bases (for example, nitrous acid).
- **Alkylating agents** - add alkyl groups to bases (for example, ethyl methane sulphonate EMS, nitrosoguanidine NTG, mustard gas).
- **Intercalating agents** - insert between bases and cause frameshift mutations (for example, acridine orange, ethidium bromide).
- **Hydroxylamine** - reacts with cytosine.

27.4.3 Biological mutagens

- **Transposons and insertion sequences** - move within the genome and disrupt genes.
- **Bacteriophages** - when they integrate, they may disrupt a gene; they may also carry genes.
- **Plasmids** - can insert genes and cause changes.

27.5 Significance of mutation

- Mutation is the source of genetic variation, which is the raw material of evolution and of natural selection.
- In industry, mutations are induced to obtain high-yielding strains of antibiotic-producing organisms (for example, high-yielding *Penicillium* for penicillin).
- In agriculture, mutations are induced to obtain improved crop varieties and improved biofertilizer and biocontrol strains.
- Mutation is also responsible for the development of antibiotic resistance in pathogens.
- In the Ames test, mutation in bacteria is used to test whether a chemical is a potential carcinogen.

Table 27.1 - Mutagens and their action

Type	Agent	Mode of action	Example of effect
Physical	UV light	Forms pyrimidine dimers	Thymine dimers
Physical	X-rays, gamma rays	Strand breaks, free radicals	Chromosomal damage
Chemical	Base analogues	Mispaired bases	5-bromouracil
Chemical	Deaminating agents	Remove amino groups	Nitrous acid
Chemical	Alkylating agents	Add alkyl groups	EMS, NTG
Chemical	Intercalating agents	Insert between bases	Acridine orange (frameshift)
Biological	Transposons	Insert into genes	Gene disruption
Biological	Bacteriophages	Integration into genome	Insertional mutation

Worked example 27.1 - Mutation frequency

Problem: In a culture of 10^8 bacteria, 25 cells are found to be resistant to a particular antibiotic, while the rest are sensitive. Calculate the mutation frequency and the mutation rate per cell per generation (assume one generation).

Step 1: Mutation frequency = number of mutants / total number of cells.

Step 2: Substitute values. Mutation frequency = 25 / 10^8 = 2.5 x 10^-7.

Step 3: If the population has passed through one generation since the mutation, the mutation rate is about the same, 2.5 x 10^-7 per cell per generation.

Answer: Mutation frequency = 2.5 x 10^-7 (about 1 in 4 million cells).

Exam focus

Key points to remember

- A mutation is a permanent, heritable change in the base sequence of DNA.
- Mutations may be spontaneous or induced; they may be point, frameshift, deletion, insertion or inversion.
- Silent, missense, nonsense and frameshift mutations differ in their effect on the protein.
- Physical mutagens - UV light, X-rays, gamma rays; chemical - base analogues, alkylating agents, intercalating agents; biological - transposons and phages.
- Mutation is the source of genetic variation and of antibiotic resistance.
- Induced mutation is used to obtain high-yielding industrial and agricultural strains.

Short-answer questions with model answers

1. Define mutation and give its types.

A mutation is a permanent, heritable change in the base sequence of DNA. Types include point (substitution) mutations, frameshift mutations, deletions, insertions, duplications and inversions. On the basis of effect they are silent, missense, nonsense or frameshift.

2. What are mutagens? Give two examples of each type.

Mutagens are agents that increase the rate of mutation. Physical - UV light and X-rays; chemical - base analogues (5-bromouracil) and alkylating agents (EMS); biological - transposons and bacteriophages.

3. How does UV light cause mutation?

UV light is absorbed by DNA and forms pyrimidine dimers (thymine dimers), which distort the DNA and block replication. If not repaired correctly, this leads to a mutation.

Long-answer questions with model answer outlines

1. Explain the principles and types of mutation in bacteria.

Outline: (i) Definition of mutation. (ii) Spontaneous vs induced. (iii) Point mutation - transition, transversion. (iv) Frameshift, deletion, insertion, inversion. (v) Effect on protein - silent, missense, nonsense. (vi) Reversion and suppression. (vii) Significance.

2. Describe the physical, chemical and biological mutagens with their modes of action.

Outline: (i) Definition of a mutagen. (ii) Physical - UV, ionising radiation, heat. (iii) Chemical - base analogues, deaminating agents, alkylating agents, intercalating agents. (iv) Biological - transposons, phages, plasmids. (v) Table. (vi) Industrial and agricultural applications and the Ames test.

Lecture 28
Genetic Recombination - Competency - Transformation

Genetic Recombination - Competency - Transformation

28.1 Genetic recombination

Genetic recombination is the process by which genetic material from one cell is transferred to another and becomes part of the recipient's genome, producing a **recombinant** cell with a new combination of genes. In bacteria, recombination occurs by three main mechanisms: **transformation**, **conjugation** and **transduction**. In all three, a fragment of donor DNA enters the recipient and recombines with the recipient chromosome by replacing a homologous segment.

- Recombination requires that the donor DNA be taken up by the recipient, that it be integrated (usually by a process requiring homologous sequences), and that the recombinant be detected by a change in characteristics.
- Recombination does not create new genes; it creates new combinations of existing genes.

28.2 Competency

Competency is the ability of a bacterial cell to take up free DNA from the environment and become transformed. Not all bacteria are naturally competent.

- Natural competency is found in *Bacillus subtilis*, *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Neisseria* and *Agrobacterium*.
- Competency develops at a particular stage of growth (usually late log phase) and is controlled by genes; it is associated with the production of a **competence factor** (a small protein) and with special proteins on the cell surface that bind DNA.
- Competency can also be induced artificially by treating the cells with calcium chloride and heat shock, or by electroporation, which is widely used in the laboratory to introduce plasmid DNA.

28.3 Transformation

Transformation is the uptake of free (naked) DNA from the environment by a competent cell, followed by its integration into the recipient genome. It was discovered by Frederick Griffith in 1928 in *Streptococcus pneumoniae*.

- **Griffith's experiment** - heat-killed virulent (smooth, S) pneumococci, when mixed with live non-virulent (rough, R) pneumococci, transformed the R cells into virulent S cells. The "transforming principle" was later shown by Avery, MacLeod and McCarty (1944) to be DNA.
- Steps of transformation:

– The donor DNA is released into the environment, usually when a cell dies and lyses.

– A competent recipient cell binds the DNA at its surface.

– One strand of the DNA enters the cell; the other is degraded.

– The single strand pairs with a homologous region of the recipient chromosome.

– The donor segment is integrated by recombination, replacing the corresponding recipient segment.

  - The cell now carries the donor gene and, when it divides, the new character is expressed.
- Transformation is used in the laboratory to introduce plasmids and to study genes. It is also the basis of some naturally occurring changes, such as the spread of penicillin resistance in pneumococci.

28.4 Agricultural relevance

- Transformation is a basic tool in genetic engineering, used to introduce useful genes into bacteria and, with special methods, into plants.
- Natural transformation contributes to the spread of genes, including antibiotic-resistance genes, among soil and plant-associated bacteria.
- Competent *Agrobacterium* and *Bacillus* are important in plant biotechnology and in biofertilizer strain improvement.

Exam focus

Key points to remember

- Genetic recombination produces a new combination of genes; in bacteria it occurs by transformation, conjugation and transduction.
- Competency is the ability to take up free DNA; it is natural in some bacteria and can be induced artificially.
- Transformation is the uptake of naked DNA and its integration into the recipient genome.
- Griffith discovered transformation in *Streptococcus pneumoniae*; Avery, MacLeod and McCarty proved that the transforming principle is DNA.
- Only a single strand of the donor DNA usually enters the cell.

Short-answer questions with model answers

1. Define transformation and list its steps.

Transformation is the uptake of free DNA from the environment by a competent cell and its integration into the recipient genome. Steps: release of donor DNA; binding by a competent cell; entry of a single strand; pairing with the homologous region of the chromosome; integration by recombination; expression of the new character.

2. What is competency? How is it induced in the laboratory?

Competency is the ability of a cell to take up free DNA. It is natural in some bacteria at a particular growth stage and is induced artificially by treatment with calcium chloride and heat shock, or by electroporation.

3. Describe Griffith's experiment.

Griffith mixed heat-killed virulent (S) pneumococci with live non-virulent (R) pneumococci and found that the R cells became virulent (S). He concluded that a "transforming principle" from the dead S cells had changed the R cells. Avery, MacLeod and McCarty later showed this principle to be DNA.

Long-answer questions with model answer outlines

1. Explain genetic recombination and the process of transformation.

Outline: (i) Meaning of recombination. (ii) Three mechanisms in bacteria. (iii) Competency - natural and induced. (iv) Transformation - discovery by Griffith and proof by Avery. (v) Steps of transformation. (vi)

Significance in nature and in the laboratory.

2. Describe the three modes of genetic recombination in bacteria.

Outline: (i) Introduction and common features. (ii) Transformation - competent cell takes up free DNA. (iii) Conjugation - DNA transferred through a pilus from donor to recipient. (iv) Transduction - DNA carried by a bacteriophage. (v) Comparison table and diagram. (vi) Significance in evolution and in genetic engineering.

Lecture 29
Genetic Recombination by Conjugation - Concept of Hfr

Genetic Recombination by Conjugation - Concept of Hfr

29.1 Conjugation

Conjugation is the transfer of genetic material from one bacterial cell (the donor) to another (the recipient) by **direct cell-to-cell contact**, usually through a tube-like structure called the **sex pilus** or mating bridge. It was discovered by Joshua Lederberg and Edward Tatum (1946) in *Escherichia coli*.

29.2 The F factor

- The ability to act as a donor is determined by a plasmid called the **F (fertility) factor**.
- A cell carrying the F factor is called **F+** (donor); a cell without it is **F-** (recipient).
- The F factor carries genes for the synthesis of the sex pilus and for the transfer of DNA.
- During conjugation, the F factor is replicated by the rolling-circle mechanism; one copy is transferred to the recipient through the pilus.
- After transfer, the recipient becomes **F+** and can itself act as a donor. The donor remains F+.

29.3 Steps of conjugation

– The F+ cell produces a sex pilus that attaches to the F- cell, forming a mating bridge.

– The F factor is nicked at a specific point (origin of transfer) and one strand is transferred into the recipient, while the other strand is copied in the donor.

– In the recipient, the complementary strand is synthesised, so the F factor becomes double-stranded again.

– The recipient now carries the F factor and is F+.

29.4 Hfr strains

- Sometimes the F factor becomes **integrated into the bacterial chromosome** by recombination between homologous insertion sequences.
- Such a cell is called an **Hfr** strain - **H**igh **f**requency of **r**ecombination, because it transfers chromosomal genes at a high frequency.
- When an Hfr cell conjugates with an F- cell, the chromosome is transferred in a **linear order**, starting from the point where the F factor is integrated.
- Because the chromosome is large, the mating bridge usually breaks before the whole chromosome is transferred; therefore the recipient receives only part of the chromosome (a few genes to most of the chromosome, depending on the time of mating) and **rarely receives the complete F factor**.
- Hence, in an Hfr x F- cross, the recipient usually becomes a **recombinant** for chromosomal genes but **remains F-**.

- By interrupting the mating at different times and finding which genes have been transferred, the order of genes on the chromosome can be mapped - this is **interrupted mating** mapping, developed by Wollman and Jacob.

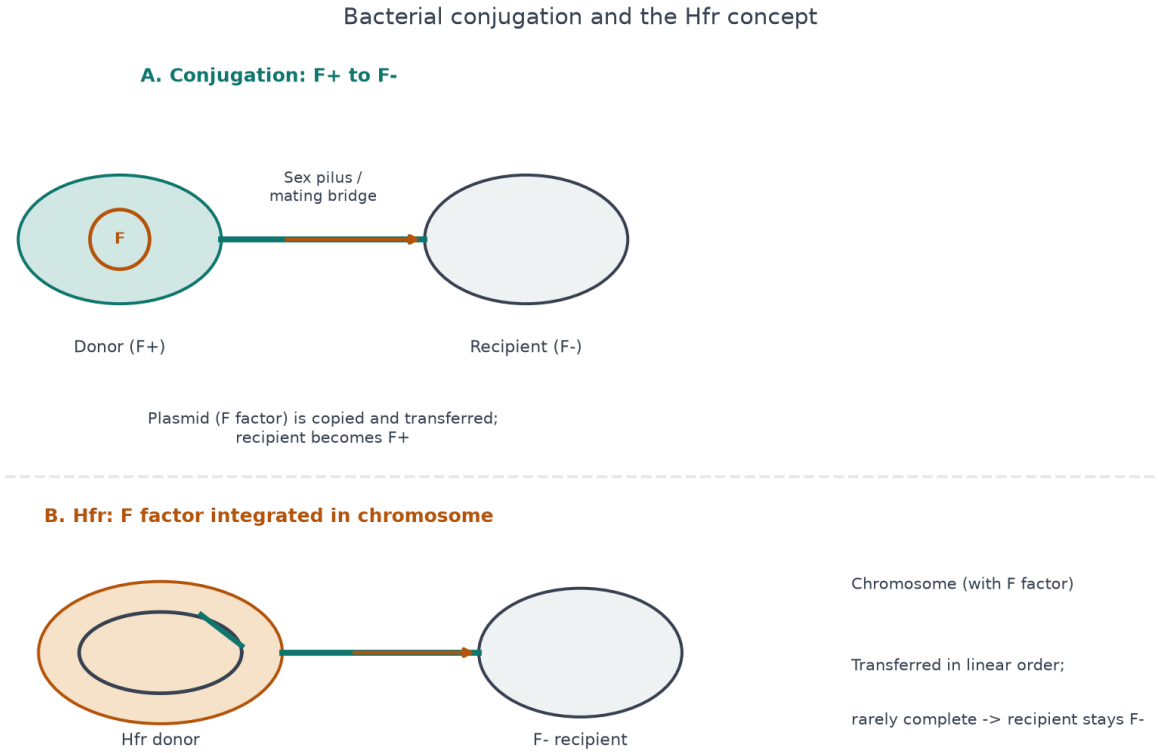


Figure 29.1 - Bacterial conjugation: transfer of the F plasmid from an F+ donor to an F- recipient (panel A), and the Hfr concept where the integrated F factor carries chromosomal DNA into the recipient (panel B).

29.5 F' (prime) factors

- If an integrated F factor is excised incorrectly, it may carry a small piece of the chromosomal DNA with it. Such an F factor is called an **F' (prime) factor**.
- When an F' cell conjugates, it transfers both the F factor and the attached chromosomal genes, so the recipient becomes a **merodiploid** (partially diploid) for those genes.
- F' factors are useful for genetic studies and for constructing partial diploids.

Table 29.1 - F+, F- and Hfr cells

Feature	F+ cell	F- cell	Hfr cell
F factor	Present, free	Absent	Present, integrated in chromosome
Role	Donor	Recipient	High-frequency donor
Pilus	Present	Absent	Present
DNA transferred	F factor only	None	Chromosome (linear, part)
Recipient becomes	F+	-	Recombinant, usually F-
Frequency of gene transfer	Low	-	High

29.6 Agricultural and medical relevance

- Conjugation spreads R-plasmids carrying antibiotic-resistance genes among bacteria, including pathogens of animals and plants; this is a serious problem in veterinary and human medicine.
- Conjugation is used in the laboratory to construct improved strains of biofertilizer bacteria and to transfer useful plasmids.
- The Ti plasmid of *Agrobacterium* can be transferred by conjugation and is the basis of many plant transformation systems.

Exam focus

Key points to remember

- Conjugation is the transfer of DNA through direct cell-to-cell contact via a sex pilus.
- The F factor determines donor ability; F+ is the donor and F- the recipient.
- In an F+ x F- cross, the recipient becomes F+.
- In an Hfr strain the F factor is integrated into the chromosome, and chromosomal DNA is transferred linearly at high frequency.
- In an Hfr x F- cross the recipient usually becomes a recombinant but stays F-.
- Interrupted mating is used to map genes on the bacterial chromosome.

Short-answer questions with model answers

1. Define conjugation and describe the role of the F factor.

Conjugation is the transfer of genetic material from a donor to a recipient cell by direct contact through a sex pilus. The F factor is a plasmid that carries genes for the sex pilus and for DNA transfer; cells with the F factor (F+) are donors and those without (F-) are recipients.

2. What is an Hfr strain?

An Hfr (high frequency of recombination) strain is one in which the F factor is integrated into the bacterial chromosome. When it conjugates, it transfers chromosomal DNA in linear order at high frequency, so the recipient frequently becomes a recombinant for chromosomal genes.

3. Why does the recipient usually remain F- in an Hfr cross?

Because the chromosome is long and the mating bridge usually breaks before the whole chromosome - including the integrated F factor, which lies at the end - is transferred. The recipient therefore receives only part of the chromosome and remains F-.

Long-answer questions with model answer outlines

1. Explain genetic recombination by conjugation with the concept of Hfr.

Outline: (i) Definition and discovery. (ii) F factor and F+/F- cells. (iii) Steps of conjugation. (iv) Integration of F factor to give Hfr. (v) Linear transfer of the chromosome and recombination. (vi) Interrupted mating and gene mapping. (vii) F' factors. (viii) Diagram and significance.

2. Describe the F factor, F+ and Hfr strains and their importance.

Outline: (i) Nature of the F factor. (ii) F+ x F- conjugation and conversion to F+. (iii) Hfr formation and transfer of chromosomal genes. (iv) Frequency of transfer. (v) F' factors and merodiploids. (vi) Table. (vii)

Importance in genetics and in the spread of antibiotic resistance.

Lecture 30
Genetic Recombination by Transduction - Generalized and Specialized

Genetic Recombination by Transduction - Generalized and Specialized

30.1 Transduction

Transduction is the transfer of genetic material from one bacterium to another **by a bacteriophage**. It was discovered by Norton Zinder and Joshua Lederberg (1952) in *Salmonella*. In transduction, the phage acts as a carrier of bacterial DNA and introduces it into a recipient cell, where it may recombine with the chromosome.

30.2 Generalized transduction

- In generalized transduction, **any** gene of the donor chromosome can be transferred.
- It occurs during the **lytic cycle**. When new phage particles are assembled, a small piece of the host (bacterial) DNA may be packed into the phage head instead of phage DNA. This is called a **transducing particle**.
- Such a particle cannot reproduce (it has no phage DNA), but it can inject the bacterial DNA into a new host cell.
- The injected DNA recombines with the recipient chromosome by homologous recombination, producing a recombinant cell.
- Only a small fraction of the phage particles are transducing particles, so the frequency of transduction is low.
- Example: phage P1 in *E. coli* and phage P22 in *Salmonella*.

30.3 Specialized (restricted) transduction

- In specialized transduction, only **specific genes** located near the site where the prophage is integrated are transferred.
- It occurs when a **temperate phage** is induced to leave the chromosome (during the lysogenic cycle). Excision is sometimes inaccurate, and a piece of the adjacent bacterial DNA is carried away with the phage DNA.
- The resulting phage carries both phage genes and a few bacterial genes.
- When this phage infects a new cell, the bacterial genes are introduced. The transducing phage is usually defective (cannot complete the lytic cycle alone).
- Example: phage lambda in *E. coli* transfers the *gal* (galactose) and *bio* (biotin) genes, which lie next to the lambda attachment site.

30.4 Comparison of the three modes of recombination

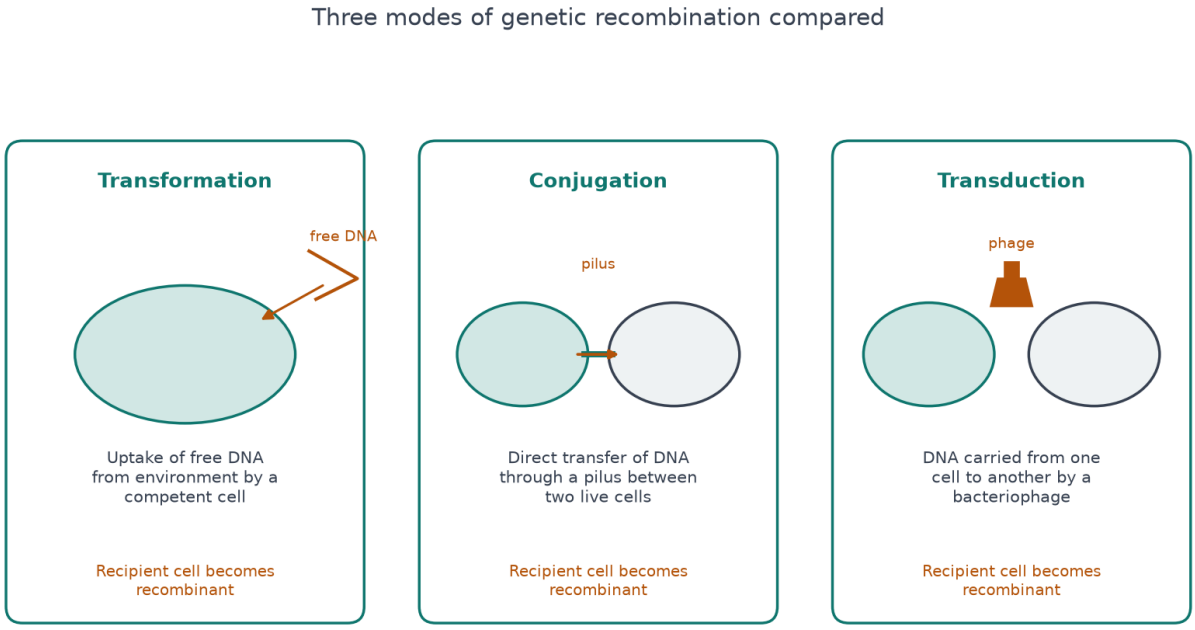


Figure 30.1 - Transformation, conjugation and transduction compared: uptake of free DNA, transfer through a pilus, and carriage by a bacteriophage.

Table 30.1 - Transformation vs conjugation vs transduction

Feature	Transformation	Conjugation	Transduction
Discovery	Griffith (1928)	Lederberg and Tatum (1946)	Zinder and Lederberg (1952)
Donor DNA source	Free DNA in environment	Donor cell	Bacteriophage
Contact required	No	Yes (sex pilus)	No
Competency needed	Yes	No	No (phage receptor needed)
DNA transferred	Any homologous fragment	F factor, or chromosome in Hfr	Any gene (generalized) or specific genes (specialized)
Recipient becomes	Recombinant	F+ or recombinant	Recombinant
Enzyme/protein carrier	None	Pilus	Phage coat

30.5 Significance

- Transduction is a natural means of gene transfer and contributes to genetic variation and evolution of bacteria.
- It is used in the laboratory for gene mapping and for transferring genes between bacteria.
- Transduction can transfer virulence genes and antibiotic-resistance genes, for example the toxin genes of *Corynebacterium diphtheriae* and *Vibrio cholerae*, which are carried by phages.

Exam focus

Key points to remember

- Transduction is the transfer of bacterial DNA by a bacteriophage.
- Generalized transduction transfers any gene and occurs during the lytic cycle (phage P1, P22).
- Specialized transduction transfers only specific genes and occurs during the lysogenic cycle (phage lambda).
- A transducing particle carries bacterial DNA instead of phage DNA and cannot reproduce.
- Transduction contributes to genetic variation and to the spread of virulence and resistance genes.

Short-answer questions with model answers

1. Define transduction and name its two types.

Transduction is the transfer of genetic material from one bacterium to another by a bacteriophage. The two types are generalized transduction, in which any gene can be transferred, and specialized transduction, in which only specific genes near the prophage site are transferred.

2. Distinguish generalized from specialized transduction.

Generalized transduction occurs during the lytic cycle and can transfer any gene; a piece of host DNA is accidentally packed into a phage head. Specialized transduction occurs when a temperate phage is induced, excises inaccurately and carries a specific adjacent bacterial gene with it.

3. How does a transducing particle differ from a normal phage?

A transducing particle contains bacterial DNA in place of phage DNA; it can inject this DNA into a new host but cannot reproduce to form new phage particles because it lacks phage genes.

Long-answer questions with model answer outlines

1. Explain genetic recombination by transduction with its two types.

Outline: (i) Definition and discovery. (ii) Role of the bacteriophage. (iii) Generalized transduction - mechanism, lytic cycle, P1/P22. (iv) Specialized transduction - mechanism, induction, lambda. (v) Transducing particle. (vi) Significance and examples.

2. Compare transformation, conjugation and transduction in bacteria.

Outline: (i) Introduction to genetic recombination. (ii) Transformation - competent cell, free DNA. (iii) Conjugation - pilus, F factor, Hfr. (iv) Transduction - phage-mediated. (v) Comparison table. (vi) Diagram. (vii) Significance in evolution, genetics and genetic engineering.

PART VI - GENETIC ENGINEERING, IMMUNOLOGY AND APPLICATIONS

Lecture 31
Microorganisms as Tools in Genetic Engineering

Microorganisms as Tools in Genetic Engineering

31.1 What is genetic engineering

Genetic engineering (recombinant DNA technology) is the deliberate modification of the genetic material of an organism by introducing, removing or altering specific genes. Microorganisms are the main tools of genetic engineering because they grow fast, are easy to culture, have simple genomes, and can be manipulated easily.

31.2 Tools of genetic engineering

31.2.1 Restriction enzymes (restriction endonucleases)

- Enzymes that cut DNA at specific base sequences called recognition sites.
- They were discovered in bacteria, where they protect the cell by cutting foreign (phage) DNA.
- They cut the DNA in a staggered manner, producing "sticky ends" that can pair with complementary ends of another DNA fragment.
- Examples: EcoRI (from *E. coli*), HindIII, BamHI, SmaI. They are named after the organism from which they are obtained.
- They are used to cut the gene of interest and the vector so that the two can be joined.

31.2.2 DNA ligase

- An enzyme that joins (seals) two DNA fragments by forming phosphodiester bonds between them.
- It is used to join the gene of interest to the vector, forming recombinant DNA.

31.2.3 Vectors (carriers)

- A vector is a DNA molecule that can carry a foreign gene into a host cell and replicate there.
- Common vectors: **plasmids** (pBR322, pUC19), **bacteriophages** (lambda), **cosmids**, **bacterial artificial chromosomes (BACs)**, **yeast artificial chromosomes (YACs)** and, for plants, the **Ti plasmid** of *Agrobacterium tumefaciens*.
- A good vector should be small, able to replicate in the host, carry a selectable marker (such as an antibiotic-resistance gene) and have unique restriction sites.

31.2.4 Host organisms

- The host is the cell in which the recombinant DNA is multiplied and expressed.
- *Escherichia coli* is the most widely used host because it grows rapidly and its genetics are well known.
- Other hosts: *Bacillus subtilis*, *Saccharomyces cerevisiae* (yeast), and plant cells via *Agrobacterium*.

31.3 Steps in genetic engineering

– **Isolation of the gene of interest** - the desired gene is identified and cut out with a restriction enzyme.

– **Cutting the vector** - the vector is cut with the same restriction enzyme, producing complementary sticky ends.

– **Joining (ligation)** - the gene and the vector are joined by DNA ligase to form recombinant DNA.

– **Introduction into the host** - the recombinant DNA is introduced into the host cell by transformation, transduction or electroporation.

– **Selection and screening** - cells that have taken up the recombinant DNA are identified using selectable markers.

– **Multiplication and expression** - the host cell multiplies and expresses the gene, producing the desired product.

– **Harvesting and purification** - the product is recovered and purified.

31.4 Products and applications

- **Human insulin** - produced by *E. coli* carrying the human insulin gene; the first genetically engineered medicine, approved in 1982.
- **Human growth hormone, interferon, erythropoietin, hepatitis B vaccine and clot-dissolving agents** - all produced using microorganisms.
- **Enzymes** - amylases, proteases, lipases, cellulases produced by engineered microbes for industry, detergents and food processing.
- **Biofertilizers and biocontrol agents** - improved strains of *Rhizobium*, *Azotobacter* and *Bacillus*.
- **Transgenic plants** - genes for insect resistance (*Bt* gene from *Bacillus thuringiensis*), herbicide tolerance, drought tolerance and disease resistance are introduced using *Agrobacterium*; Bt cotton is widely grown in India.
- **Bioremediation** - engineered bacteria that degrade pesticides, oil and other pollutants.
- **Gene therapy** - correction of genetic diseases by introducing normal genes.
- **Vaccines** - recombinant vaccines such as the hepatitis B vaccine.

31.5 Agricultural relevance

- Bt cotton and Bt brinjal carry the *Bt* gene from *Bacillus thuringiensis*, which makes the plant resistant to certain insect pests.
- *Agrobacterium tumefaciens* and its Ti plasmid are the main tools for transferring genes into crop plants.
- Genetically improved biofertilizer and biocontrol strains give better crop yields and disease control.
- Genetically engineered microbes are used to produce enzymes and vitamins for animal feed.

Table 31.1 - Tools of genetic engineering and their functions

Tool	Example	Function
Restriction enzyme	EcoRI	Cuts DNA at specific sites
DNA ligase	T4 ligase	Joins DNA fragments
Plasmid vector	pBR322	Carries the gene into the host
Phage vector	Lambda	Carries and delivers DNA
Ti plasmid	*Agrobacterium*	Transfers genes into plants
Host cell	*E. coli*	Multiplies and expresses the gene
Selectable marker	Antibiotic-resistance gene	Identifies transformed cells

Exam focus

Key points to remember

- Genetic engineering is the deliberate alteration of genetic material using recombinant DNA technology.
- Restriction enzymes cut DNA; DNA ligase joins DNA; vectors carry DNA into the host.
- *E. coli* is the most common host; the Ti plasmid is used for plant transformation.
- Steps: isolation, cutting, ligation, introduction, selection, expression, harvesting.
- Products include human insulin, growth hormone, vaccines and enzymes.
- Bt cotton is a transgenic crop produced with the *Bt* gene from *Bacillus thuringiensis*.

Short-answer questions with model answers

1. What are restriction enzymes? Give an example and their use.

Restriction enzymes (restriction endonucleases) are enzymes that cut DNA at specific base sequences. Example: EcoRI from *E. coli*. They are used to cut the gene of interest and the vector so that they can be joined to form recombinant DNA.

2. What is a vector? Give two examples.

A vector is a DNA molecule that can carry a foreign gene into a host cell and replicate there. Examples: the plasmid pBR322 and the bacteriophage lambda. The Ti plasmid of *Agrobacterium tumefaciens* is a vector for plants.

3. List the steps of genetic engineering.

Isolation of the gene of interest, cutting the vector with a restriction enzyme, joining the two with DNA ligase, introducing the recombinant DNA into the host, selection and screening of transformed cells, multiplication and expression of the gene, and harvesting and purification of the product.

Long-answer questions with model answer outlines

1. Describe the tools and steps of genetic engineering.

Outline: (i) Meaning of genetic engineering. (ii) Restriction enzymes. (iii) DNA ligase. (iv) Vectors - plasmid, phage, Ti plasmid. (v) Host organisms. (vi) Steps of the process. (vii) Products and applications.

2. Discuss the applications of genetic engineering in agriculture.

Outline: (i) Introduction. (ii) Transgenic crops - Bt cotton, Bt brinjal, herbicide tolerance. (iii) *Agrobacterium* and the Ti plasmid. (iv) Improved biofertilizers and biocontrol agents. (v) Engineered enzymes and feed additives. (vi) Bioremediation. (vii) Issues and biosafety.

Lecture 32
Immunology - Principles - Specific and Non-Specific Defence

Immunology - Principles - Specific and Non-Specific Defence

32.1 Introduction to immunology

Immunology is the study of the defence mechanisms of the body against infection. **Immunity** is the ability of the body to resist infection and to neutralise foreign substances. The substances that trigger an immune response are called **antigens**; the proteins produced by the body in response are called **antibodies**.

32.2 Non-specific (innate) defence

Non-specific defence mechanisms act against all invaders in the same way and are present from birth. They are the first line of defence.

32.2.1 First line of defence - barriers

- **Skin** - an intact, dry, slightly acidic barrier that prevents the entry of most microbes.
- **Mucous membranes** - line the respiratory, digestive and reproductive tracts; mucus traps microbes, and cilia move them out.
- **Secretions** - tears and saliva contain **lysozyme**, which destroys bacterial cell walls; stomach acid kills many microbes; sweat and sebaceous secretions contain antimicrobial substances.
- **Normal (resident) microflora** - harmless bacteria on the skin and in the gut compete with pathogens for nutrients and space.

32.2.2 Second line of defence - internal defences

- **Phagocytosis** - cells such as neutrophils and macrophages engulf and destroy microbes.
- **Inflammation** - redness, heat, swelling and pain; increases blood flow, brings phagocytes to the site and limits the spread of infection.
- **Fever** - raised body temperature inhibits the growth of many pathogens and speeds up the immune response.
- **Complement system** - a group of plasma proteins that lyse microbes, enhance phagocytosis and promote inflammation.
- **Interferons** - proteins produced by virus-infected cells that make neighbouring cells resistant to viral infection.
- **Natural killer (NK) cells** - destroy virus-infected cells and tumour cells.

32.3 Specific (acquired) defence

Specific defence is directed against a particular antigen and is acquired during life. It has two main features - **specificity** (it acts only against the antigen that triggered it) and **memory** (a second exposure produces a faster and stronger response).

32.3.1 Humoral immunity (antibody-mediated)

- Mediated by **B lymphocytes (B cells)**.
- When a B cell meets its specific antigen, it is activated, multiplies (clonal selection) and differentiates into **plasma cells**, which secrete antibodies, and **memory cells**, which remain for a long time.
- Antibodies circulate in the blood and lymph and neutralise antigens, bacteria and viruses.
- This is the basis of most **vaccines**.

32.3.2 Cell-mediated immunity

- Mediated by **T lymphocytes (T cells)**.
- **Helper T cells (CD4)** activate B cells and macrophages and coordinate the immune response.
- **Cytotoxic T cells (CD8)** destroy infected cells, cancer cells and foreign tissue cells.
- **Suppressor (regulatory) T cells** switch off the immune response when the infection is over.
- Cell-mediated immunity is important against viruses, fungi and intracellular bacteria, and in the rejection of transplants.

32.3.3 Active and passive immunity

- **Active immunity** - the body produces its own antibodies after exposure to an antigen (natural infection or vaccination). It develops slowly but lasts a long time.
- **Passive immunity** - antibodies are given from outside (from the mother to the foetus through the placenta or in colostrum, or as an antiserum). It acts quickly but lasts only a short time.

32.4 Agricultural and veterinary relevance

- Vaccination of cattle and poultry protects them against anthrax, foot-and-mouth disease, brucellosis, fowl pox, Newcastle disease and other diseases.
- Colostrum provides passive immunity to newborn calves and piglets; colostrum feeding within the first hours of life is essential.
- In plants, there is no true immune system, but plants defend themselves by physical barriers, chemical compounds and the hypersensitive response; resistance breeding uses these mechanisms.

Table 32.1 - Specific vs non-specific defence

Feature	Non-specific (innate)	Specific (acquired)
Presence	Present from birth	Acquired during life
Action	Against all invaders	Against a specific antigen
Speed	Immediate	Slower, develops after exposure
Memory	No memory	Memory present
Cells involved	Phagocytes, NK cells	B and T lymphocytes
Examples	Skin, mucus, lysozyme, inflammation, fever	Antibody production, cytotoxic T cells
Basis of	First line of defence	Vaccination

Exam focus

Key points to remember

- Immunity is the ability to resist infection; antigens trigger it and antibodies are produced in response.
- Non-specific defence is present from birth and acts against all invaders - barriers, phagocytosis, inflammation, fever, complement, interferons.
- Specific defence is acquired, specific and shows memory; it is mediated by B and T lymphocytes.
- Humoral immunity is mediated by B cells and antibodies; cell-mediated immunity by T cells.
- Active immunity is produced by the body and lasts long; passive immunity is given from outside and lasts short.
- Vaccination is based on active immunity.

Short-answer questions with model answers

1. Distinguish specific from non-specific defence.

Non-specific defence is present from birth, acts against all invaders in the same way and has no memory (for example, skin, phagocytosis, inflammation). Specific defence is acquired during life, acts only against a particular antigen and shows memory; it is mediated by B and T lymphocytes.

2. What is the difference between active and passive immunity?

In active immunity the body produces its own antibodies after exposure to an antigen, either by natural infection or by vaccination; it develops slowly but lasts long. In passive immunity antibodies are given from outside, as from the mother through the placenta or in colostrum, or as an antiserum; it acts quickly but lasts only a short time.

3. Name the cells of the specific immune system and their functions.

B lymphocytes produce plasma cells that secrete antibodies and memory cells that give long-lasting protection. T lymphocytes include helper T cells (activate B cells and macrophages), cytotoxic T cells (destroy infected cells) and suppressor T cells (switch off the response).

Long-answer questions with model answer outlines

1. Describe the non-specific defence mechanisms of the body.

Outline: (i) Meaning of non-specific defence. (ii) First line - skin, mucous membranes, secretions, normal flora. (iii) Second line - phagocytosis, inflammation, fever, complement, interferons, NK cells. (iv) Their role in preventing infection. (v) Comparison with specific defence.

2. Explain specific immunity and the basis of vaccination.

Outline: (i) Meaning of specific immunity. (ii) Humoral immunity - B cells, plasma cells, antibodies, memory. (iii) Cell-mediated immunity - helper, cytotoxic and suppressor T cells. (iv) Primary and secondary responses. (v) Active and passive immunity. (vi) Basis of vaccination and its veterinary importance.

Lecture 33
Antigen-Antibody Reactions - Vaccines - Applications

Antigen-Antibody Reactions - Vaccines - Applications

33.1 Antigens and antibodies

- An **antigen** is any substance (usually a foreign protein or polysaccharide) that, when introduced into the body, stimulates the production of antibodies and reacts specifically with them. The part of the antigen that is recognised is called the **epitope** or antigenic determinant.
- An **antibody (immunoglobulin)** is a protein produced by plasma cells in response to an antigen and which binds specifically to it. Antibodies are of five classes - IgG, IgM, IgA, IgE and IgD.
- An antibody molecule has a Y shape and consists of two heavy and two light polypeptide chains. It has a **variable region** (which forms the antigen-binding site and is specific for the antigen) and a **constant region** (which determines the class and biological function).

33.2 Antigen-antibody reaction

The antigen-antibody reaction is a highly specific **lock-and-key** binding between the antigen-binding site of the antibody and the epitope of the antigen. The binding is by weak forces (hydrogen bonds, ionic bonds, hydrophobic and van der Waals forces), but it is strong because many such bonds form at the same time.

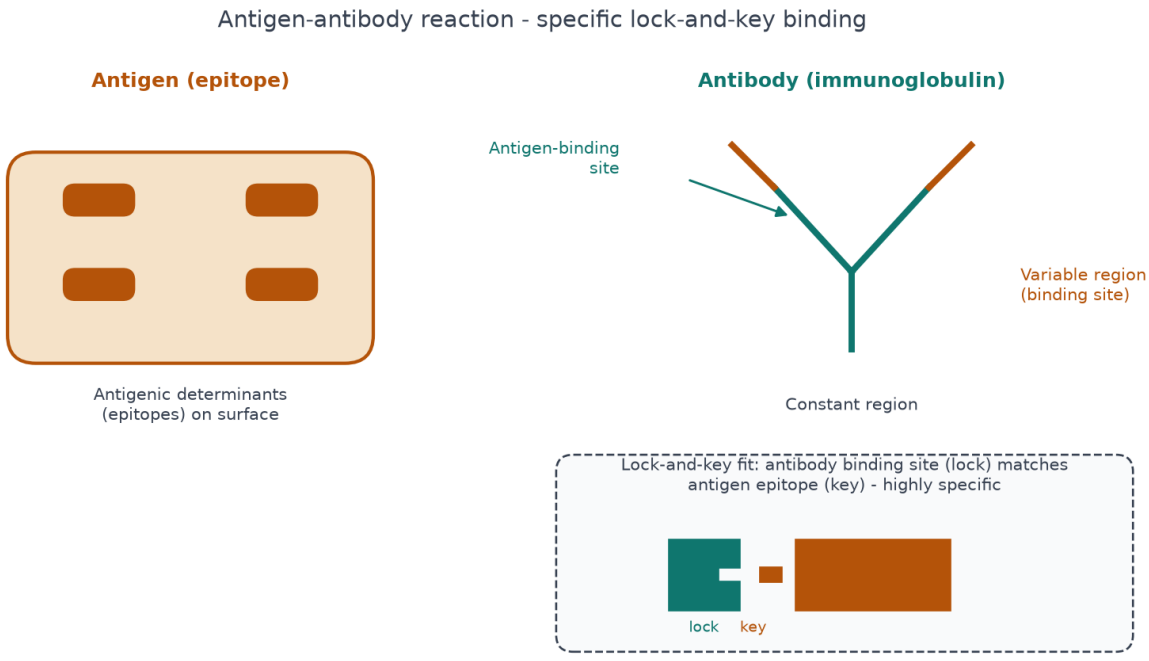


Figure 33.1 - Antigen-antibody reaction: the lock-and-key binding between the antibody binding site and the antigen epitope, with a labelled antibody showing variable and constant regions.

- Consequences of the reaction: neutralisation of toxins and viruses, agglutination of bacteria, precipitation of soluble antigens, activation of complement, and enhanced phagocytosis (opsonisation).

- **Agglutination** - antibodies clump bacteria or cells together; the basis of the Widal test for typhoid and blood grouping.
- **Precipitation** - soluble antigen and antibody form a visible precipitate in the correct proportions.
- **Neutralisation** - antibodies block toxins and viruses.

33.3 Types of antigen-antibody reactions used in diagnosis

- **Agglutination tests** - Widal test (typhoid), blood grouping, bacterial agglutination.
- **Precipitation tests** - ring test, gel diffusion (Ouchterlony).
- **Complement fixation test** - used in the diagnosis of many infections.
- **ELISA** (enzyme-linked immunosorbent assay) - widely used to detect antigens or antibodies, including plant viruses.
- **Immunofluorescence** - fluorescent antibodies are used to identify pathogens in tissues.
- **Radioimmunoassay (RIA)** and **Western blotting** - highly sensitive techniques.

33.4 Vaccines

A **vaccine** is a preparation of an antigen that, when administered, stimulates the body to produce antibodies and memory cells, giving active immunity against a disease without causing the disease itself.

- **Live attenuated vaccines** - contain weakened but living organisms (for example, BCG for tuberculosis, oral polio vaccine, MMR, smallpox).
- **Killed (inactivated) vaccines** - contain killed organisms (for example, Salk polio vaccine, whole-cell pertussis, rabies, cholera).
- **Toxoids** - inactivated toxins (for example, tetanus and diphtheria toxoids).
- **Subunit vaccines** - contain only part of the organism (for example, hepatitis B surface antigen, acellular pertussis).
- **Recombinant vaccines** - produced by genetic engineering (for example, hepatitis B vaccine).
- **Conjugate vaccines** - polysaccharide antigen attached to a carrier protein (for example, *Haemophilus influenzae* type b vaccine).
- **DNA vaccines** - plasmid DNA carrying a gene for an antigen; still largely experimental.

33.5 Applications

- **Control of human diseases** - smallpox has been eradicated; polio, measles, tetanus and diphtheria are greatly reduced.
- **Veterinary vaccines** - against anthrax, foot-and-mouth disease, brucellosis, rabies, Newcastle disease, fowl pox and mastitis, protecting livestock and poultry.
- **Diagnosis** - antigen-antibody tests are used to diagnose disease in humans, animals and plants.
- **Detection of plant viruses** - ELISA and immunofluorescence are routine in seed testing and plant quarantine.
- **Blood grouping and tissue typing** - based on antigen-antibody reactions.

- **Pregnancy testing and hormone assays** - use antibodies.

Table 33.1 - Major vaccines and their types

Vaccine	Type	Disease prevented
BCG	Live attenuated	Tuberculosis
Oral polio (Sabin)	Live attenuated	Poliomyelitis
MMR	Live attenuated	Measles, mumps, rubella
Salk polio	Killed	Poliomyelitis
DPT	Toxoid + killed	Diphtheria, pertussis, tetanus
Hepatitis B	Recombinant subunit	Hepatitis B
Anthrax (veterinary)	Live attenuated (spore)	Anthrax in animals
FMD vaccine	Killed	Foot-and-mouth disease in cattle

Exam focus

Key points to remember

- An antigen stimulates antibody production; the antibody binds to the antigen in a lock-and-key manner.
- An antibody is a Y-shaped protein with variable and constant regions; classes are IgG, IgM, IgA, IgE, IgD.
- The reaction is highly specific and results in neutralisation, agglutination, precipitation or complement activation.
- ELISA and immunofluorescence are important diagnostic tests.
- Vaccines give active immunity; they may be live attenuated, killed, toxoid, subunit, recombinant or conjugate.
- Vaccination is used in human, veterinary and plant disease management.

Short-answer questions with model answers

1. Define antigen and antibody. Describe the antigen-antibody reaction.

An antigen is a foreign substance that stimulates the production of antibodies and reacts with them; an antibody is a protein produced by plasma cells that binds specifically to an antigen. The reaction is a highly specific lock-and-key binding between the antigen-binding site of the antibody and the epitope of the antigen, held by weak forces. It results in neutralisation, agglutination, precipitation or complement activation.

2. What is a vaccine? Name the types.

A vaccine is a preparation of antigen that stimulates active immunity without causing the disease. Types: live attenuated, killed, toxoid, subunit, recombinant, conjugate and DNA vaccines.

3. Give two applications of antigen-antibody reactions in agriculture.

ELISA is used to detect plant viruses in seed testing and quarantine; immunofluorescence helps identify plant pathogens such as *Xanthomonas*. Veterinary vaccines based on antigens protect livestock against diseases such as anthrax and foot-and-mouth disease.

Long-answer questions with model answer outlines

1. Explain antigen-antibody reactions and their diagnostic applications.

Outline: (i) Antigens and epitopes. (ii) Antibody structure and classes. (iii) Lock-and-key reaction and forces involved. (iv) Consequences - neutralisation, agglutination, precipitation, complement activation. (v) Diagnostic tests - Widal, ELISA, immunofluorescence. (vi) Diagram and applications.

2. Describe vaccines and their applications in agriculture and public health.

Outline: (i) Meaning and principle of vaccination. (ii) Types of vaccines with examples. (iii) Mechanism of active immunity and memory. (iv) Human vaccines and eradication of smallpox. (v) Veterinary vaccines. (vi) Plant virus detection. (vii) Conclusion.

BACK MATTER - GLOSSARY AND REFERENCES

Glossary of Important Terms

Glossary

- **Abiogenesis** - the theory that living organisms arise from non-living matter; spontaneous generation.
- **Aerobic respiration** - oxidation of a substrate with oxygen as the final electron acceptor.
- **Agar** - a polysaccharide from seaweed used to solidify culture media.
- **Agglutination** - clumping of cells or bacteria by antibodies.
- **Antibiotic** - a substance produced by a microorganism that inhibits or kills other microorganisms.
- **Antibody** - a protein produced by plasma cells that binds specifically to an antigen.
- **Antigen** - a foreign substance that stimulates the production of antibodies and reacts with them.
- **Antiseptic** - a chemical applied to living tissue to destroy or inhibit microbes.
- **Autoclave** - equipment that sterilises with steam under pressure (121 C, 15 psi, 15 min).
- **Axenic culture** - a pure culture containing only one organism.
- **Bacteriophage** - a virus that infects bacteria.
- **Biofertilizer** - a preparation containing living microorganisms that improves soil fertility.
- **Biogenesis** - the theory that living organisms arise only from pre-existing living organisms.
- **Capsid** - the protein coat of a virus.
- **Capsule** - a gelatinous layer outside the bacterial cell wall.
- **Chemostat** - a continuous culture device in which growth rate is controlled by a limiting nutrient.
- **Chemotaxis** - movement of a cell towards or away from a chemical.
- **Clone** - a population of genetically identical cells derived from one parent.
- **Colony** - a visible mass of cells arising from one cell on a solid medium.
- **Competency** - the ability of a bacterial cell to take up free DNA.
- **Conjugation** - transfer of DNA between bacteria through direct cell contact.
- **Diauxic growth** - two growth phases separated by a lag when two sugars are used sequentially.
- **Endospore** - a highly resistant dormant structure formed by *Bacillus* and *Clostridium*.
- **Enrichment culture** - a culture designed to favour the growth of a particular microorganism.
- **Epitope** - the part of an antigen recognised by an antibody.
- **Fermentation** - energy generation in which an organic compound is the final electron acceptor.
- **Flagellum** - a long appendage used for motility.
- **Generation time** - the time taken for a population to double.
- **Gram stain** - a differential stain that divides bacteria into Gram-positive and Gram-negative.
- **Hfr strain** - a bacterium with the F factor integrated into the chromosome; a high-frequency donor.

- **Immunity** - the ability of the body to resist infection.
- **Lyophilisation** - preservation by freeze-drying.
- **Lysogeny** - the state in which phage DNA is integrated into the host chromosome as a prophage.
- **Mesophile** - an organism with an optimum growth temperature of 25-40 C.
- **Mutagen** - an agent that increases the rate of mutation.
- **Mutation** - a permanent, heritable change in the base sequence of DNA.
- **Mycelium** - a mass of fungal hyphae.
- **Nitrification** - the conversion of ammonia to nitrite and then to nitrate.
- **Nitrogen fixation** - conversion of atmospheric nitrogen to ammonia.
- **Nucleoid** - the region of a prokaryotic cell containing the chromosome.
- **Pasteurisation** - heating to 63 C for 30 min or 72 C for 15 s to kill pathogens.
- **Peptidoglycan** - the polymer of the bacterial cell wall.
- **Plasmid** - a small, circular, extrachromosomal DNA molecule.
- **Prophage** - phage DNA integrated into the bacterial chromosome.
- **Pure culture** - a culture containing only one type of organism.
- **Recombination** - the production of a new combination of genes.
- **Resolving power** - the smallest distance at which two points can be seen as separate.
- **Sterilisation** - the destruction of all microbial life including spores.
- **Transduction** - transfer of bacterial DNA by a bacteriophage.
- **Transformation** - uptake of free DNA by a competent cell.
- **Transposon** - a DNA segment that can move within the genome.
- **Turbidostat** - a continuous culture device in which cell density is kept constant.
- **Vaccine** - a preparation of antigen that gives active immunity without causing disease.
- **Vector** - a DNA molecule that carries a foreign gene into a host cell.
- **Viable count** - the number of living cells capable of forming colonies.

Reference List and Further Reading

Standard textbooks

- Pelczar, M. J., Chan, E. C. S. and Krieg, N. R. *Microbiology* (5th edition). McGraw-Hill, New Delhi. - A comprehensive introductory text covering all branches of microbiology.
- Prescott, L. M., Harley, J. P. and Klein, D. A. *Microbiology* (latest edition). McGraw-Hill. - Strong on microbial physiology, genetics and immunology.
- Tortora, G. J., Funke, B. R. and Case, C. L. *Microbiology - An Introduction* (latest edition). Pearson. - Clear explanations with excellent diagrams and applications.
- Dubey, R. C. and Maheshwari, D. K. *A Textbook of Microbiology*. S. Chand, New Delhi. - Widely used in Indian agricultural universities; covers soil, agricultural and industrial microbiology.
- Salle, A. J. *Fundamental Principles of Bacteriology*. Tata McGraw-Hill. - A classical reference on staining, sterilisation and culture methods.
- Frobisher, M. *Fundamentals of Microbiology*. Saunders. - A classical text on general microbiology.

Agricultural and soil microbiology

- Subba Rao, N. S. *Soil Microorganisms and Plant Growth*. Oxford and IBH, New Delhi.
- Alexander, M. *Introduction to Soil Microbiology*. Wiley Eastern, New Delhi.
- Rangaswami, G. and Bagyaraj, D. J. *Agricultural Microbiology*. PHI Learning, New Delhi. - Especially useful for biofertilizers, plant pathogens and soil processes.
- Mehrotra, R. S. *Plant Pathology*. Tata McGraw-Hill, New Delhi. - For fungal and bacterial plant diseases.

Genetics, molecular biology and immunology

- Freifelder, D. *Molecular Biology* and *Microbial Genetics*. Narosa, New Delhi.
- Watson, J. D. and others. *Molecular Biology of the Gene*. Pearson.
- Kuby, J. *Immunology*. W. H. Freeman.
- Bergey's Manual of Determinative Bacteriology and Bergey's Manual of Systematic Bacteriology. Williams and Wilkins / Springer. - The standard reference for bacterial classification.

Course material

- ICAR e-Course material on Fundamentals of Microbiology, Agricultural Microbiology and Soil Microbiology (Indian Council of Agricultural Research, New Delhi).
- TNAU e-Course material and lecture notes, Tamil Nadu Agricultural University, Coimbatore.
- NCERT and ICAR undergraduate agriculture textbooks on general microbiology.

How to use these notes

- Read each lecture with the diagram and the table in front of you; the table summarises the differences that are usually asked in examinations.
- Practise every worked example by covering the solution and doing the calculation again.
- Use the Exam-focus section of each lecture as a checklist before the examination; if any key point is unclear, go back to the relevant heading.
- The glossary at the end is useful for quick revision of definitions, and the references above are for deeper reading of any topic.